

Grade 8 Mathematics — Complete Curriculum

Jamaican National Standards Curriculum (NSC / APSE I)

Terms 1–3 • 38 lessons • Complete practice and examination preparation

Terms 1–3 Curriculum span	38 Complete lessons	570 Practice questions	190 Exam-style questions
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Simple explanations • Fully worked examples • Scaffolded practice • Complete answer keys

Student message: Take one small step at a time. Read, copy, practise, check—and try again. You can do this!

Designed for learners who benefit from slower pacing, repetition, explicit steps and frequent checks.

How to Use This Curriculum

Each lesson is a self-contained learning packet. It may be taught in one extended block or divided into two to four shorter sessions. Mastery matters more than speed.

- Start with the three-question prerequisite check.
- Read the Main Idea and copy the notebook visual before attempting examples.
- Follow every worked example one line at a time; say why each step is used.
- Complete Guided Practice and check immediately.
- Attempt Independent and Exam-Style questions before opening their answer keys.
- Use the one-page cheat sheet for revision, then retry missed questions without looking.

For a supporting adult: If an answer is wrong, locate the first incorrect step. Ask the learner to explain that step aloud, model one similar item, and then let the learner retry.

What Is Included in Every Lesson

3+	5	10	5
worked examples	guided questions	independent questions	exam-style questions

Alignment Note

This independently developed study resource covers the Grade 8 objectives in the curriculum outline and uses the current APSE I term/unit sequence as a cross-check. School pacing and lesson order may vary. Teachers should follow current Ministry and school guidance.

Complete Three-Term Curriculum Map

The table below shows the lesson sequence and its direct curriculum objective coverage.

No.	Unit / strand	Lesson	NSC objective coverage
T1.1	Unit 1: Numbers Numbers	Indices, Powers and Evaluating Expressions	State the meaning of a^m , where a and m are whole numbers. • Evaluate a^m , $a^m \times b^n$ and $a^m \div b^n$ where the letters represent whole numbers.
T1.2	Unit 1: Numbers Numbers	Writing Large Numbers in Standard Form	Write numbers greater than or equal to 10 in standard form.
T1.3	Unit 1: Numbers Numbers	Decimal Places and Significant Figures	Write a number to a given number of decimal places and significant figures.
T1.4	Unit 1: Numbers Numbers	Place Value in Bases Other Than Ten	Express place values of digits in all bases including base 10.
T1.5	Unit 1: Numbers Numbers	Converting Between Base Ten and Other Bases	Convert numbers written in base n , where $1 < n < 10$, to base 10 and vice versa.
T1.6	Unit 1: Numbers Numbers	Adding, Subtracting and Multiplying in Other Bases	Add, subtract and multiply numbers written in base n , where $1 < n < 10$.
T1.7	Unit 2: Numbers Numbers	Closure, Reflexive, Symmetric, Transitive and Trichotomy Properties	Identify and use closure, reflexive, symmetry, transitive and trichotomy concepts.
T1.8	Unit 3: Algebra Algebra	Changing the Subject of a Formula	Change the subject of a simple formula.
T1.9	Unit 3: Algebra Algebra	Writing, Solving and Graphing Linear Inequalities	Write inequalities for word problems. • Illustrate inequalities on a number line. • Solve simple linear inequalities and represent the solution on a number line.
T1.10	Unit 4: Geometry Geometry	Angles Formed by Transversals and Intersecting Lines	Investigate relationships among angles formed by a transversal of parallel lines and by intersecting non-parallel lines.
T1.11	Unit 4: Geometry Geometry	Top, Front and Side Views of Solids	Sketch different views of solids and use dotted lines for unseen edges.
T1.12	Unit 4: Geometry Geometry	Translations, Reflections and Rotations on the Coordinate Plane	Perform translations and find translation vectors. • Reflect objects in the x - or y -axis. • Rotate objects 90° , 180° and 270° about the origin.
T2.1	Unit 1: Geometry Geometry	Properties and Classification of Polygons	Determine the properties of n -sided polygons, where $3 \leq n \leq 10$.
T2.2	Unit 1: Geometry Geometry	Interior and Exterior Angles of Polygons	Find the sum of the interior angles of polygons with 3 to 10 sides.

No.	Unit / strand	Lesson	NSC objective coverage
T2.3	Unit 2: Statistics Statistics	Mean, Median and Mode from Frequency Tables	Determine the mode, median and mean from a frequency table.
T2.4	Unit 2: Statistics Statistics	Using Mean, Median and Mode to Interpret Information	Use the mode, median and mean to interpret information.
T2.5	Unit 2: Statistics Statistics	Pictographs, Bar Charts, Pie Charts and Line Graphs	Read, interpret and construct pictographs, bar charts, pie charts and line graphs.
T2.6	Unit 3: Numbers Numbers	Ratio, Proportion and Sharing	Solve simple problems involving ratio and proportion.
T2.7	Unit 4: Numbers Consumer Arithmetic	Bank Accounts, Deposit and Withdrawal Forms, and Cheques	Identify bank-account types and their features. • Complete withdrawal and deposit forms. • Complete information accurately on a cheque leaf.
T2.8	Unit 4: Numbers Consumer Arithmetic	Simple Interest on Loans and Deposits	Use proportions involving principal, rate and time to develop simple interest. • Calculate simple interest on loans and deposits.
T2.9	Unit 4: Numbers Consumer Arithmetic	Compound Interest by the Recursive Method	Calculate compound interest using a calculator and a recursive method.
T2.10	Unit 4: Numbers Consumer Arithmetic	Hire Purchase Cost and Comparison	Calculate total cost in a hire-purchase agreement. • Compare hire-purchase price and cash price.
T2.11	Unit 4: Numbers Consumer Arithmetic	Discounts and Taxes from Given Instructions	Calculate discounts and taxes from given instructions.
T2.12	Unit 5: Measurement Measurement	Areas of Parallelograms, Triangles and Trapezia	Derive and use formulae for the area of parallelograms, triangles and trapezia.
T2.13	Unit 5: Measurement Measurement	Total Surface Area of Prisms and Cylinders	Compute total surface area of cubes, cuboids, cylinders and triangular prisms.
T2.14	Unit 5: Measurement Measurement	Converting Square and Cubic Units	Perform conversions within units up to cubed units.
T2.15	Unit 5: Measurement Measurement	Volume and Capacity of Prisms, Cylinders and Composite Objects	Establish volume formulae. • Estimate and calculate volume and capacity of cubes, cuboids, prisms, cylinders and composite objects.
T3.1	Unit 1: Algebra Relations, Functions and Graphs	Mappings, Ordered Pairs and Coordinate Graphs	Plot the ordered pairs of a mapping as a graph.
T3.2	Unit 1: Algebra Relations, Functions and Graphs	Tables and Straight-Line Graphs	Draw straight-line graphs by plotting points and by using gradient and intercept.
T3.3	Unit 1: Algebra Relations, Functions and Graphs	Gradient and Intercepts of Straight Lines	Determine gradients and intercepts of straight-line graphs.

No.	Unit / strand	Lesson	NSC objective coverage
T3.4	Unit 1: Algebra Relations, Functions and Graphs	Gradient as a Rate of Change	Relate the gradient of a graph to the rate of change of quantities.
T3.5	Unit 1: Algebra Relations, Functions and Graphs	Finding the Equation of a Straight Line	Find the equation of any straight line.
T3.6	Unit 1: Algebra Relations, Functions and Graphs	Solving Two Linear Equations by Graphing	Plot two linear equations on the same axes and interpret their point or points of intersection, if any.
T3.7	Unit 1: Algebra Relations, Functions and Graphs	Graphing Linear Inequalities on the Coordinate Plane	Graph linear inequalities on the coordinate plane and identify regions.
T3.8	Unit 2: Geometry Constructions	Geometric Constructions with Ruler and Compasses	Construct angle bisectors, angles of 90° , 45° , 60° and 30° , and triangles using appropriate geometric instruments.
T3.9	Unit 3: Numbers Sets	Sets, Subsets and the Number of Subsets	Determine the number of subsets of a given set. • List every possible subset when the original set has no more than four elements.
T3.10	Unit 3: Numbers Sets	Two-Set Venn Diagrams and Problem Solving	Solve simple problems involving at most two subsets of the universal set.
T3.11	Unit 3: Numbers Logic	Propositions, Logical Connectives and Truth Tables	Understand propositions and use negation, conjunction, disjunction, conditional and equivalence. • Differentiate simple and compound propositions. • Express propositions symbolically and in words.

TERM 1

Numbers, Algebra and Geometry Foundations

12 complete lessons

Term goal: Build confidence through clear models, repeated practice and cumulative exam preparation.

Term 1 Lesson Map

The table below shows the lesson sequence and its direct curriculum objective coverage.

No.	Unit / strand	Lesson	NSC objective coverage
T1.1	Unit 1: Numbers Numbers	Indices, Powers and Evaluating Expressions	State the meaning of a^m , where a and m are whole numbers. • Evaluate a^m , $a^m \times b^n$ and $a^m \div b^n$ where the letters represent whole numbers.
T1.2	Unit 1: Numbers Numbers	Writing Large Numbers in Standard Form	Write numbers greater than or equal to 10 in standard form.
T1.3	Unit 1: Numbers Numbers	Decimal Places and Significant Figures	Write a number to a given number of decimal places and significant figures.
T1.4	Unit 1: Numbers Numbers	Place Value in Bases Other Than Ten	Express place values of digits in all bases including base 10.
T1.5	Unit 1: Numbers Numbers	Converting Between Base Ten and Other Bases	Convert numbers written in base n , where $1 < n < 10$, to base 10 and vice versa.
T1.6	Unit 1: Numbers Numbers	Adding, Subtracting and Multiplying in Other Bases	Add, subtract and multiply numbers written in base n , where $1 < n < 10$.
T1.7	Unit 2: Numbers Numbers	Closure, Reflexive, Symmetric, Transitive and Trichotomy Properties	Identify and use closure, reflexive, symmetry, transitive and trichotomy concepts.
T1.8	Unit 3: Algebra Algebra	Changing the Subject of a Formula	Change the subject of a simple formula.
T1.9	Unit 3: Algebra Algebra	Writing, Solving and Graphing Linear Inequalities	Write inequalities for word problems. • Illustrate inequalities on a number line. • Solve simple linear inequalities and represent the solution on a number line.
T1.10	Unit 4: Geometry Geometry	Angles Formed by Transversals and Intersecting Lines	Investigate relationships among angles formed by a transversal of parallel lines and by intersecting non-parallel lines.
T1.11	Unit 4: Geometry Geometry	Top, Front and Side Views of Solids	Sketch different views of solids and use dotted lines for unseen edges.
T1.12	Unit 4: Geometry Geometry	Translations, Reflections and Rotations on the Coordinate Plane	Perform translations and find translation vectors. • Reflect objects in the x - or y -axis. • Rotate objects 90° , 180° and 270° about the origin.

TERM 1 • LESSON 1

Indices, Powers and Evaluating Expressions

Unit 1: Numbers | Numbers

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 1: Numbers
Strand	Numbers
Lesson	1: Indices, Powers and Evaluating Expressions
NSC objective(s)	State the meaning of a^m , where a and m are whole numbers. • Evaluate a^m , $a^m \times b^n$ and $a^m \div b^n$ where the letters represent whole numbers.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Index notation is a short way to show repeated multiplication. It is used in science, computing, measurement and later algebra.

Learning Objectives — I Can

- I can identify the base and exponent in a power.
- I can expand a power as repeated multiplication.
- I can evaluate expressions containing powers.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Base	The number that is multiplied repeatedly.
Exponent / index	The small raised number that tells how many equal factors are used.
Power	An expression made from a base and an exponent, such as 3^4 .
Factor	A number being multiplied.
Evaluate	Work out the numerical value.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

In a^m , a is the base and m is the exponent. The exponent counts factors, not the number of multiplication signs.

For example, 4^3 means $4 \times 4 \times 4$. It has three factors of 4 and two multiplication signs.

When different bases appear, evaluate each power first, then complete the multiplication or division.

Small Steps to Follow

1. Circle each base and underline each exponent.
2. Expand each power using the correct number of equal factors.
3. Calculate each power.
4. Complete any multiplication or division that remains.
5. Estimate or reverse the operation to check the answer.

Notebook visual: Draw two boxes. In the first, write BASE = big number. In the second, write EXPONENT = small raised number. Draw an arrow from 3^4 to four factors: $3 \times 3 \times 3 \times 3$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Evaluate 3^4 .

1. Write four factors of 3: $3 \times 3 \times 3 \times 3$.
2. $3 \times 3 = 9$ and $3 \times 3 = 9$.
3. $9 \times 9 = 81$.

Answer: $3^4 = 81$.

Worked Example 2

Problem: Evaluate $2^3 \times 5^2$.

1. $2^3 = 2 \times 2 \times 2 = 8$.
2. $5^2 = 5 \times 5 = 25$.
3. Multiply the two values: $8 \times 25 = 200$.

Answer: $2^3 \times 5^2 = 200$.

Worked Example 3

Problem: Evaluate $6^3 \div 3^2$.

1. $6^3 = 6 \times 6 \times 6 = 216$.
2. $3^2 = 3 \times 3 = 9$.
3. Divide: $216 \div 9 = 24$.

Answer: $6^3 \div 3^2 = 24$.

Common Mistakes and Fixes

Common mistake	How to fix it
Multiplying the base by the exponent	Remember: 5^3 is $5 \times 5 \times 5$, not 5×3 .
Counting multiplication signs	The exponent counts factors. Four factors have only three multiplication signs.
Doing the outside operation first	Evaluate every power before multiplying or dividing the results.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Expand 5^3 .

Check: $5^3 = 5 \times 5 \times 5$.

2. Evaluate 2^5 .

Check: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.

3. Evaluate $4^2 \times 3^2$.

Check: $4^2 \times 3^2 = 16 \times 9 = 144$.

4. Evaluate $10^3 \div 5^2$.

Check: $10^3 \div 5^2 = 1,000 \div 25 = 40$.

5. In 7^4 , state the base and exponent.

Check: Base = 7; exponent = 4.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Which is the correct expansion of 6^4 ? A. 6×4 B. $6+6+6+6$ C. $6 \times 6 \times 6 \times 6$ D. $4 \times 4 \times 4 \times 4$

Working: _____

2. Evaluate 5^3 .

Working: _____

3. Evaluate 3^5 .

Working: _____

4. Evaluate $2^4 \times 3^2$.

Working: _____

5. Evaluate $8^2 \div 2^3$.

Working: _____

6. Find the missing exponent: $4 \square = 64$.

Working: _____

7. A square display uses 12^2 tiles. How many tiles are used?

Working: _____

8. A cube has side 4 cm. Its volume is 4^3 cm^3 . Find the volume.

Working: _____

9. Explain the mistake: $2^4 = 2 \times 4 = 8$.

Working: _____

10. Evaluate $(3^2 \times 4^3) \div 2^4$.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $C. 6^4 = 6 \times 6 \times 6 \times 6$.

2: $5^3 = 5 \times 5 \times 5 = 125$.

3: $3^5 = 3 \times 3 \times 3 \times 3 \times 3 = 243$.

4: $2^4 \times 3^2 = 16 \times 9 = 144$.

5: $8^2 \div 2^3 = 64 \div 8 = 8$.

6: $4^3 = 64$, so the exponent is 3.

7: $12^2 = 12 \times 12 = 144$ tiles.

8: $4^3 = 64$, so the volume is 64 cm^3 .

9: The exponent counts factors. $2^4 = 2 \times 2 \times 2 \times 2 = 16$.

10: $3^2 = 9$, $4^3 = 64$ and $2^4 = 16$. Therefore $(9 \times 64) \div 16 = 576 \div 16 = 36$.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A student says 7^3 has three multiplication signs. Correct the statement and evaluate 7^3 . [3 marks]

Working space: _____

2. Evaluate $3^4 \times 2^5$. Show each power separately. [4 marks]

Working space: _____

3. Evaluate $12^2 \div 2^3$, then explain why the answer is smaller than 12^2 . [4 marks]

Working space: _____

4. A school stacks 5^3 boxes in each of 4 rooms. Find the total number of boxes. [4 marks]

Working space: _____

5. Calculate $(2^6 \times 5^2) \div 4^2$ and check by reversing the final division. [5 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 3 marks

Model solution: 7^3 has three factors and two multiplication signs: $7 \times 7 \times 7 = 343$. (3)

Question 2 — 4 marks

Model solution: $3^4 = 81$ and $2^5 = 32$. Then $81 \times 32 = 2,592$. (4)

Question 3 — 4 marks

Model solution: $12^2 = 144$; $2^3 = 8$; $144 \div 8 = 18$. Division by 8 makes the value smaller. (4)

Question 4 — 4 marks

Model solution: $5^3 = 125$ boxes per room. $125 \times 4 = 500$ boxes. (4)

Question 5 — 5 marks

Model solution: $2^6 = 64$, $5^2 = 25$ and $4^2 = 16$. $(64 \times 25) \div 16 = 1,600 \div 16 = 100$. Check: $100 \times 16 = 1,600$. (5)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: a^m means a multiplied by itself m times.

Remember: The base is the large number; the exponent is the small raised number.

Remember: Evaluate powers before other multiplication or division.

Remember: Expand a difficult power to check the number of factors.

Final Self-Check

- ☐ I can identify the base and exponent in a power.
- ☐ I can expand a power as repeated multiplication.
- ☐ I can evaluate expressions containing powers.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 2

Writing Large Numbers in Standard Form

Unit 1: Numbers | Numbers

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 1: Numbers
Strand	Numbers
Lesson	2: Writing Large Numbers in Standard Form
NSC objective(s)	Write numbers greater than or equal to 10 in standard form.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Standard form makes very large values easier to read, compare and use in mathematics, science, technology and Jamaican budget reports.

Learning Objectives — I Can

- I can write a large ordinary number in standard form.
- I can change standard form back to an ordinary number.
- I can check that the coefficient is at least 1 but less than 10.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Standard form	A number written as $a \times 10^n$, where $1 \leq a < 10$.
Coefficient	The number before $\times 10$.
Exponent	The small raised number on 10.
Power of ten	A value such as 10, 100, 1,000 or 10,000.
Decimal point	The dot that separates whole-number and decimal places.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A large number is in standard form when it is written as $a \times 10^n$ and the coefficient a is at least 1 but less than 10.

For numbers 10 or greater, move the decimal point left until one non-zero digit remains before it. The number of moves becomes the positive exponent.

To change back, move the decimal point right by the exponent and add zeros when needed.

Small Steps to Follow

1. Place the decimal point after the first non-zero digit.
2. Keep all important digits, including zeros between non-zero digits.
3. Count the moves to the left.
4. Use the move count as the exponent on 10.
5. Check $1 \leq \text{coefficient} < 10$ by expanding the answer.

Notebook visual: Write 63,000 with an invisible decimal at the end. Draw four curved arrows moving it left to 6.3. The four arrows show the exponent 4.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Write 56,000 in standard form.

1. Move the decimal after 5: 5.6.
2. The decimal moved 4 places left.
3. Attach $\times 10^4$.

Answer: $56,000 = 5.6 \times 10^4$.

Worked Example 2

Problem: Write 7,350,000 in standard form.

1. Keep 7, 3 and 5: 7.35.
2. The decimal moved 6 places left.
3. Use exponent 6.

Answer: $7,350,000 = 7.35 \times 10^6$.

Worked Example 3

Problem: Write 4.27×10^6 as an ordinary number.

1. The exponent is 6.
2. Move the decimal 6 places right.
3. Add zeros and commas.

Answer: $4.27 \times 10^6 = 4,270,000$.

Common Mistakes and Fixes

Common mistake	How to fix it
Coefficient is 10 or more	Move the decimal farther left and increase the exponent.
Wrong exponent	Count decimal-point moves, not only visible zeros.
Important zero dropped	Keep zeros that lie between non-zero digits.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Write 62,000 in standard form.

Check: 62,000 $\rightarrow$ 6.2; 4 moves. Answer: 6.2×10^4 .

2. Write 900,000 in standard form.

Check: 900,000 $\rightarrow$ 9; 5 moves. Answer: 9×10^5 .

3. Write 4,580,000 in standard form.

Check: 4,580,000 $\rightarrow$ 4.58; 6 moves. Answer: 4.58×10^6 .

4. Write 70,400 in standard form.

Check: Keep the middle zero: $70,400 = 7.04 \times 10^4$.

5. Write 8.3×10^5 as an ordinary number.

Check: Move right 5 places: 830,000.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Which correctly writes 34,000? A. 34×10^3 B. 3.4×10^4 C. 3.4×10^3 D. 0.34×10^5

Working: _____

2. Write 600,000 in standard form.

Working: _____

3. Write 91,200 in standard form.

Working: _____

4. Write 5.07×10^6 as an ordinary number.

Working: _____

5. Correct 12.5×10^3 so it is in standard form.

Working: _____

6. Correct: $408,000 = 4.8 \times 10^5$.

Working: _____

7. Which is greater: 7.2×10^5 or 690,000?

Working: _____

8. Write J\$8,450,000 in standard form.

Working: _____

9. Arrange 3.2×10^5 , 295,000 and 4.01×10^5 from smallest to largest.

Working: _____

10. Write 82,500,000 in standard form and justify the exponent.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: B. $34,000 = 3.4 \times 10^4$.

2: $600,000 = 6 \times 10^5$.

3: $91,200 = 9.12 \times 10^4$.

4: $5.07 \times 10^6 = 5,070,000$.

5: $12.5 \times 10^3 = 1.25 \times 10^4$.

6: The middle zero was dropped. $408,000 = 4.08 \times 10^5$.

7: $7.2 \times 10^5 = 720,000$, so it is greater.

8: J\$8,450,000 = J\$8.45 $\times 10^6$.

9: 295,000; 3.2×10^5 ; 4.01×10^5 .

10: $82,500,000 = 8.25 \times 10^7$ because the decimal moves 7 places left.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Write 68,500,000 in standard form. State the coefficient and exponent. [4 marks]

Working space: _____

2. A programme collected 4.2×10^5 bottles and another collected 385,000. Find the difference in standard form. [5 marks]

Working space: _____

3. Correct 12.6×10^6 to standard form, then write half of the corrected value in standard form. [5 marks]

Working space: _____

4. Order 7.05×10^6 , 6,950,000 and 72×10^5 from least to greatest. [5 marks]

Working space: _____

5. A plant processes 125,000 items monthly for 24 months and rejects 250,000. Write the accepted amount in standard form. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 4 marks

Model solution: $68,500,000 = 6.85 \times 10^7$. Coefficient = 6.85; exponent = 7. (4)

Question 2 — 5 marks

Model solution: $4.2 \times 10^5 = 420,000$. Difference = $420,000 - 385,000 = 35,000 = 3.5 \times 10^4$ bottles. (5)

Question 3 — 5 marks

Model solution: $12.6 \times 10^6 = 1.26 \times 10^7$. Half is 6.3×10^6 . (5)

Question 4 — 5 marks

Model solution: They are 7,050,000; 6,950,000; 7,200,000. Order: 6,950,000; 7.05×10^6 ; 72×10^5 . (5)

Question 5 — 6 marks

Model solution: Processed = $125,000 \times 24 = 3,000,000$. Accepted = $3,000,000 - 250,000 = 2,750,000 = 2.75 \times 10^6$ items. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Standard form: $a \times 10^n$, where $1 \leq a < 10$.

Remember: Large number to standard form: move left and count moves.

Remember: Standard form to ordinary number: move right by the exponent.

Remember: Expand to check that the original number returns.

Final Self-Check

- ☐ I can write a large ordinary number in standard form.
- ☐ I can change standard form back to an ordinary number.
- ☐ I can check that the coefficient is at least 1 but less than 10.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 3

Decimal Places and Significant Figures

Unit 1: Numbers | Numbers

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 1: Numbers
Strand	Numbers
Lesson	3: Decimal Places and Significant Figures
NSC objective(s)	Write a number to a given number of decimal places and significant figures.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Rounding gives a useful level of accuracy for money, measurement, population reports and calculator answers.

Learning Objectives — I Can

- I can round to a stated number of decimal places.
- I can round to a stated number of significant figures.
- I can explain which digit decides whether to round up.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Round	Replace a number with a nearby value that is easier to use.
Decimal place	A position to the right of the decimal point.
Significant figure	A meaningful digit counted from the first non-zero digit.
Rounding digit	The last digit kept in the rounded answer.
Deciding digit	The digit immediately after the rounding digit.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

For decimal places, start counting immediately after the decimal point. For significant figures, start at the first non-zero digit.

Look one digit to the right of the last digit being kept. If it is 0–4, keep the rounding digit. If it is 5–9, increase the rounding digit by 1.

Zeros between significant digits count. Leading zeros in a small decimal do not count.

Small Steps to Follow

1. Mark the digit that must be kept last.
2. Look at the next digit to the right.
3. Keep the marked digit for 0–4 or increase it for 5–9.
4. Remove later digits; use zeros as place holders in whole numbers.
5. Check that the rounded answer has the requested number of places or figures.

Notebook visual: Draw a traffic light: 0–4 under red means STOP and keep; 5–9 under green means GO UP by one.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Round 8.746 to 2 decimal places.

1. The second decimal digit is 4.
2. The deciding digit is 6.
3. 6 means round 4 up to 5.

Answer: $8.746 \approx 8.75$ to 2 decimal places.

Worked Example 2

Problem: Round 0.004786 to 2 significant figures.

1. The first non-zero digit is 4; the second is 7.
2. The deciding digit is 8.
3. Round 7 up to 8.

Answer: $0.004786 \approx 0.0048$ to 2 significant figures.

Worked Example 3

Problem: Round 4,895,000 to 3 significant figures.

1. Keep 4, 8 and 9.
2. The deciding digit is 5, so increase 9. This creates a carry.
3. 4,89 becomes 4,90 and later places become zeros.

Answer: $4,895,000 \approx 4,900,000$ to 3 significant figures.

Common Mistakes and Fixes

Common mistake	How to fix it
Counting significant figures from the decimal point	Start significant figures at the first non-zero digit.
Looking at the wrong deciding digit	The deciding digit is immediately to the right of the last digit kept.
Dropping place-holder zeros	Whole-number zeros may be needed to keep the correct place value.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Round 6.283 to 1 decimal place.

Check: Keep 2; next digit 8 rounds up. Answer: 6.3.

2. Round 14.956 to 2 decimal places.

Check: Keep 5; next digit 6 rounds up. Answer: 14.96.

3. Round 7,482 to 2 significant figures.

Check: Keep 7 and 4; next digit 8 rounds up. Answer: 7,500.

4. Round 0.06791 to 3 significant figures.

Check: Keep 6, 7, 9; next digit 1. Answer: 0.0679.

5. Round J\$3,449.75 to the nearest dollar.

Check: 75 cents rounds up. Answer: J\$3,450.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Round 5.641 to 2 decimal places.

Working: _____

2. Round 23.987 to 1 decimal place.

Working: _____

3. Round 0.43856 to 3 decimal places.

Working: _____

4. Round 68,751 to 2 significant figures.

Working: _____

5. Round 905,400 to 3 significant figures.

Working: _____

6. Round 0.003084 to 2 significant figures.

Working: _____

7. Which is 43.276 to 2 d.p.? A. 43.27 B. 43.28 C. 43.3 D. 44

Working: _____

8. A road distance is 126.78 km. Give it to the nearest kilometre.

Working: _____

9. A calculator shows 8.333333. Give the answer to 3 significant figures.

Working: _____

10. Explain why 0.00502 has three significant figures.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: 5.64; the deciding digit is 1.
- 2: 24.0; 8 rounds 9 up and the carry changes 23.9 to 24.0.
- 3: 0.439.
- 4: 69,000.
- 5: 905,000; the zero between 9 and 5 is significant.
- 6: 0.0031.
- 7: B. 43.28.
- 8: 127 km.
- 9: 8.33.
- 10: The leading zeros only locate the decimal. The digits 5, 0 and 2 are significant because the zero lies between non-zero digits.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Round 9.9956 to 3 decimal places and explain the carrying. [3 marks]

Working space: _____

2. Round 0.0007849 to 2 significant figures. [3 marks]

Working space: _____

3. A school orders 18.76 m of fabric but the shop sells whole metres. Estimate the amount that must be bought and explain. [4 marks]

Working space: _____

4. A parish population estimate is 72,649. Give it to 3 significant figures and to the nearest thousand. [4 marks]

Working space: _____

5. A bill is J\$12,487.65. Round it to the nearest dollar, nearest hundred dollars and 2 significant figures. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 3 marks

Model solution: Keep 9.995; next digit 6 rounds the last 5 up. The carry gives 9.996. (3)

Question 2 — 3 marks

Model solution: First two significant digits are 7 and 8; next digit 4. Answer: 0.00078. (3)

Question 3 — 4 marks

Model solution: Round up to 19 m. Buying only 18 m would be too little, so the practical context requires 19 m. (4)

Question 4 — 4 marks

Model solution: 3 s.f.: 72,600. Nearest thousand: 73,000. (4)

Question 5 — 6 marks

Model solution: Nearest dollar: J\$12,488. Nearest hundred: J\$12,500. Two significant figures: J\$12,000. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Decimal places: count right from the decimal point.

Remember: Significant figures: count from the first non-zero digit.

Remember: 0–4 keep; 5–9 round up.

Remember: Write place-holder zeros when rounding whole numbers.

Final Self-Check

- ☐ I can round to a stated number of decimal places.
- ☐ I can round to a stated number of significant figures.
- ☐ I can explain which digit decides whether to round up.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 4

Place Value in Bases Other Than Ten

Unit 1: Numbers | Numbers

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 1: Numbers
Strand	Numbers
Lesson	4: Place Value in Bases Other Than Ten
NSC objective(s)	Express place values of digits in all bases including base 10.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Different bases are used in computers and coding. Understanding them strengthens place-value thinking.

Learning Objectives — I Can

- I can list the digits allowed in a base.
- I can identify the place values in a base-n number.
- I can expand a number written in another base.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Base	The number of different digits used before regrouping.
Digit	A single symbol used to write a number.
Place value	The value a digit has because of its position.
Expanded form	A sum showing each digit multiplied by its place value.
Subscript	The small lower number that names the base, such as the 5 in 324_5 .

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

Base n uses only the digits 0 through $n-1$. Base 5 therefore uses 0, 1, 2, 3 and 4; it cannot contain a digit 5.

Place values are powers of the base. From right to left they are n^0 , n^1 , n^2 , n^3 and so on.

For 324_5 , the places are 5^2 , 5^1 and 5^0 , so its expanded form is $3 \times 25 + 2 \times 5 + 4 \times 1$.

Small Steps to Follow

1. Read the subscript to identify the base.
2. Check that every digit is less than the base.
3. Write powers of the base above the digits, beginning with base^0 on the right.
4. Multiply each digit by its place value.
5. Add the parts only if a base-ten value is requested.

Notebook visual: Draw a place-value chart for base 5 with columns 5^3 , 5^2 , 5^1 , 5^0 . Under them write 125, 25, 5, 1.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Expand 324_5 .

1. Base 5 places are 5^2 , 5^1 , 5^0 .
2. Match digits: $3 \times 5^2 + 2 \times 5^1 + 4 \times 5^0$.
3. Replace powers: $3 \times 25 + 2 \times 5 + 4 \times 1$.

Answer: $324_5 = 3 \times 25 + 2 \times 5 + 4 \times 1$.

Worked Example 2

Problem: State the value of the leftmost digit 1 in 2101_3 .

1. From right, the places are 3^0 , 3^1 , 3^2 , 3^3 .
2. The leftmost 1 is in the 3^3 place.
3. $3^3 = 27$.

Answer: The leftmost digit 1 is worth 27 in base ten.

Worked Example 3

Problem: Decide whether 248_8 is a valid base-8 number.

1. Base 8 allows digits 0 through 7.
2. The numeral contains digit 8.
3. 8 is not less than the base.

Answer: 248_8 is invalid.

Common Mistakes and Fixes

Common mistake	How to fix it
Using base-ten place values	Use powers of the stated base, not 1, 10, 100.
Allowing a digit equal to the base	The largest digit is always base minus 1.
Starting with base ¹ on the right	The rightmost place is base ⁰ , which equals 1.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. List all digits allowed in base 4.

Check: 0, 1, 2 and 3.

2. Write the place values for a four-digit base-3 number.

Check: 3^3 , 3^2 , 3^1 , 3^0 ; that is 27, 9, 3, 1.

3. Expand 132_4 .

Check: $132_4 = 1 \times 4^2 + 3 \times 4^1 + 2 \times 4^0$.

4. What is the value of the 2 in 201_5 ?

Check: The 2 is in the 5^2 place, so it is worth $2 \times 25 = 50$.

5. Is 560_6 valid? Explain.

Check: It is valid because base 6 allows digits 0 through 5.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. List the digits used in base 2.

Working: _____

2. Which numeral is invalid in base 5? A. 304_5 B. 142_5 C. 520_5 D. 413_5

Working: _____

3. Expand 1011_2 .

Working: _____

4. Expand 243_6 .

Working: _____

5. State the value of the 3 in 312_4 .

Working: _____

6. State the value of the middle 1 in 111_7 .

Working: _____

7. Write $4 \times 5^3 + 2 \times 5 + 1$ in base-5 numeral form.

Working: _____

8. Write the place-value chart for 2304_5 .

Working: _____

9. Explain why 999_{10} is valid but 999_9 is invalid.

Working: _____

10. Find the base-ten contribution of each digit in 321_4 .

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: 0 and 1.

2: C, because digit 5 is not allowed in base 5.

3: $1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$.

4: $2 \times 6^2 + 4 \times 6^1 + 3 \times 6^0$.

5: $3 \times 4^2 = 3 \times 16 = 48$.

6: $1 \times 7^1 = 7$.

7: 4021_5 .

8: Digits 2,3,0,4 sit under $5^3, 5^2, 5^1, 5^0$ respectively.

9: Base 10 allows digits 0–9. Base 9 allows only 0–8, so digit 9 is invalid.

10: $3 \times 16 = 48$; $2 \times 4 = 8$; $1 \times 1 = 1$.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. For 2301_4 , write the place values, expanded form and contribution of each non-zero digit. [5 marks]

Working space: _____

2. A student writes 645_6 . Identify every problem and give a valid three-digit base-6 numeral using similar digits. [4 marks]

Working space: _____

3. Compare the place occupied by the leftmost digit in 1000_2 and 1000_5 . [4 marks]

Working space: _____

4. Write $3 \times 7^3 + 5 \times 7^2 + 0 \times 7 + 6$ as one base-7 numeral and verify its digits. [4 marks]

Working space: _____

5. A code is written as 12021_3 . Make a place-value table and find the value represented by each digit. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 5 marks

Model solution: Places: $4^3, 4^2, 4^1, 4^0$. Expanded: $2 \times 64 + 3 \times 16 + 0 \times 4 + 1 \times 1$. Contributions: 128, 48 and 1. (5)

Question 2 — 4 marks

Model solution: Digits 6 and 5? Base 6 allows 0–5, so 6 is invalid while 5 is allowed. One valid answer is 545_6 . (4)

Question 3 — 4 marks

Model solution: In 1000_2 it is $2^3 = 8$. In 1000_5 it is $5^3 = 125$. The positions look the same but their values differ. (4)

Question 4 — 4 marks

Model solution: The coefficients form 3506₇. All digits 3, 5, 0, 6 are less than 7, so it is valid. (4)

Question 5 — 6 marks

Model solution: Places: $3^4, 3^3, 3^2, 3^1, 3^0 = 81, 27, 9, 3, 1$. Contributions: $1 \times 81 = 81$, $2 \times 27 = 54$, 0 , $2 \times 3 = 6$, $1 \times 1 = 1$. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Base n uses digits 0 to $n-1$.

Remember: Places from right: n^0 , n^1 , n^2 , n^3 ...

Remember: Expanded form = digit $\times$ place value for every position.

Remember: The subscript tells the base and is not part of the numeral.

Final Self-Check

- ☐ I can list the digits allowed in a base.
- ☐ I can identify the place values in a base- n number.
- ☐ I can expand a number written in another base.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 5

Converting Between Base Ten and Other Bases

Unit 1: Numbers | Numbers

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 1: Numbers
Strand	Numbers
Lesson	5: Converting Between Base Ten and Other Bases
NSC objective(s)	Convert numbers written in base n , where $1 < n < 10$, to base 10 and vice versa.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Base conversion connects ordinary counting to computer number systems and builds strong place-value reasoning.

Learning Objectives — I Can

- I can convert a base- n numeral to base ten.
- I can convert a base-ten number to another base.
- I can check a conversion by reversing it.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Convert	Change the form without changing the value.
Remainder	The amount left after division.
Repeated division	Dividing again and again by the new base.
Place-value expansion	Writing each digit times a power of the base.
Equivalent	Having the same value.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

To change base n to base ten, expand the numeral using powers of n and add the contributions.

To change base ten to base n , divide repeatedly by n , record each remainder, then read the remainders from bottom to top.

A reliable check is to convert the answer back to the starting base.

Small Steps to Follow

1. For base n to base 10: write powers of n from right to left.
2. Multiply each digit by its power and add.
3. For base 10 to base n : divide by n and record the remainder.
4. Continue with each quotient until the quotient is zero.
5. Read remainders upward and reverse-convert to check.

Notebook visual: Draw a repeated-division ladder. Put 43 at the top, divide by 2, and write each quotient on the next line with the remainder at the side. Read side remainders upward.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Convert 324_5 to base ten.

1. $324_5 = 3 \times 5^2 + 2 \times 5 + 4$.
2. $= 3 \times 25 + 10 + 4$.
3. $= 75 + 10 + 4 = 89$.

Answer: $324_5 = 89_{10}$.

Worked Example 2

Problem: Convert 1011_2 to base ten.

1. $1011_2 = 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$.
2. Add the non-zero contributions: $8 + 2 + 1$.
3. The sum is 11.

Answer: $1011_2 = 11_{10}$.

Worked Example 3

Problem: Convert 43_{10} to base 2.

1. $43 \div 2 = 21$ r1; $21 \div 2 = 10$ r1; $10 \div 2 = 5$ r0.
2. $5 \div 2 = 2$ r1; $2 \div 2 = 1$ r0; $1 \div 2 = 0$ r1.
3. Read remainders upward: 101011.

Answer: $43_{10} = 101011_2$.

Common Mistakes and Fixes

Common mistake	How to fix it
Reading remainders top to bottom	For repeated division, read remainders from the last one upward.
Using powers of ten	When expanding base n , use powers of n .
Not checking digits	Every digit in the final numeral must be less than the base.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Convert 23_4 to base ten.

Check: $2 \times 4 + 3 = 11$, so $23_4 = 11_{10}$.

2. Convert 210_3 to base ten.

Check: $2 \times 9 + 1 \times 3 + 0 = 21$, so $210_3 = 21_{10}$.

3. Convert 11101_2 to base ten.

Check: $16 + 8 + 4 + 0 + 1 = 29$.

4. Convert 19_{10} to base 3.

Check: $19 \div 3 = 6 \text{ r}1$; $6 \div 3 = 2 \text{ r}0$; $2 \div 3 = 0 \text{ r}2$. Read up: 201_3 .

5. Convert 54_{10} to base 5.

Check: $54 \div 5 = 10 \text{ r}4$; $10 \div 5 = 2 \text{ r}0$; $2 \div 5 = 0 \text{ r}2$. Answer: 204_5 .

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Convert 132_4 to base ten.

Working: _____

2. Convert 243_6 to base ten.

Working: _____

3. Convert 100101_2 to base ten.

Working: _____

4. Convert 71_8 to base ten.

Working: _____

5. Convert 25_{10} to base 2.

Working: _____

6. Convert 38_{10} to base 4.

Working: _____

7. Convert 100_{10} to base 5.

Working: _____

8. Which equals 17_{10} ? A. 10001_2 B. 10010_2 C. 1111_2 D. 10101_2

Working: _____

9. A student converts 14_{10} to 112_3 . Check the answer.

Working: _____

10. Convert 321_4 to base ten, then convert the result to base 2.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $1 \times 16 + 3 \times 4 + 2 = 30$.

2: $2 \times 36 + 4 \times 6 + 3 = 99$.

3: $32 + 4 + 1 = 37$.

4: $7 \times 8 + 1 = 57$.

5: $25 \div 2$ remainders read upward give 11001_2 .

6: $38 \div 4 = 9$ r2; $9 \div 4 = 2$ r1; $2 \div 4 = 0$ r2. Answer: 212_4 .

7: $100 = 4 \times 25$, so 400_5 .

8: A. $10001_2 = 16 + 1 = 17$.

9: $112_3 = 1 \times 9 + 1 \times 3 + 2 = 14$, so it is correct.

10: $321_4 = 3 \times 16 + 2 \times 4 + 1 = 57$. $57_{10} = 111001_2$.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Convert 2301_4 to base ten, showing every place value. [4 marks]

Working space: _____

2. Convert 86_{10} to base 3 using repeated division. [5 marks]

Working space: _____

3. A computer code 110110_2 must be written in base ten and base 4. [6 marks]

Working space: _____

4. Show that 203_5 and 11010_2 are not equivalent. [4 marks]

Working space: _____

5. Find the base-6 numeral equivalent to 145_{10} and verify by expansion. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 4 marks

Model solution: $2 \times 4^3 + 3 \times 4^2 + 0 \times 4 + 1 = 128 + 48 + 0 + 1 = 177$. (4)

Question 2 — 5 marks

Model solution: $86 \div 3 = 28$ r2; $28 \div 3 = 9$ r1; $9 \div 3 = 3$ r0; $3 \div 3 = 1$ r0; $1 \div 3 = 0$ r1. Answer: 10012_3 . (5)

Question 3 — 6 marks

Model solution: $110110_2 = 32 + 16 + 4 + 2 = 54$. Then $54 \div 4 = 13$ r2; $13 \div 4 = 3$ r1; $3 \div 4 = 0$ r3, so 312_4 . (6)

Question 4 — 4 marks

Model solution: $203_5 = 2 \times 25 + 3 = 53$. $11010_2 = 16 + 8 + 2 = 26$. Since $53 \neq 26$, they are not equivalent. (4)

Question 5 — 6 marks

Model solution: $145 \div 6 = 24$ r1; $24 \div 6 = 4$ r0; $4 \div 6 = 0$ r4, so 401_6 . Check: $4 \times 36 + 0 \times 6 + 1 = 145$. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Base $n \rightarrow$ base 10: expand with powers of n and add.

Remember: Base 10 $\rightarrow$ base n : repeatedly divide by n .

Remember: Read remainders from bottom to top.

Remember: Reverse the conversion to check.

Final Self-Check

- ☐ I can convert a base- n numeral to base ten.
- ☐ I can convert a base-ten number to another base.
- ☐ I can check a conversion by reversing it.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 6

Adding, Subtracting and Multiplying in Other Bases

Unit 1: Numbers | Numbers

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 1: Numbers
Strand	Numbers
Lesson	6: Adding, Subtracting and Multiplying in Other Bases
NSC objective(s)	Add, subtract and multiply numbers written in base n , where $1 < n < 10$.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Arithmetic in other bases shows why carrying and borrowing work and supports computer-number systems.

Learning Objectives — I Can

- I can add in another base and carry groups equal to the base.
- I can subtract by borrowing one group of the base.
- I can multiply base- n numerals and check in base ten.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Carry	Regroup one complete group of the base into the next place.
Borrow	Exchange one higher-place unit for base units in the next place.
Regroup	Rewrite a value using different place-value groups.
Partial product	One result formed during multiplication.
Check	Use base-ten conversion to test an answer.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

The column methods are like base ten, but regrouping happens when a column reaches the stated base. In base 5, five ones become one five and zero ones.

When subtracting in base n , borrowing 1 from the next place adds n to the current digit.

For multiplication, multiply each digit, regroup in the base, and align partial products by place.

Small Steps to Follow

1. Align digits by place value and confirm both numbers use the same base.
2. Work from right to left.
3. For addition, carry whenever a column total is at least the base.
4. For subtraction, borrow one base-group when the top digit is too small.
5. Convert all numbers to base ten to check the operation.

Notebook visual: Draw five bottle caps as one bundle in base 5. Show 4 caps + 3 caps becoming one bundle of 5 with 2 caps left: $4 + 3 = 12_5$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Add $243_5 + 132_5$.

1. Ones: $3+2=5_{10}=10_5$. Write 0, carry 1.
2. Fives: $4+3+1=8_{10}=13_5$. Write 3, carry 1.
3. Twenty-fives: $2+1+1=4$.

Answer: $243_5 + 132_5 = 430_5$.

Worked Example 2

Problem: Subtract $403_5 - 124_5$.

1. Ones: 3 cannot subtract 4; borrow one five, giving $8-4=4$.
2. The middle 0 had to borrow from 4 first; after passing a group right it becomes 4, then $4-2=2$.
3. Left place becomes 3; $3-1=2$.

Answer: $403_5 - 124_5 = 224_5$.

Worked Example 3

Problem: Multiply $23_4 \times 2_4$.

1. $3 \times 2 = 6_{10} = 12_4$. Write 2, carry 1.
2. $2 \times 2 + 1 = 5_{10} = 11_4$.
3. Write 11_4 before the final digit 2.

Answer: $23_4 \times 2_4 = 112_4$.

Common Mistakes and Fixes

Common mistake	How to fix it
Carrying at ten	Carry when the column reaches the stated base.
Borrowing ten	Borrowing one higher-place unit adds the base, not always 10.
Writing illegal digits	Every answer digit must be less than the base.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Add $13_4 + 22_4$.

Check: $3+2=5=11_4$; write 1 carry 1. Then $1+2+1=4=10_4$. Answer: 101_4 .

2. Add $1011_2 + 110_2$.

Check: $1011_2 + 0110_2 = 10001_2$.

3. Subtract $32_5 - 14_5$.

Check: Borrow: $2+5-4=3$; left becomes 2 and $2-1=1$. Answer: 13_5 .

4. Subtract $1000_2 - 1_2$.

Check: $1000_2 - 0001_2 = 111_2$.

5. Multiply $12_3 \times 2_3$.

Check: $2 \times 2 = 4 = 11_3$, write 1 carry 1; $1 \times 2 + 1 = 3 = 10_3$. Answer: 101_3 .

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Add $24_5 + 13_5$.

Working: _____

2. Add $132_4 + 213_4$.

Working: _____

3. Add $1101_2 + 111_2$.

Working: _____

4. Subtract $43_5 - 21_5$.

Working: _____

5. Subtract $302_4 - 123_4$.

Working: _____

6. Subtract $10110_2 - 101_2$.

Working: _____

7. Multiply $21_3 \times 2_3$.

Working: _____

8. Multiply $32_5 \times 3_5$.

Working: _____

9. Check whether $23_4 + 12_4 = 101_4$.

Working: _____

10. A base-5 counter shows 143_5 , then 24_5 more items arrive. Find the new count.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $24_5 + 13_5 = 42_5$.

2: $132_4 + 213_4 = 1011_4$.

3: $1101_2 + 0111_2 = 10100_2$.

4: 22_5 .

5: 113_4 .

6: 10001_2 .

7: $21_3 \times 2_3 = 112_3$.

8: $2 \times 3 = 6 = 11_5$; write 1 carry 1. $3 \times 3 + 1 = 10 = 20_5$. Answer: 201_5 .

9: $23_4 = 11$, $12_4 = 6$; sum = 17. $101_4 = 17$. The statement is correct.

10: $143_5 + 024_5 = 222_5$.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Calculate $324_5 + 243_5$ and verify in base ten. [6 marks]

Working space: _____

2. Calculate $1002_3 - 121_3$ and check by addition. [6 marks]

Working space: _____

3. Multiply 123_4 by 3_4 using base-4 regrouping. [6 marks]

Working space: _____

4. A student writes $44_5 + 13_5 = 57_5$. Explain two errors and give the correct answer. [5 marks]

Working space: _____

5. Find $(132_5 + 24_5) - 43_5$ and verify using base ten. [7 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 6 marks

Model solution: Base-5 addition gives 1122_5 . Check: $324_5=89$, $243_5=73$; $\text{sum}=162$. $1122_5=125+25+10+2=162$. (6)

Question 2 — 6 marks

Model solution: $1002_3=29$; $121_3=16$; $\text{difference}=13=111_3$. Check: $111_3 + 121_3 = 1002_3$. (6)

Question 3 — 6 marks

Model solution: $3 \times 3 = 9 = 21_4$; write 1 carry 2. $2 \times 3 + 2 = 8 = 20_4$; write 0 carry 2. $1 \times 3 + 2 = 5 = 11_4$. Answer: 1101_4 . (6)

Question 4 — 5 marks

Model solution: Digits 5 and 7 are illegal in base 5. Correct addition: $4+3=7=12_5$, write 2 carry 1; $4+1+1=6=11_5$. Answer 112_5 . (5)

Question 5 — 7 marks

Model solution: $132_5 + 024_5 = 211_5$; $211_5 - 43_5 = 113_5$. Check: $132_5=42$, $24_5=14$, $43_5=23$; $42+14-23=33$. $113_5=25+5+3=33$. (7)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Addition: regroup a column total in groups of the base.

Remember: Subtraction: one borrow adds the base to the current place.

Remember: Multiplication: regroup each product in the base.

Remember: Convert to base ten to check every answer.

Final Self-Check

- ☐ I can add in another base and carry groups equal to the base.
- ☐ I can subtract by borrowing one group of the base.
- ☐ I can multiply base- n numerals and check in base ten.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 7

Closure, Reflexive, Symmetric, Transitive and Trichotomy Properties

Unit 2: Numbers | Numbers

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 2: Numbers
Strand	Numbers
Lesson	7: Closure, Reflexive, Symmetric, Transitive and Trichotomy Properties
NSC objective(s)	Identify and use closure, reflexive, symmetry, transitive and trichotomy concepts.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Mathematical properties explain why number statements and comparisons are trustworthy. They support algebra and proof.

Learning Objectives — I Can

- I can name a property shown by a statement.
- I can decide whether a number set is closed under an operation.
- I can use trichotomy to compare two real numbers.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Closure	An operation stays inside a stated set.
Reflexive	Every quantity is equal to itself: $a = a$.
Symmetric	If $a = b$, then $b = a$.
Transitive	If $a = b$ and $b = c$, then $a = c$.
Trichotomy	Exactly one of $a < b$, $a = b$ or $a > b$ is true.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

Closure always names both a set and an operation. Whole numbers are closed under addition because adding any two whole numbers gives another whole number. They are not closed under subtraction because $3 - 8 = -5$, which is not a whole number.

Reflexive, symmetric and transitive properties describe equality. Trichotomy describes order and says only one comparison can be true for a pair of real numbers.

A counterexample is one example that proves a closure claim false.

Small Steps to Follow

1. Identify the set, operation or equality statement.
2. For closure, test several examples and look for a counterexample.
3. For equality, match the statement to reflexive, symmetric or transitive form.
4. For a comparison, write exactly one of $<$, $=$ or $>$.
5. Explain the property in a complete sentence.

Notebook visual: Draw five labelled cards: closure = stay inside; reflexive = mirror to self; symmetric = two-way arrow; transitive = chain; trichotomy = three doors with only one open.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Are integers closed under subtraction?

1. Choose any two integers, such as 4 and 9.
2. $4 - 9 = -5$.
3. -5 is still an integer.

Answer: Yes. Integers are closed under subtraction.

Worked Example 2

Problem: Name the property: if $x = 12$, then $12 = x$.

1. The order of the equal quantities is reversed.
2. The value of the statement does not change.
3. This matches: if $a=b$, then $b=a$.

Answer: Symmetric property.

Worked Example 3

Problem: Use trichotomy to compare 3^2 and 10.

1. Evaluate $3^2 = 9$.
2. Compare 9 and 10.
3. Only the less-than statement is true.

Answer: $3^2 < 10$.

Common Mistakes and Fixes

Common mistake	How to fix it
Calling one successful example proof of closure	A closure statement must work for every pair in the set.
Confusing symmetry with transitivity	Symmetry reverses; transitivity chains.
Writing two trichotomy relations	Exactly one of $<$, $=$ and $>$ is true for two real values.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Name the property shown by $7 = 7$.

Check: Reflexive property.

2. If $a=b$ and $b=c$, what can you conclude?

Check: $a=c$ by the transitive property.

3. Are natural numbers closed under multiplication?

Check: Yes. The product of any two natural numbers is a natural number.

4. Give a counterexample showing whole numbers are not closed under division.

Check: $1 \div 2 = 0.5$, and 0.5 is not a whole number.

5. Insert $<$, $=$ or $>$: 5^3 ____ 12^2 .

Check: $5^3=125$ and $12^2=144$, so $5^3 < 12^2$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Which property is $a=a$? A. closure B. reflexive C. symmetric D. transitive

Working: _____

2. Name the property: if $2x=18$, then $18=2x$.

Working: _____

3. Name the property: $p=q$, $q=r$, therefore $p=r$.

Working: _____

4. Are even numbers closed under addition? Explain.

Working: _____

5. Are odd numbers closed under addition? Give evidence.

Working: _____

6. Are integers closed under division?

Working: _____

7. Use trichotomy to compare -4 and -9 .

Working: _____

8. Can both $a < b$ and $a > b$ be true for the same real numbers?

Working: _____

9. A student says closure means the answer is positive. Correct the student.

Working: _____

10. If J\$500 equals the price of a bag and the bag price equals the price of a book, compare the two prices.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: B. Reflexive.

2: Symmetric property.

3: Transitive property.

4: Yes. Even + even is always even.

5: No. For example, $3+5=8$, which is even, not odd.

6: No. $1\div 2=0.5$ is not an integer.

7: $-4 > -9$.

8: No. Trichotomy allows exactly one of $<$, $=$ or $>$.

9: Closure means the answer remains in the stated set; it does not necessarily mean positive.

10: By transitivity, J\$500 equals the book price.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Test whether whole numbers are closed under addition, subtraction and multiplication. Give a reason or counterexample for each. [6 marks]

Working space: _____

2. Explain the difference between the symmetric and transitive properties using symbols and examples. [6 marks]

Working space: _____

3. A learner claims natural numbers are closed under division because $12\div 3=4$. Evaluate the claim. [5 marks]

Working space: _____

4. Use trichotomy to compare 4^3 and 8^2 , then explain what happens. [4 marks]

Working space: _____

5. Given $a=b$, $b=2c$ and $2c=20$, use named properties to connect a to 20. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 6 marks

Model solution: Addition: closed. Multiplication: closed. Subtraction: not closed because, for example, $3-5=-2$ is not whole. (6)

Question 2 — 6 marks

Model solution: Symmetric reverses one equality: $a=b \Rightarrow b=a$, e.g. $x=4 \Rightarrow 4=x$. Transitive chains two equalities: $a=b$ and $b=c \Rightarrow a=c$, e.g. $x=4$ and $4=y \Rightarrow x=y$. (6)

Question 3 — 5 marks

Model solution: One supporting example cannot prove closure. Counterexample: $3 \div 2 = 1.5$, not natural. Therefore the set is not closed under division. (5)

Question 4 — 4 marks

Model solution: $4^3=64$ and $8^2=64$, so $4^3=8^2$. The equality case is the one true relation. (4)

Question 5 — 6 marks

Model solution: From $b=2c$ and $2c=20$, transitivity gives $b=20$. From $a=b$ and $b=20$, transitivity gives $a=20$. Symmetry could also write $20=a$. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Closure: the result stays in the stated set.

Remember: Reflexive: $a=a$.

Remember: Symmetric: $a=b \Rightarrow b=a$.

Remember: Transitive: $a=b$ and $b=c \Rightarrow a=c$.

Remember: Trichotomy: exactly one of $a < b$, $a = b$, $a > b$.

Final Self-Check

- ☐ I can name a property shown by a statement.
- ☐ I can decide whether a number set is closed under an operation.
- ☐ I can use trichotomy to compare two real numbers.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 8

Changing the Subject of a Formula

Unit 3: Algebra | Algebra

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 3: Algebra
Strand	Algebra
Lesson	8: Changing the Subject of a Formula
NSC objective(s)	Change the subject of a simple formula.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Changing the subject lets you solve a formula for the quantity you need in science, measurement, business and everyday calculations.

Learning Objectives — I Can

- I can identify the subject of a formula.
- I can use inverse operations to isolate a new subject.
- I can check a rearranged formula by substituting values.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Formula	A rule written with symbols.
Subject	The single variable alone on one side of a formula.
Inverse operation	An operation that undoes another operation.
Isolate	Get a variable alone.
Transpose	Rearrange a formula while keeping it balanced.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A formula is like a balanced scale. Whatever operation is applied to one side must also be applied to the other side.

Undo operations in reverse order. Addition is undone by subtraction; multiplication by division; squares by square roots when appropriate.

The target subject must appear alone, usually with coefficient 1.

Small Steps to Follow

1. Circle the variable that must become the subject.
2. Identify the operations attached to it.
3. Undo those operations one at a time in reverse order on both sides.
4. Simplify and write the target variable alone.
5. Substitute easy values into the original and rearranged formulas to check.

Notebook visual: Draw a balance scale. Write C on the left and $2\pi r$ on the right. Show both sides divided by 2π so r is left alone.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Make r the subject of $C=2\pi r$.

1. r is multiplied by 2π .
2. Divide both sides by 2π : $C \div 2\pi = r$.
3. Write the subject first.

Answer: $r = C/(2\pi)$.

Worked Example 2

Problem: Make w the subject of $A=lw$.

1. w is multiplied by l .
2. Divide both sides by l .
3. $A/l = w$.

Answer: $w = A/l$.

Worked Example 3

Problem: Make t the subject of $v=u+at$.

1. Subtract u from both sides: $v-u=at$.
2. Divide both sides by a .
3. t is isolated.

Answer: $t = (v-u)/a$.

Common Mistakes and Fixes

Common mistake	How to fix it
Moving a term without changing its operation	Use the same inverse operation on both sides.
Undoing in the wrong order	Work backward from the operations closest to the outside.
Leaving the target with a coefficient	The new subject must be alone with coefficient 1.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Make x the subject of $y=x+5$.

Check: Subtract 5: $x=y-5$.

2. Make b the subject of $P=2l+2b$.

Check: $P-2l=2b$; divide by 2: $b=(P-2l)/2$.

3. Make h the subject of $A=\frac{1}{2}bh$.

Check: Multiply by 2: $2A=bh$; divide by b : $h=2A/b$.

4. Make d the subject of $s=d/t$.

Check: Multiply both sides by t : $d=st$.

5. Make p the subject of $q=3p-7$.

Check: $q+7=3p$; divide by 3: $p=(q+7)/3$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. What is the subject of $V=lwh$?

Working: _____

2. Make m the subject of $c=m+9$.

Working: _____

3. Make x the subject of $y=4x$.

Working: _____

4. Make r the subject of $A=\pi r^2$.

Working: _____

5. Make l the subject of $P=2l+2w$.

Working: _____

6. Make a the subject of $v=u+at$.

Working: _____

7. Make F the subject of $C=(F-32)\times 5/9$.

Working: _____

8. Which makes x the subject of $2x+3=y$? A. $x=y-3/2$ B. $x=(y-3)/2$ C. $x=2y-3$ D. $x=y/5$

Working: _____

9. Check $h=2A/b$ using $A=30$ and $b=5$.

Working: _____

10. Explain why $x=y-5$ is not the correct rearrangement of $y=5x$.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: V is the subject because it is alone.
- 2: $m=c-9$.
- 3: $x=y/4$.
- 4: $r^2=A/\pi$, so $r=\sqrt{(A/\pi)}$ for a positive radius.
- 5: $P-2w=2l$; $l=(P-2w)/2$.
- 6: $v-u=at$; $a=(v-u)/t$.
- 7: Multiply by $9/5$: $9C/5=F-32$; add 32: $F=9C/5+32$.
- 8: B. $x=(y-3)/2$.
- 9: $h=2 \times 30/5=12$. Original: $\frac{1}{2} \times 5 \times 12=30$, so it checks.
- 10: The 5 multiplies x , so it must be undone by division: $x=y/5$.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Make h the subject of $V=\frac{1}{3}\pi r^2 h$, showing every inverse operation. [5 marks]

Working space: _____

2. Make w the subject of $A=2lw+2lh+2wh$. [6 marks]

Working space: _____

3. The formula $d=vt$ gives distance. Make v the subject and find v when $d=180$ km and $t=3$ h. [4 marks]

Working space: _____

4. Make x the subject of $y=(3x-4)/5$ and check with $x=8$. [6 marks]

Working space: _____

5. $A=\frac{1}{2}(a+b)h$. Make b the subject, then find b when $A=54$, $a=6$ and $h=9$. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 5 marks

Model solution: Multiply by 3: $3V = \pi r^2 h$. Divide by πr^2 : $h = 3V / (\pi r^2)$. (5)

Question 2 — 6 marks

Model solution: Group w terms: $A - 2lh = 2lw + 2wh = 2w(l + h)$. Divide: $w = (A - 2lh) / [2(l + h)]$. (6)

Question 3 — 4 marks

Model solution: $v = d/t$. Substitute: $v = 180/3 = 60$ km/h. (4)

Question 4 — 6 marks

Model solution: $5y = 3x - 4$; $5y + 4 = 3x$; $x = (5y + 4)/3$. If $x = 8$, $y = (24 - 4)/5 = 4$; rearranged gives $(20 + 4)/3 = 8$. (6)

Question 5 — 6 marks

Model solution: $2A = (a + b)h$; $2A/h = a + b$; $b = 2A/h - a$. Substitute: $108/9 - 6 = 12 - 6 = 6$. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: A formula stays balanced when the same operation is used on both sides.

Remember: Undo $+$ with $-$ and $\times$ with $\div$.

Remember: Undo operations in reverse order.

Remember: Check by substituting values into both forms.

Final Self-Check

- ☐ I can identify the subject of a formula.
- ☐ I can use inverse operations to isolate a new subject.
- ☐ I can check a rearranged formula by substituting values.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 9

Writing, Solving and Graphing Linear Inequalities

Unit 3: Algebra | Algebra

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 3: Algebra
Strand	Algebra
Lesson	9: Writing, Solving and Graphing Linear Inequalities
NSC objective(s)	Write inequalities for word problems. • Illustrate inequalities on a number line. • Solve simple linear inequalities and represent the solution on a number line.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Inequalities describe limits such as age rules, budgets, capacity and minimum exam scores.

Learning Objectives — I Can

- I can translate words into inequality symbols.
- I can solve a simple linear inequality.
- I can graph a solution using an open or closed circle.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Inequality	A statement comparing values that may not be equal.
Solution set	All values that make an inequality true.
Open circle	An endpoint that is not included.
Closed circle	An endpoint that is included.
Boundary	The endpoint separating solutions from non-solutions.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

Use $<$ or $>$ when the boundary is not included. Use $\leq$ or $\geq$ when words such as at most, at least, no more than or no less than include the boundary.

Solve an inequality like an equation by using inverse operations. If both sides are multiplied or divided by a negative number, reverse the inequality sign.

On a number line, shade toward the smaller values for $<$ or $\leq$ and toward larger values for $>$ or $\geq$.

Small Steps to Follow

1. Choose a variable and translate the words into a symbol statement.
2. Use inverse operations to isolate the variable.
3. Reverse the sign only if multiplying or dividing by a negative number.
4. Draw an open circle for $<$ or $>$ and a closed circle for $\leq$ or $\geq$.
5. Test one shaded value in the original inequality.

Notebook visual: Draw four mini number lines: $x < 3$ open-left; $x \leq 3$ closed-left; $x > 3$ open-right; $x \geq 3$ closed-right.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: A taxi holds at most 4 passengers. Write an inequality for p .

1. At most means no more than.
2. The boundary 4 is allowed.
3. Use the symbol $\leq$.

Answer: $p \leq 4$.

Worked Example 2

Problem: Solve $3x + 5 < 20$.

1. Subtract 5: $3x < 15$.
2. Divide by positive 3; the sign stays the same.
3. $x < 5$.

Answer: $x < 5$; graph an open circle at 5 and shade left.

Worked Example 3

Problem: Solve $-2x \leq 8$.

1. Divide both sides by -2 .
2. Division by a negative reverses $\leq$ to $\geq$.
3. $8 \div -2 = -4$.

Answer: $x \geq -4$; closed circle at -4 , shade right.

Common Mistakes and Fixes

Common mistake	How to fix it
Confusing at least and at most	At least means $\geq$; at most means $\leq$.
Always reversing the sign	Reverse only when multiplying or dividing by a negative.
Using the wrong endpoint circle	$\leq$ and $\geq$ include the boundary, so use a closed circle.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Write: a score s is at least 60.

Check: $s \geq 60$.

2. Write: a parcel has mass less than 10 kg.

Check: $m < 10$.

3. Solve $x + 7 > 12$.

Check: $x > 5$.

4. Solve $4x \leq 28$.

Check: $x \leq 7$.

5. Describe the graph of $y \geq -2$.

Check: Closed circle at -2 ; shade to the right.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Which means no more than 25? A. $x < 25$ B. $x \leq 25$ C. $x > 25$ D. $x \geq 25$

Working: _____

2. Write: students must be older than 12.

Working: _____

3. Write: the bag can carry up to and including 15 kg.

Working: _____

4. Solve $x - 9 \geq 4$.

Working: _____

5. Solve $5x < 45$.

Working: _____

6. Solve $2x + 3 \leq 17$.

Working: _____

7. Solve $18 - 3x > 6$.

Working: _____

8. Graph $x > -1$ in words.

Working: _____

9. Is $x=5$ a solution of $2x+1 \leq 11$?

Working: _____

10. A student shades left for $x \geq 3$. Correct the graph.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: B. $x \leq 25$.

2: $a > 12$.

3: $m \leq 15$.

4: $x \geq 13$.

5: $x < 9$.

6: $2x \leq 14$, so $x \leq 7$.

7: $-3x > -12$; divide by -3 and reverse: $x < 4$.

8: Open circle at -1 , shade right.

9: Yes. $2(5)+1=11$, and $11 \leq 11$ is true.

10: Use a closed circle at 3 and shade right because solutions are 3 or greater.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A minibus can carry no more than 30 passengers. It already has 18. Write and solve an inequality for the additional passengers x . [5 marks]

Working space: _____

2. Solve $4x - 7 \leq 21$ and show the solution on a number line. [4 marks]

Working space: _____

3. Solve $5 - 2x < 13$. Explain the sign reversal. [5 marks]

Working space: _____

4. A student needs at least 70 marks. She has 46 and one task worth x marks. Form and solve an inequality. [5 marks]

Working space: _____

5. Solve $-3 \leq 2x + 1 < 9$ and represent the combined solution. [7 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 5 marks

Model solution: $18 + x \leq 30$, so $x \leq 12$. At most 12 more passengers can enter. (5)

Question 2 — 4 marks

Model solution: $4x \leq 28$; $x \leq 7$. Closed circle at 7, shade left. (4)

Question 3 — 5 marks

Model solution: $-2x < 8$. Dividing by -2 reverses the sign, so $x > -4$. (5)

Question 4 — 5 marks

Model solution: $46 + x \geq 70$, so $x \geq 24$. She needs at least 24 marks. (5)

Question 5 — 7 marks

Model solution: Subtract 1: $-4 \leq 2x < 8$. Divide by 2: $-2 \leq x < 4$. Closed at -2 , open at 4, shade between. (7)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: $<$ or $>$: open circle.

Remember: $\leq$ or $\geq$: closed circle.

Remember: Shade left for less; right for greater.

Remember: Reverse the sign when multiplying or dividing by a negative.

Final Self-Check

- ☐ I can translate words into inequality symbols.
- ☐ I can solve a simple linear inequality.
- ☐ I can graph a solution using an open or closed circle.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 10

Angles Formed by Transversals and Intersecting Lines

Unit 4: Geometry | Geometry

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 4: Geometry
Strand	Geometry
Lesson	10: Angles Formed by Transversals and Intersecting Lines
NSC objective(s)	Investigate relationships among angles formed by a transversal of parallel lines and by intersecting non-parallel lines.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Angle relationships are used in road design, construction, surveying and geometric proof.

Learning Objectives — I Can

- I can identify corresponding, alternate and co-interior angles.
- I can use vertical and straight-line angle facts.
- I can calculate unknown angles and explain the relationship used.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Transversal	A line crossing two or more other lines.
Corresponding angles	Matching-corner angles; equal when lines are parallel.
Alternate angles	Z-shaped angles; equal when lines are parallel.
Co-interior angles	C-shaped inside angles; sum to 180° for parallel lines.
Vertically opposite	Opposite angles at an intersection; they are equal.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

When a transversal cuts parallel lines, corresponding angles are equal, alternate interior angles are equal and co-interior angles total 180° .

At any intersection, vertically opposite angles are equal. Adjacent angles on a straight line total 180° . Angles around a point total 360° .

Always name the relationship before calculating; this earns reasoning marks.

Small Steps to Follow

1. Mark parallel lines with matching arrows.
2. Look for F (corresponding), Z (alternate) or C (co-interior) shapes.
3. Write equal or sum-to-180 relationships.
4. Use straight-line, vertical or around-a-point facts as needed.
5. Substitute the answer back into the angle relationship.

Notebook visual: Draw two horizontal parallel lines crossed by a sloping transversal. Colour an F pair blue, a Z pair green and a C pair gold.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: One corresponding angle is 68° . Find its partner.

1. The lines are parallel.
2. Corresponding angles are equal.
3. Copy the measure.

Answer: The partner is 68° .

Worked Example 2

Problem: A co-interior angle is 117° . Find the other.

1. Co-interior angles total 180° .
2. Let the other angle be x : $x+117=180$.
3. $x=63$.

Answer: 63° .

Worked Example 3

Problem: Two intersecting lines form one angle of 132° . Find the other three.

1. The vertical angle is also 132° .
2. Each adjacent angle is $180 - 132 = 48^\circ$.
3. The second vertical angle is 48° .

Answer: The angles are 132° , 48° , 132° , 48° .

Common Mistakes and Fixes

Common mistake	How to fix it
Using angle facts without parallel marks	F, Z and C facts require the relevant lines to be parallel.
Saying co-interior angles are equal	They add to 180° .
Forgetting to name the reason	Write the angle relationship before the calculation.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. An alternate interior angle to 75° is what size?

Check: 75° , because alternate angles are equal for parallel lines.

2. Find x if x and 104° are co-interior.

Check: $x=180-104=76^\circ$.

3. Find the vertically opposite angle to 39° .

Check: 39° .

4. Find an adjacent angle on a straight line to 146° .

Check: $180-146=34^\circ$.

5. Angles around a point are 90° , 110° , 75° and x . Find x .

Check: $x=360-(90+110+75)=85^\circ$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Which angle pair makes an F shape?

Working: _____

2. Which angle pair makes a Z shape?

Working: _____

3. Find x : corresponding angles are $(3x+5)^\circ$ and 80° .

Working: _____

4. Find x : co-interior angles are $2x^\circ$ and 60° .

Working: _____

5. At an intersection, one angle is 27° . Find an adjacent angle.

Working: _____

6. At an intersection, angles are $(4x+2)^\circ$ and $(6x-38)^\circ$ vertically opposite. Find x .

Working: _____

7. Do equal corresponding angles prove two lines are parallel?

Working: _____

8. Find x if three angles on a straight line are 45° , x° and 72° .

Working: _____

9. A learner says co-interior angles are equal. Correct the statement.

Working: _____

10. Two lines cross and one angle is twice its adjacent angle. Find both.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: Corresponding angles.

2: Alternate angles.

3: $3x+5=80$; $3x=75$; $x=25$.

4: $2x+60=180$; $2x=120$; $x=60$.

5: 153° .

6: $4x+2=6x-38$; $40=2x$; $x=20$.

7: Yes, equal corresponding angles are a converse condition for parallel lines.

8: $x=180-45-72=63^\circ$.

9: Co-interior angles on parallel lines sum to 180° ; they are not generally equal.

10: Let small= x , large= $2x$. $3x=180$, $x=60$. Angles are 60° and 120° .

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Parallel lines are cut by a transversal. A corresponding pair is $(5x-12)^\circ$ and $(3x+28)^\circ$. Find x and the angle. [5 marks]

Working space: _____

2. One co-interior angle is $(2x+15)^\circ$ and the other is $(3x-5)^\circ$. Find x and both angles. [6 marks]

Working space: _____

3. At an intersection, one angle is $(7x-9)^\circ$ and its adjacent angle is $(5x+21)^\circ$. Find all four angles. [6 marks]

Working space: _____

4. A road crosses two parallel streets. The acute crossing angle is 58° . Determine all acute and obtuse angles formed at both intersections. [5 marks]

Working space: _____

5. Explain two different routes for finding an angle diagonally across the two parallel intersections from a 116° angle. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 5 marks

Model solution: $5x - 12 = 3x + 28$; $2x = 40$; $x = 20$. Angle $= 5(20) - 12 = 88^\circ$. (5)

Question 2 — 6 marks

Model solution: Their sum is 180: $5x + 10 = 180$; $x = 34$. Angles: 83° and 97° . (6)

Question 3 — 6 marks

Model solution: They sum to 180: $12x + 12 = 180$; $x = 14$. Angles: 89° and 91° , repeated vertically. (6)

Question 4 — 5 marks

Model solution: All four acute angles are 58° . Each obtuse angle is $180 - 58 = 122^\circ$, so all four obtuse angles are 122° . (5)

Question 5 — 6 marks

Model solution: Route 1: corresponding angle equals 116° , then vertical angle equals 116° . Route 2: vertical angle first equals 116° , then corresponding angle equals 116° . Both use equality properties. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: F: corresponding angles are equal.

Remember: Z: alternate angles are equal.

Remember: C: co-interior angles total 180° .

Remember: Vertical angles are equal; a straight line totals 180° ; around a point totals 360° .

Final Self-Check

- ☐ I can identify corresponding, alternate and co-interior angles.
- ☐ I can use vertical and straight-line angle facts.
- ☐ I can calculate unknown angles and explain the relationship used.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 11

Top, Front and Side Views of Solids

Unit 4: Geometry | Geometry

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 4: Geometry
Strand	Geometry
Lesson	11: Top, Front and Side Views of Solids
NSC objective(s)	Sketch different views of solids and use dotted lines for unseen edges.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Plans and elevations help builders, engineers and designers show three-dimensional objects on flat paper.

Learning Objectives — I Can

- I can identify top, front and side views.
- I can draw a simple solid using visible and hidden edges.
- I can use cube stacks to connect a 3-D object with 2-D views.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Plan	The view looking straight down from the top.
Front elevation	The view seen directly from the front.
Side elevation	The view seen directly from one side.
Hidden edge	An edge blocked from sight and drawn with a dotted line.
Depth	How far an object extends from front to back.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A view shows only two dimensions at a time. The top view shows width and depth; the front shows width and height; the side shows depth and height.

Visible edges use solid lines. Edges behind a face use short, even dotted lines.

For cube stacks, the top view shows occupied floor positions while front and side views show the greatest height in each column.

Small Steps to Follow

1. Mark which direction is called the front.
2. Imagine your eye looking straight at the object, not from an angle.
3. Project the width, height or depth onto a flat grid.
4. Draw visible outlines with solid lines and hidden edges with dotted lines.
5. Compare dimensions across views to check consistency.

Notebook visual: Draw a cuboid. Beside it draw three rectangles labelled TOP (width×depth), FRONT (width×height) and SIDE (depth×height).

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: A cuboid is 6 cm wide, 4 cm high and 3 cm deep. State each rectangular view.

1. Top uses width and depth: 6 by 3.
2. Front uses width and height: 6 by 4.
3. Side uses depth and height: 3 by 4.

Answer: Top 6×3; front 6×4; side 3×4.

Worked Example 2

Problem: Cube-stack heights from left to right are 1, 3 and 2 in one row. Describe the front view.

1. The front view records the height of each column.
2. Draw three adjacent grid columns.
3. Their heights are 1, 3 and 2.

Answer: A stepped outline with column heights 1, 3, 2.

Worked Example 3

Problem: When should a dotted line be used on a solid sketch?

1. Look for an edge behind a visible face.
2. The edge exists but cannot be seen directly.
3. Use short, evenly spaced dashes.

Answer: Use a dotted line for every hidden edge needed to explain the shape.

Common Mistakes and Fixes

Common mistake	How to fix it
Mixing up width and depth	Label all three dimensions before drawing.
Drawing perspective	Plans and elevations are straight-on orthographic views.
Using solid lines for hidden edges	Hidden edges use evenly spaced dots or dashes.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Which dimensions appear in a top view?

Check: Width and depth.

2. Which dimensions appear in a front view?

Check: Width and height.

3. Which dimensions appear in a side view?

Check: Depth and height.

4. A cube has side 5 cm. What shape and size is every main view?

Check: A 5 cm by 5 cm square.

5. Why are hidden edges dotted?

Check: To show that the edges exist but are blocked from direct view.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Name the view seen from above.

Working: _____

2. A box is 8 wide, 2 high and 5 deep. Give the front view dimensions.

Working: _____

3. Give the side view dimensions for the same box.

Working: _____

4. Give the top view dimensions for the same box.

Working: _____

5. Which line style shows an unseen edge? A. thick B. dotted C. curved D. double

Working: _____

6. A front view has column heights 2,2,4,1. Describe its outline.

Working: _____

7. Can two different solids have the same front view? Explain.

Working: _____

8. A top view has four occupied squares in a 2×2 block. What does this tell you?

Working: _____

9. Explain why an angled perspective drawing is not a front elevation.

Working: _____

10. A stack has top footprint 3 by 1 and heights 2,1,3. State the number of cubes.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: Top view or plan.

2: 8 by 2.

3: 5 by 2.

4: 8 by 5.

5: B. Dotted.

6: Four adjacent columns with heights 2, 2, 4 and 1.

7: Yes. They may have different depths or arrangements hidden behind the front outline.

8: At least one cube stands on each of the four floor positions; it does not show their heights.

9: An elevation is viewed straight on and shows only width and height without perspective depth.

10: $2+1+3=6$ cubes.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A storage tank is a cuboid 10 m long, 4 m high and 6 m deep. State the scale dimensions of top, front and right-side views using 1 cm:2 m. [6 marks]

Working space: _____

2. A one-row cube model has column heights 3,1,2,3. Describe the top, front and side views. [6 marks]

Working space: _____

3. A 2×2 footprint has stack heights $[[1,3],[2,0]]$. Find total cubes and describe the top view. [5 marks]

Working space: _____

4. Explain how hidden lines can distinguish a hollow rectangular frame from a solid cuboid in an engineering sketch. [5 marks]

Working space: _____

5. A model's front view has heights 2,4,1 and its top view has a 3×2 full footprint. What can and cannot be concluded? [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 6 marks

Model solution: Top $5\text{ cm} \times 3\text{ cm}$; front $5\text{ cm} \times 2\text{ cm}$; side $3\text{ cm} \times 2\text{ cm}$. (6)

Question 2 — 6 marks

Model solution: Top: four squares in a row. Front: heights 3,1,2,3. From the end side: one column of height 3, the maximum height. (6)

Question 3 — 5 marks

Model solution: $\text{Total} = 1 + 3 + 2 + 0 = 6$. Top view has three occupied squares in an L shape; the zero position is blank. (5)

Question 4 — 5 marks

Model solution: Dotted lines can show rear or internal edges blocked by front faces. A solid cuboid needs only outer hidden edges, while a hollow frame requires internal hidden boundaries to show the opening. (5)

Question 5 — 6 marks

Model solution: Every one of six floor positions has at least one cube. In each of the three width columns, the maximum depth height is 2,4,1. Exact individual back-row heights cannot be known from only these views. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Top = width $\times$ depth.

Remember: Front = width $\times$ height.

Remember: Side = depth $\times$ height.

Remember: Solid lines are visible; dotted lines are hidden.

Remember: Top view shows footprint, not stack height.

Final Self-Check

- ☐ I can identify top, front and side views.
- ☐ I can draw a simple solid using visible and hidden edges.
- ☐ I can use cube stacks to connect a 3-D object with 2-D views.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 1 • LESSON 12

Translations, Reflections and Rotations on the Coordinate Plane

Unit 4: Geometry | Geometry

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 1
Unit	Unit 4: Geometry
Strand	Geometry
Lesson	12: Translations, Reflections and Rotations on the Coordinate Plane
NSC objective(s)	Perform translations and find translation vectors. • Reflect objects in the x- or y-axis. • Rotate objects 90° , 180° and 270° about the origin.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Transformations describe movement in maps, animation, design, robotics and computer graphics.

Learning Objectives — I Can

- I can translate a point and find a translation vector.
- I can reflect points in the coordinate axes.
- I can rotate points about the origin and identify the image.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Object	The original shape before a transformation.
Image	The new shape after a transformation.
Vector	A movement written as horizontal then vertical change.
Mirror line	The line across which a reflection occurs.
Centre of rotation	The fixed point around which a shape turns.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A translation slides every point by the same vector. A vector (a,b) changes (x,y) to $(x+a,y+b)$.

Reflection in the x -axis changes (x,y) to $(x,-y)$; reflection in the y -axis changes it to $(-x,y)$.

For anticlockwise rotations about the origin: 90° gives $(-y,x)$; 180° gives $(-x,-y)$; 270° gives $(y,-x)$. Shape and size are preserved.

Small Steps to Follow

1. Write the coordinate rule before moving any point.
2. Transform every vertex separately.
3. Label images with primes, such as A' .
4. Plot the transformed points and join them in the original order.
5. Check that distances and angles are unchanged.

Notebook visual: Create a rule card: translate $(x+a,y+b)$; x -axis $(x,-y)$; y -axis $(-x,y)$; 90° ACW $(-y,x)$; 180° $(-x,-y)$; 270° ACW $(y,-x)$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Translate $A(2,-1)$ by vector $(3,4)$.

1. Add 3 to x : $2+3=5$.
2. Add 4 to y : $-1+4=3$.
3. Write the image coordinate.

Answer: $A'(5,3)$.

Worked Example 2

Problem: Reflect $B(-4,2)$ in the y -axis.

1. The y -axis reflection changes the sign of x .
2. -4 becomes 4 ; y stays 2 .
3. The point is equally far on the other side.

Answer: $B'(4,2)$.

Worked Example 3

Problem: Rotate $C(3,1)$ 90° anticlockwise about the origin.

1. Use $(x,y) \rightarrow (-y,x)$.
2. $-y = -1$ and $x = 3$.
3. Write the ordered pair.

Answer: $C'(-1,3)$.

Common Mistakes and Fixes

Common mistake	How to fix it
Adding the vector in the wrong order	Horizontal change affects x; vertical change affects y.
Changing both signs for every reflection	x-axis changes y only; y-axis changes x only.
Mixing rotation directions	Write and use the coordinate rule before substituting.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Translate $(1,2)$ by $(4,-3)$.

Check: $(1+4,2-3)=(5,-1)$.

2. Find the vector from $P(-2,1)$ to $P'(3,5)$.

Check: $(3-(-2),5-1)=(5,4)$.

3. Reflect $(6,-2)$ in the x-axis.

Check: $(6,2)$.

4. Rotate $(-2,5)$ 180° about the origin.

Check: $(2,-5)$.

5. Rotate $(4,-1)$ 270° anticlockwise.

Check: Use $(y,-x)$: $(-1,-4)$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Translate $A(0,3)$ by $(-2,5)$.

Working: _____

2. Find the vector from $(4,-3)$ to $(-1,2)$.

Working: _____

3. Reflect $(-7,4)$ in the x-axis.

Working: _____

4. Reflect $(-7,4)$ in the y-axis.

Working: _____

5. Rotate $(2,6)$ 90° anticlockwise.

Working: _____

6. Rotate $(2,6)$ 180° .

Working: _____

7. Rotate $(2,6)$ 270° anticlockwise.

Working: _____

8. Which transformation maps $(3, -4)$ to $(-3, -4)$?

Working: _____

9. Triangle vertices are $(1, 1), (4, 1), (1, 3)$. Translate by $(2, -1)$.

Working: _____

10. Explain why a reflection does not change side lengths.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $A'(-2,8)$.

2: $(-5,5)$.

3: $(-7,-4)$.

4: $(7,4)$.

5: $(-6,2)$.

6: $(-2,-6)$.

7: $(6,-2)$.

8: Reflection in the y-axis.

9: Images: $(3,0), (6,0), (3,2)$.

10: Each point is placed the same perpendicular distance on the opposite side of the mirror line, so distances between points are preserved.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Triangle $A(-1,2)$, $B(2,2)$, $C(1,5)$ is translated by $(4,-3)$. Find all image coordinates and the vector from B to B' . [6 marks]

Working space: _____

2. Reflect $P(-3,4)$, $Q(1,4)$, $R(1,-2)$ in the x-axis, then state one invariant. [6 marks]

Working space: _____

3. Rotate point $(5,-2)$ through 90° , 180° and 270° anticlockwise about the origin. [6 marks]

Working space: _____

4. An image of $A(2,-5)$ is $A'(-2,5)$. Identify the single rotation about the origin and justify. [4 marks]

Working space: _____

5. A point is translated from $(-4,1)$ to $(3,-2)$, reflected in the y-axis, then rotated 90° anticlockwise. Find the final point. [7 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 6 marks

Model solution: $A'(3, -1)$, $B'(6, -1)$, $C'(5, 2)$. Vector $B \rightarrow B' = (4, -3)$. (6)

Question 2 — 6 marks

Model solution: $P'(-3, -4)$, $Q'(1, -4)$, $R'(1, 2)$. Side lengths and angles are unchanged (also area). (6)

Question 3 — 6 marks

Model solution: 90° : $(2, 5)$; 180° : $(-5, 2)$; 270° : $(-2, -5)$. (6)

Question 4 — 4 marks

Model solution: $A' = (-x, -y)$, so the transformation is a 180° rotation about the origin. (4)

Question 5 — 7 marks

Model solution: Translation image is $(3, -2)$, with vector $(7, -3)$. Reflect y-axis: $(-3, -2)$. Rotate 90° ACW: $(2, -3)$. (7)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Translation (a,b) : $(x,y) \rightarrow (x+a,y+b)$.

Remember: Reflect x-axis: $(x,-y)$; y-axis: $(-x,y)$.

Remember: 90° ACW: $(-y,x)$; 180° : $(-x,-y)$; 270° ACW: $(y,-x)$.

Remember: Find a vector by image minus object.

Final Self-Check

- ☐ I can translate a point and find a translation vector.
- ☐ I can reflect points in the coordinate axes.
- ☐ I can rotate points about the origin and identify the image.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2

Geometry, Statistics, Consumer Arithmetic and Measurement

15 complete lessons

Term goal: Build confidence through clear models, repeated practice and cumulative exam preparation.

Term 2 Lesson Map

The table below shows the lesson sequence and its direct curriculum objective coverage.

No.	Unit / strand	Lesson	NSC objective coverage
T2.1	Unit 1: Geometry Geometry	Properties and Classification of Polygons	Determine the properties of n-sided polygons, where $3 \leq n \leq 10$.
T2.2	Unit 1: Geometry Geometry	Interior and Exterior Angles of Polygons	Find the sum of the interior angles of polygons with 3 to 10 sides.
T2.3	Unit 2: Statistics Statistics	Mean, Median and Mode from Frequency Tables	Determine the mode, median and mean from a frequency table.
T2.4	Unit 2: Statistics Statistics	Using Mean, Median and Mode to Interpret Information	Use the mode, median and mean to interpret information.
T2.5	Unit 2: Statistics Statistics	Pictographs, Bar Charts, Pie Charts and Line Graphs	Read, interpret and construct pictographs, bar charts, pie charts and line graphs.
T2.6	Unit 3: Numbers Numbers	Ratio, Proportion and Sharing	Solve simple problems involving ratio and proportion.
T2.7	Unit 4: Numbers Consumer Arithmetic	Bank Accounts, Deposit and Withdrawal Forms, and Cheques	Identify bank-account types and their features. • Complete withdrawal and deposit forms. • Complete information accurately on a cheque leaf.
T2.8	Unit 4: Numbers Consumer Arithmetic	Simple Interest on Loans and Deposits	Use proportions involving principal, rate and time to develop simple interest. • Calculate simple interest on loans and deposits.
T2.9	Unit 4: Numbers Consumer Arithmetic	Compound Interest by the Recursive Method	Calculate compound interest using a calculator and a recursive method.
T2.10	Unit 4: Numbers Consumer Arithmetic	Hire Purchase Cost and Comparison	Calculate total cost in a hire-purchase agreement. • Compare hire-purchase price and cash price.
T2.11	Unit 4: Numbers Consumer Arithmetic	Discounts and Taxes from Given Instructions	Calculate discounts and taxes from given instructions.
T2.12	Unit 5: Measurement Measurement	Areas of Parallelograms, Triangles and Trapezia	Derive and use formulae for the area of parallelograms, triangles and trapezia.
T2.13	Unit 5: Measurement Measurement	Total Surface Area of Prisms and Cylinders	Compute total surface area of cubes, cuboids, cylinders and triangular prisms.
T2.14	Unit 5: Measurement Measurement	Converting Square and Cubic Units	Perform conversions within units up to cubed units.
T2.15	Unit 5: Measurement Measurement	Volume and Capacity of Prisms, Cylinders and Composite Objects	Establish volume formulae. • Estimate and calculate volume and capacity of cubes, cuboids, prisms, cylinders and composite objects.

TERM 2 • LESSON 1

Properties and Classification of Polygons

Unit 1: Geometry | Geometry

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 1: Geometry
Strand	Geometry
Lesson	1: Properties and Classification of Polygons
NSC objective(s)	Determine the properties of n-sided polygons, where $3 \leq n \leq 10$.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Polygons appear in buildings, road signs, artwork, maps and design. Their properties help us classify and construct shapes accurately.

Learning Objectives — I Can

- I can name polygons with 3 to 10 sides.
- I can distinguish regular, irregular, convex and concave polygons.
- I can use side and angle properties to classify triangles and quadrilaterals.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Polygon	A closed flat shape made only from straight line segments.
Vertex	A corner where two sides meet.
Regular polygon	A polygon with all sides and all interior angles equal.
Convex	A polygon with every interior angle less than 180° .
Concave	A polygon with at least one interior angle greater than 180° .

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A polygon must be closed, flat and made only of straight sides. A circle is not a polygon, and an open chain of segments is not a polygon.

Names are based on the number of sides: triangle 3, quadrilateral 4, pentagon 5, hexagon 6, heptagon 7, octagon 8, nonagon 9 and decagon 10.

Regular means both equal sides and equal angles. A shape with only equal sides is not automatically regular unless its angles are also equal.

Small Steps to Follow

1. Check that the shape is closed and all edges are straight.
2. Count sides and vertices; these counts are equal for a polygon.
3. Use the side count to name the polygon.
4. Check equal-marked sides and angles to decide regular or irregular.
5. Look for an inward dent to decide concave or convex.

Notebook visual: Draw a table with columns: sides, name, sketch. Fill rows 3 through 10 and colour regular shapes purple and concave examples gold.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Classify a closed shape with 6 straight sides, all equal, and all equal angles.

1. Six sides means hexagon.
2. All sides and all angles are equal.
3. No inward dent is stated.

Answer: A regular convex hexagon.

Worked Example 2

Problem: A quadrilateral has one pair of parallel sides. Name its most specific common class.

1. It has four sides, so it is a quadrilateral.
2. One pair of parallel sides matches a trapezium in Jamaican/British usage.
3. No extra equal-side information is given.

Answer: A trapezium.

Worked Example 3

Problem: Explain why a five-point inward-dented shape can still be a polygon.

1. It is closed.
2. Every boundary part is a straight segment.
3. The inward dent makes an angle greater than 180° .

Answer: It is a concave polygon.

Common Mistakes and Fixes

Common mistake	How to fix it
Counting vertices twice	Trace once around the boundary and mark the starting point.
Calling equal sides alone regular	Regular requires equal sides and equal angles.
Treating a curved boundary as a side	Polygon sides must be straight segments.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Name a polygon with 7 sides.

Check: A heptagon.

2. Name a polygon with 10 sides.

Check: A decagon.

3. What two conditions make a polygon regular?

Check: All sides equal and all interior angles equal.

4. Is a circle a polygon? Explain.

Check: No. A polygon must have straight sides.

5. What feature immediately shows a polygon is concave?

Check: An inward dent, giving at least one interior angle greater than 180° .

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. How many vertices does an octagon have?

Working: _____

2. Name a 9-sided polygon.

Working: _____

3. Which is not a polygon? A. triangle B. circle C. pentagon D. rectangle

Working: _____

4. A shape has 5 sides of different lengths. Classify it.

Working: _____

5. A polygon has 4 equal sides and 4 right angles. Name it.

Working: _____

6. A quadrilateral has both pairs of opposite sides parallel. Name the broad class.

Working: _____

7. Can a regular polygon be concave? Explain.

Working: _____

8. A triangle has two equal sides. Name it.

Working: _____

9. A triangle has one 90° angle. Name it.

Working: _____

10. Explain the difference between a rhombus and a square.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: 8 vertices.
- 2: A nonagon.
- 3: B. Circle.
- 4: An irregular pentagon.
- 5: A square.
- 6: A parallelogram.
- 7: No at this level. A regular simple polygon has equal angles less than 180° and is convex.
- 8: An isosceles triangle.
- 9: A right-angled triangle.
- 10: Both have four equal sides, but a square must have four right angles; a rhombus need not.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Create a classification for a polygon with 8 equal sides, 8 equal angles and no inward dent. Justify every word. [6 marks]

Working space: _____

2. Compare a rectangle, rhombus and square using parallel sides, equal sides and right angles. [6 marks]

Working space: _____

3. A student calls every four-sided shape a rectangle. Correct the claim and give three other quadrilateral classes. [5 marks]

Working space: _____

4. A logo outline has 10 straight sides but crosses itself. Explain why classification may need care. [5 marks]

Working space: _____

5. A quadrilateral has diagonals that bisect each other, opposite sides parallel and all sides equal, but no right angles. Identify it and state two properties. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 6 marks

Model solution: It is an octagon because it has 8 sides; regular because all sides and angles are equal; convex because there is no inward dent and all interior angles are below 180° . (6)

Question 2 — 6 marks

Model solution: All have two pairs of parallel sides. Rectangle: opposite sides equal, four right angles. Rhombus: all sides equal, opposite angles equal. Square: all sides equal and four right angles, so it is both a rectangle and rhombus. (6)

Question 3 — 5 marks

Model solution: Four-sided shapes are quadrilaterals. Rectangles are only those with four right angles. Other classes include trapezium, parallelogram, rhombus, square or kite. (5)

Question 4 — 5 marks

Model solution: The Grade 8 polygon definition normally assumes a simple closed boundary that does not cross itself. Although ten segments are present, a self-crossing outline is not a simple decagon and its interior is ambiguous. (5)

Question 5 — 6 marks

Model solution: It is a rhombus. Properties include all sides equal, opposite sides parallel, opposite angles equal, and diagonals bisect at right angles. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: 3 triangle; 4 quadrilateral; 5 pentagon; 6 hexagon.

Remember: 7 heptagon; 8 octagon; 9 nonagon; 10 decagon.

Remember: Regular = all sides and all angles equal.

Remember: Concave has an inward dent; convex does not.

Final Self-Check

- ☐ I can name polygons with 3 to 10 sides.
- ☐ I can distinguish regular, irregular, convex and concave polygons.
- ☐ I can use side and angle properties to classify triangles and quadrilaterals.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 2

Interior and Exterior Angles of Polygons

Unit 1: Geometry | Geometry

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 1: Geometry
Strand	Geometry
Lesson	2: Interior and Exterior Angles of Polygons
NSC objective(s)	Find the sum of the interior angles of polygons with 3 to 10 sides.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Angle sums help designers calculate turns, build frames and check whether polygon measurements are possible.

Learning Objectives — I Can

- I can calculate the interior-angle sum of a polygon.
- I can find a missing interior angle.
- I can find each interior or exterior angle of a regular polygon.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Interior angle	An angle inside a polygon at a vertex.
Exterior angle	The turning angle formed outside a polygon.
Angle sum	The total of a set of angles.
Regular	Having all equal sides and angles.
Triangulate	Divide a polygon into non-overlapping triangles.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

An n -sided polygon can be divided from one vertex into $n-2$ triangles. Each triangle totals 180° , so the interior-angle sum is $(n-2) \times 180^\circ$.

For a regular polygon, each interior angle is the total divided by n . The exterior angles of any simple polygon make one full turn and total 360° .

At one vertex, an interior angle and its adjacent exterior angle total 180° .

Small Steps to Follow

1. Count the number of sides n .
2. Calculate $n-2$.
3. Multiply by 180° for the interior sum.
4. For a regular polygon, divide the sum by n .
5. Use exterior = 180° - interior or $360^\circ \div n$ for a regular polygon.

Notebook visual: Draw a pentagon and diagonals from one vertex to form three triangles. Write $3 \times 180^\circ = 540^\circ$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Find the interior-angle sum of a hexagon.

1. A hexagon has $n=6$.
2. $n-2=4$ triangles.
3. $4 \times 180^\circ = 720^\circ$.

Answer: 720° .

Worked Example 2

Problem: Find each interior angle of a regular octagon.

1. Sum = $(8-2) \times 180 = 1,080^\circ$.
2. There are 8 equal angles.
3. $1,080 \div 8 = 135^\circ$.

Answer: Each interior angle is 135° .

Worked Example 3

Problem: A quadrilateral has angles 90° , 82° , 104° and x . Find x .

1. Quadrilateral sum $= (4-2) \times 180 = 360^\circ$.
2. Known total $= 90 + 82 + 104 = 276^\circ$.
3. $x = 360 - 276 = 84^\circ$.

Answer: $x = 84^\circ$.

Common Mistakes and Fixes

Common mistake	How to fix it
Using $n \times 180$	A polygon forms $n-2$ triangles, not n .
Dividing for an irregular polygon	Only regular polygons have equal angles.
Confusing interior and exterior	Adjacent interior and exterior angles total 180° .

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Find the interior sum of a pentagon.

Check: $(5-2) \times 180 = 540^\circ$.

2. Find the interior sum of a decagon.

Check: $(10-2) \times 180 = 1,440^\circ$.

3. Find each interior angle of a regular hexagon.

Check: $720 \div 6 = 120^\circ$.

4. Find each exterior angle of a regular nonagon.

Check: $360 \div 9 = 40^\circ$.

5. A triangle has angles 48° , 67° and x . Find x .

Check: $x = 180 - 48 - 67 = 65^\circ$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Find the interior sum of a heptagon.

Working: _____

2. Find the interior sum of a 9-sided polygon.

Working: _____

3. Which formula gives the sum? A. $n \times 180$ B. $(n-2) \times 180$ C. $360/n$ D. $180/n$

Working: _____

4. Find each interior angle of a regular pentagon.

Working: _____

5. Find each exterior angle of a regular decagon.

Working: _____

6. A regular polygon has exterior angle 45° . Find n .

Working: _____

7. A pentagon has angles 100° , 110° , 95° , 120° and x . Find x .

Working: _____

8. Can a quadrilateral have four angles of 100° ?

Working: _____

9. Find the exterior angle adjacent to an interior angle of 132° .

Working: _____

10. Explain why a decagon forms eight triangles from one vertex.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $(7-2) \times 180 = 900^\circ$.

2: $(9-2) \times 180 = 1,260^\circ$.

3: B. $(n-2) \times 180^\circ$.

4: $540 \div 5 = 108^\circ$.

5: $360 \div 10 = 36^\circ$.

6: $n = 360 \div 45 = 8$, so it is an octagon.

7: $x = 540 - 425 = 115^\circ$.

8: No. Their total would be 400° , but a quadrilateral must total 360° .

9: $180 - 132 = 48^\circ$.

10: The two sides meeting the chosen vertex are already boundaries, and diagonals to the remaining non-adjacent vertices divide it into $n-2=8$ triangles.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Find the interior sum and each interior angle of a regular heptagon. [5 marks]

Working space: _____

2. The angles of a pentagon are x , $x+10$, $x+20$, $x+30$ and $x+40$ degrees. Find all angles. [7 marks]

Working space: _____

3. A regular polygon has each interior angle 150° . Determine its number of sides. [5 marks]

Working space: _____

4. Three exterior angles of a polygon are 70° , 85° and 90° . The remaining two are equal. Find them. [5 marks]

Working space: _____

5. A builder proposes a regular 9-sided frame with each interior angle 140° . Check the proposal. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 5 marks

Model solution: $\text{Sum} = (7-2) \times 180 = 900^\circ$. $\text{Each} = 900 \div 7 = 128 \frac{4}{7}^\circ \approx 128.57^\circ$. (5)

Question 2 — 7 marks

Model solution: $\text{Sum} = 540$. $5x + 100 = 540$, so $x = 88$. Angles: $88^\circ, 98^\circ, 108^\circ, 118^\circ, 128^\circ$. (7)

Question 3 — 5 marks

Model solution: $\text{Exterior} = 180 - 150 = 30^\circ$. $n = 360 \div 30 = 12$ sides. (5)

Question 4 — 5 marks

Model solution: $\text{Exterior total} = 360$. $\text{Known} = 245$, $\text{remaining} = 115$. $\text{Each} = 57.5^\circ$. (5)

Question 5 — 6 marks

Model solution: A regular nonagon has sum $1,260^\circ$, so each angle $= 1,260 \div 9 = 140^\circ$. The proposal is correct; exterior angle is 40° and $9 \times 40 = 360^\circ$. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Interior sum $= (n-2) \times 180^\circ$.

Remember: Regular interior angle $= [(n-2) \times 180^\circ] / n$.

Remember: Exterior angles total 360° .

Remember: Regular exterior angle $= 360^\circ / n$.

Remember: Interior + adjacent exterior $= 180^\circ$.

Final Self-Check

- ☐ I can calculate the interior-angle sum of a polygon.
- ☐ I can find a missing interior angle.
- ☐ I can find each interior or exterior angle of a regular polygon.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 3

Mean, Median and Mode from Frequency Tables

Unit 2: Statistics | Statistics

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 2: Statistics
Strand	Statistics
Lesson	3: Mean, Median and Mode from Frequency Tables
NSC objective(s)	Determine the mode, median and mean from a frequency table.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Frequency tables summarize repeated data, such as attendance, rainfall or test scores, without listing every observation separately.

Learning Objectives — I Can

- I can find total frequency and modal value.
- I can locate the median using cumulative frequency.
- I can calculate the mean using an fx column.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Frequency	How many times a value occurs.
Mode	The value with the highest frequency.
Median	The middle value after data are ordered.
Mean	Total of all values divided by the number of values.
Cumulative frequency	A running total of frequencies.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

In a frequency table, the mode is the value beside the largest frequency—not the frequency itself.

The median position depends on the total frequency. For odd N , use position $(N+1)/2$. For even N , average the values at positions $N/2$ and $N/2+1$.

For the mean, create an fx column by multiplying each value x by its frequency f . Then $\text{mean} = \Sigma fx / \Sigma f$.

Small Steps to Follow

1. Add all frequencies to find N .
2. Find the mode from the largest frequency.
3. Use cumulative frequency to locate the middle position or positions.
4. Multiply $x \times f$ in each row and add to find Σfx .
5. Divide Σfx by N and include sensible units.

Notebook visual: Draw a four-column table: value x , frequency f , cumulative frequency, product fx . Colour the largest f and the median position.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Scores 1,2,3 have frequencies 2,5,3. Find the mode.

1. Compare frequencies 2,5,3.
2. The largest frequency is 5.
3. It belongs to score 2.

Answer: Mode=2.

Worked Example 2

Problem: For the same table, find the median.

1. $N = 2 + 5 + 3 = 10$.
2. Middle positions are 5th and 6th.
3. Positions 3 through 7 are score 2.

Answer: Median=2.

Worked Example 3

Problem: For the same table, find the mean.

1. $\Sigma fx = 1 \times 2 + 2 \times 5 + 3 \times 3 = 2 + 10 + 9 = 21.$

2. $\Sigma f = 10.$

3. $\text{Mean} = 21 \div 10.$

Answer: Mean = 2.1.

Common Mistakes and Fixes

Common mistake	How to fix it
Choosing the largest frequency as the mode	Choose the value paired with it.
Ignoring repeated values for the median	Use cumulative positions.
Dividing Σfx by the number of rows	Divide by total frequency Σf .

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Values 2,3,4 have frequencies 1,4,2. Find N.

Check: $N=1+4+2=7$.

2. Find the mode for that table.

Check: Mode=3 because its frequency 4 is highest.

3. Find the median for that table.

Check: $N=7$; median is 4th value. Positions 2–5 are 3, so median=3.

4. Find Σfx for that table.

Check: $2 \times 1 + 3 \times 4 + 4 \times 2 = 2 + 12 + 8 = 22$.

5. Find the mean.

Check: Mean = $22 \div 7 \approx 3.14$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Values 0,1,2,3 have frequencies 1,3,5,1. Find the mode.

Working: _____

2. Find total frequency for the same table.

Working: _____

3. Find the median for the same table.

Working: _____

4. Find the mean for the same table.

Working: _____

5. Which column is needed for the mean? A. $f+x$ B. fx C. $f-x$ D. x/f

Working: _____

6. Values 5,10,15 have frequencies 2,1,1. Find the mean.

Working: _____

7. Values 1,2,3,4 have frequencies 2,2,2,2. Find the mode.

Working: _____

8. A table has $N=11$. What is the median position?

Working: _____

9. A table has $N=20$. Which positions determine the median?

Working: _____

10. Explain why the largest frequency is not automatically the mode.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: Mode=2.
- 2: $N=1+3+5+1=10$.
- 3: 5th and 6th positions both have value 2, so median=2.
- 4: $\Sigma fx=0+3+10+3=16$; mean= $16/10=1.6$.
- 5: B. fx.
- 6: $\Sigma fx=10+10+15=35$; $N=4$; mean=8.75.
- 7: There is no single mode; all values have the same frequency.
- 8: $(11+1)/2=6$ th position.
- 9: 10th and 11th positions.
- 10: The mode is the data value attached to the largest frequency; the frequency only tells how often it occurs.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Number of books 0,1,2,3,4 has frequencies 2,5,6,4,3. Find mean, median and mode. [8 marks]

Working space: _____

2. A score table has values 2,4,6,8 and frequencies 3,5,k,2. If total frequency is 14, find k and the mode. [5 marks]

Working space: _____

3. Travel times 10,20,30 minutes have frequencies 4,3,3. Calculate the mean and interpret it. [5 marks]

Working space: _____

4. A frequency table has values 1,2,3 with frequencies 2,6,2. One frequency-2 entry at value 3 is accidentally changed to value 30. Explain the effect on mode, median and mean. [6 marks]

Working space: _____

5. Construct an fx table for values 4,5,6,7 with frequencies 1,3,4,2 and verify that the mean lies between 4 and 7. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 8 marks

Model solution: $N=20$. $\Sigma fx=0+5+12+12+12=41$; $\text{mean}=2.05$. Positions 10 and 11 fall at value 2, so $\text{median}=2$. $\text{Mode}=2$. (8)

Question 2 — 5 marks

Model solution: $3+5+k+2=14$, so $k=4$. Largest frequency is 5 at value 4, so $\text{mode}=4$. (5)

Question 3 — 5 marks

Model solution: $\Sigma fx=40+60+90=190$; $N=10$; $\text{mean}=19$ minutes. The average travel time is 19 minutes. (5)

Question 4 — 6 marks

Model solution: Mode remains 2 and median remains 2 because positions barely change. Mean rises greatly because the total increases by 27 for each changed observation; outliers strongly affect mean. (6)

Question 5 — 6 marks

Model solution: fx values: 4, 15, 24, 14; $\Sigma fx=57$; $N=10$; $\text{mean}=5.7$, which lies between minimum 4 and maximum 7. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: $N = \sum f$.

Remember: Mode = value with largest f .

Remember: Median: locate middle position using cumulative frequency.

Remember: Mean = $\sum fx / \sum f$.

Remember: A mean must lie between the smallest and largest values.

Final Self-Check

- ☐ I can find total frequency and modal value.
- ☐ I can locate the median using cumulative frequency.
- ☐ I can calculate the mean using an fx column.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 4

Using Mean, Median and Mode to Interpret Information

Unit 2: Statistics | Statistics

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 2: Statistics
Strand	Statistics
Lesson	4: Using Mean, Median and Mode to Interpret Information
NSC objective(s)	Use the mode, median and mean to interpret information.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Different averages answer different questions. Choosing the right one prevents misleading conclusions about marks, income, prices and performance.

Learning Objectives — I Can

- I can choose the most useful average for a situation.
- I can explain how an outlier affects averages.
- I can compare two data sets using more than one statistic.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Typical	A value that reasonably represents a data set.
Outlier	A value much higher or lower than most others.
Distribution	How values are spread across a data set.
Representative	Giving a fair picture of the whole group.
Range	Largest value minus smallest value.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

The mean uses every value and is useful for balanced numerical data, but it is affected by outliers. The median is resistant to extreme values and often better for income or house prices.

The mode is best for the most common category, size or choice. It can be used with words as well as numbers.

A fair comparison should include a measure of centre and, when possible, a measure of spread such as range.

Small Steps to Follow

1. Read the question: most common, middle, or fair share?
2. Check for outliers and whether the data are numerical or categorical.
3. Calculate or identify the relevant averages.
4. Compare the statistics in context, with units.
5. State a cautious conclusion and mention any limitation.

Notebook visual: Draw a decision tree: most common→mode; extreme values present→median; balanced numeric data/fair share→mean.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Choose an average for shoe sizes a shop should stock most.

1. The shop wants the most common size.
2. Shoe size is a repeatable category.
3. Use the value occurring most often.

Answer: Use the mode.

Worked Example 2

Problem: Choose an average for household incomes 60k, 62k, 65k, 68k, 900k.

1. 900k is an extreme outlier.
2. It pulls the mean upward.
3. The middle income is less affected.

Answer: Use the median for a typical household.

Worked Example 3

Problem: Compare Class A mean 70, range 12 with Class B mean 70, range 32.

1. The classes have equal mean performance.
2. Class A has a much smaller range.
3. Its scores are more consistent.

Answer: Same average, but Class A is more consistent.

Common Mistakes and Fixes

Common mistake	How to fix it
Always choosing mean	Check outliers and the purpose of the question.
Comparing only one statistic	Use centre and spread when possible.
Claiming causation	Averages describe data; they do not by themselves prove why a difference exists.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Best average for the most popular lunch choice?

Check: Mode.

2. Best average for house prices with one mansion?

Check: Median.

3. Best average for sharing 50 sweets equally among 10 students?

Check: Mean=5 sweets.

4. What happens to the mean when one very high outlier is added?

Check: The mean usually increases.

5. Two teams have equal means but ranges 4 and 18. Which is more consistent?

Check: The team with range 4.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Choose mode, median or mean for the most common bus route.

Working: _____

2. Choose an average for typical waiting time when one day had a 3-hour delay.

Working: _____

3. Data 4,5,5,6,30: find median and mean.

Working: _____

4. Which better represents those data?

Working: _____

5. If every value increases by 2, what happens to mean and median?

Working: _____

6. If the highest value increases, which average is most likely unchanged?

Working: _____

7. Class X mean=64, median=72. What might this suggest?

Working: _____

8. Can data have two modes?

Working: _____

9. Why is range not an average?

Working: _____

10. A seller advertises the mean salary as J\$200,000, but most workers earn J\$80,000. What should be checked?

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: Mode.
- 2: Median, because the extreme delay affects the mean.
- 3: Median=5. Mean= $50/5=10$.
- 4: Median 5, because 30 is an outlier.
- 5: Both increase by 2.
- 6: The mode and often median; the mean changes.
- 7: Some low values may be pulling the mean below the median.
- 8: Yes. If two values share the highest frequency, the data are bimodal.
- 9: It measures spread, not the centre or typical value.
- 10: Check for a few very high salaries and compare the median and mode; the mean may be misleading.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Five weekly sales are 20,22,23,24,91. Calculate mean and median and recommend the better typical value. [6 marks]

Working space: _____

2. Two classes have: A mean 72, median 72, range 10; B mean 74, median 68, range 40. Compare them fairly. [7 marks]

Working space: _____

3. A data set has mode 5. One additional value 9 is added. Explain what can be said for certain about mean, median and mode. [6 marks]

Working space: _____

4. Explain why a clothing store uses the mode rather than mean for T-shirt sizes. [5 marks]

Working space: _____

5. Create a conclusion from two communities: both median travel time 25 min; ranges 12 and 45 min. [5 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 6 marks

Model solution: Mean= $180/5=36$. Median=23. The median is better because 91 is an outlier. (6)

Question 2 — 7 marks

Model solution: B has the higher mean, but its lower median and large range suggest extreme high marks raise the mean and performance is less consistent. A is more consistent and its mean and median agree. (7)

Question 3 — 6 marks

Model solution: Mean moves toward 9 (and rises if original mean <9). Median may or may not change depending on size/order. Mode may remain 5 unless frequencies create a tie; not enough information for exact values. (6)

Question 4 — 5 marks

Model solution: Sizes are categories and the store needs the most frequently demanded size. A numerical mean might produce a size that few or no customers wear. (5)

Question 5 — 5 marks

Model solution: Typical travel time is the same by median. The community with range 12 has more consistent travel times; the range-45 community experiences greater variation. More data would be needed to explain why. (5)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Mean: fair share; uses every value; affected by outliers.

Remember: Median: middle; resistant to outliers.

Remember: Mode: most common; works for categories.

Remember: Range: largest–smallest; measures spread.

Remember: Write conclusions in context with units.

Final Self-Check

- ☐ I can choose the most useful average for a situation.
- ☐ I can explain how an outlier affects averages.
- ☐ I can compare two data sets using more than one statistic.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 5

Pictographs, Bar Charts, Pie Charts and Line Graphs

Unit 2: Statistics | Statistics

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 2: Statistics
Strand	Statistics
Lesson	5: Pictographs, Bar Charts, Pie Charts and Line Graphs
NSC objective(s)	Read, interpret and construct pictographs, bar charts, pie charts and line graphs.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Graphs communicate information quickly in school reports, weather records, business data and public-health dashboards.

Learning Objectives — I Can

- I can choose an appropriate graph for a data set.
- I can read scales, keys and sector angles accurately.
- I can construct a labelled graph and explain its main pattern.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Scale	The value represented by equal intervals on an axis.
Key	An explanation of what a symbol represents.
Category	A named group in data.
Sector	A slice of a pie chart.
Trend	The general direction of change over time.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

Pictographs use repeated symbols and require a key. Bar charts compare separate categories and must begin with clearly labelled axes and equal scale intervals.

Pie charts show parts of a whole. $\text{Sector angle} = (\text{category frequency} \div \text{total}) \times 360^\circ$. All sectors must total 360° .

Line graphs show change over ordered time. Plot points at correct coordinates and join in order; do not claim values outside the displayed data without caution.

Small Steps to Follow

1. Identify whether the data are categories, parts of a whole or change over time.
2. Choose pictograph/bar for categories, pie for composition, line for time trend.
3. Create a clear title, labels, scale or key.
4. Plot accurately and check totals or axis positions.
5. State at least one observation supported by the graph.

Notebook visual: Draw four mini-panels labelled pictograph=symbols, bar=separate bars, pie=whole 360° , line=time points joined.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: A pictograph key says ★=4 students. What do $3\frac{1}{2}$ stars represent?

1. Three stars = $3 \times 4 = 12$.
2. Half star = 2.
3. Total = 14.

Answer: 14 students.

Worked Example 2

Problem: A pie chart category has 15 of 60 responses. Find its angle.

1. Fraction = $15/60 = 1/4$.
2. Angle = $1/4 \times 360^\circ$.
3. $360 \div 4 = 90^\circ$.

Answer: 90° .

Worked Example 3

Problem: Rainfall rises 20,35,50,42 mm over four months. Describe the line trend.

1. It rises from 20 to 50 over first three months.
2. It then falls to 42.
3. The maximum is 50 in month 3.

Answer: Overall rise followed by a small decline.

Common Mistakes and Fixes

Common mistake	How to fix it
Missing key or scale	A reader cannot interpret symbols or heights without it.
Pie angles not totaling 360°	Recheck every frequency and total.
Using a line graph for unordered categories	Lines imply an ordered or continuous progression.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Best graph for favourite subjects by category?

Check: A bar chart or pictograph.

2. Best graph for monthly temperature?

Check: A line graph.

3. Best graph for how a budget is divided?

Check: A pie chart.

4. Key $\bullet = 6$ books. What do $2\frac{1}{2}$ circles represent?

Check: $2 \times 6 + 3 = 15$ books.

5. Find the pie angle for 20 out of 80 students.

Check: $20/80 \times 360 = 90^\circ$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. What must every pictograph include?

Working: _____

2. Why should bar-chart bars usually be separated?

Working: _____

3. A bar reaches halfway between 40 and 50. Read the value.

Working: _____

4. Find the pie angle for frequency 12 out of total 48.

Working: _____

5. A sector angle is 72° in a total of 50 people. Find the frequency.

Working: _____

6. Pie sectors are $90^\circ, 120^\circ, 80^\circ$ and x . Find x .

Working: _____

7. Line graph values are 5, 8, 8, 3. When is there no change?

Working: _____

8. Which graph best shows daily exchange rate over a month?

Working: _____

9. List four required graph features.

Working: _____

10. Explain why a truncated vertical axis can mislead.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: A clear key.
- 2: They represent distinct categories, not continuous intervals.
- 3: 45, if the scale is linear.
- 4: $12/48 \times 360 = 90^\circ$.
- 5: $72/360 \times 50 = 10$ people.
- 6: $x = 360 - 290 = 70^\circ$.
- 7: Between the second and third points.
- 8: A line graph.
- 9: Suitable title, labelled axes/categories, accurate scale/key, and correctly plotted data.
- 10: Starting far above zero can exaggerate small differences in bar heights.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A survey of 120 students gives Math 42, English 36, Science 30, Arts 12. Calculate all pie-chart angles. [8 marks]

Working space: _____

2. A pictograph uses one mango icon for 8 crates. Monday has $4\frac{1}{2}$ icons, Tuesday $3\frac{3}{4}$, Wednesday 5. Find each total and the three-day total. [6 marks]

Working space: _____

3. Monthly rainfall is Jan 45, Feb 30, Mar 70, Apr 65 mm. Describe a correct line graph and two conclusions. [6 marks]

Working space: _____

4. A bar chart shows Clinic A=85 and Clinic B=90 but begins its y-axis at 80. Explain the visual risk and suggest a correction. [5 marks]

Working space: _____

5. Choose and justify a graph for (a) parish population by parish, (b) a family's spending shares, and (c) temperature each hour. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 8 marks

Model solution: Math= $42/120 \times 360 = 126^\circ$; English= 108° ; Science= 90° ; Arts= 36° . Total= 360° . (8)

Question 2 — 6 marks

Model solution: Monday 36; Tuesday 26; Wednesday 40; total=102 crates. (6)

Question 3 — 6 marks

Model solution: Plot months in order on x-axis and rainfall on a labelled equal scale y-axis. Rainfall falls Jan–Feb, rises sharply to a March maximum of 70, then falls slightly in April to 65. (6)

Question 4 — 5 marks

Model solution: The 5-unit difference may look very large because bars show only 5 and 10 units above the baseline. Start at zero or clearly mark an axis break and report exact values. (5)

Question 5 — 6 marks

Model solution: (a) bar chart for categories; (b) pie chart for parts of one total; (c) line graph for ordered time change. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Pictograph: symbols + key.

Remember: Bar chart: compare categories.

Remember: Pie angle = $\text{frequency} / \text{total} \times 360^\circ$.

Remember: Line graph: change over ordered time.

Remember: Always title, label, scale and check.

Final Self-Check

- ☐ I can choose an appropriate graph for a data set.
- ☐ I can read scales, keys and sector angles accurately.
- ☐ I can construct a labelled graph and explain its main pattern.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 6

Ratio, Proportion and Sharing

Unit 3: Numbers | Numbers

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 3: Numbers
Strand	Numbers
Lesson	6: Ratio, Proportion and Sharing
NSC objective(s)	Solve simple problems involving ratio and proportion.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Ratios compare quantities in recipes, maps, mixtures, team sizes, prices and scale drawings.

Learning Objectives — I Can

- I can simplify and form ratios in the same units.
- I can share a quantity in a given ratio.
- I can solve a direct proportion using unit rate or equivalent ratios.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Ratio	A comparison of quantities by division.
Proportion	A statement that two ratios are equal.
Unit rate	The amount for one unit.
Equivalent ratios	Ratios with the same value.
Scale factor	The multiplier connecting proportional quantities.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A ratio compares quantities in the same order and same units. Simplify by dividing every part by a common factor.

To share a total in ratio $a:b$, add the parts, find the value of one part, then multiply by a and b .

In direct proportion, multiplying one quantity by a factor multiplies the other by the same factor. Unit rate and cross multiplication are two valid methods.

Small Steps to Follow

1. Write quantities in the same units and correct order.
2. Simplify ratios by a common factor.
3. For sharing: add ratio parts and divide the total by that sum.
4. For proportion: find one-unit value or equivalent scale factor.
5. Check that shared parts total the original amount.

Notebook visual: Draw a bar split into 2 purple parts and 3 gold parts. Label total parts 5 and show how a total is divided by 5 first.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Simplify 18:24.

1. Greatest common factor is 6.
2. $18 \div 6 = 3$.
3. $24 \div 6 = 4$.

Answer: $18:24 = 3:4$.

Worked Example 2

Problem: Share J\$7,500 in ratio 2:3.

1. Total parts $= 2 + 3 = 5$.
2. One part $= 7,500 \div 5 = 1,500$.
3. Shares: $2 \times 1,500$ and $3 \times 1,500$.

Answer: J\$3,000 and J\$4,500.

Worked Example 3

Problem: If 4 books cost J\$2,600, find the cost of 7 at the same rate.

1. Unit cost= $2,600 \div 4 = 650$.
2. Seven cost 7×650 .
3. $7 \times 650 = 4,550$.

Answer: J\$4,550.

Common Mistakes and Fixes

Common mistake	How to fix it
Comparing different units	Convert units before forming the ratio.
Dividing the total by one ratio number	Divide by the sum of all ratio parts.
Reversing order	Keep the ratio in the order named in the question.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Simplify 15:25.

Check: Divide by 5: 3:5.

2. Write 2 m:50 cm in simplest form.

Check: 2 m=200 cm. 200:50=4:1.

3. Share 72 in ratio 5:3.

Check: 8 parts; one part=9. Shares=45 and 27.

4. 3 kg costs J\$1,800. Find 5 kg.

Check: Unit cost=600; 5 kg=J\$3,000.

5. If 6 workers take 4 days for a direct-proportion task measure, find the equivalent worker-days.

Check: $6 \times 4 = 24$ worker-days. Note actual time/workers is inverse proportion, so context must be interpreted carefully.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Simplify 28:42.

Working: _____

2. Simplify 0.6:1.5.

Working: _____

3. Write 3 hours:45 minutes in simplest form.

Working: _____

4. Share 240 in ratio 1:3.

Working: _____

5. Share J\$18,000 in ratio 2:3:4.

Working: _____

6. 5 notebooks cost J\$1,250. Find 8 notebooks.

Working: _____

7. A recipe uses flour:sugar=4:1. If flour is 600 g, find sugar.

Working: _____

8. Solve $x/12=5/3$.

Working: _____

9. Are 14:21 and 8:12 proportional?

Working: _____

10. A map scale is 1:50,000. What actual distance is 6 cm?

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: 2:3.

2: Multiply by 10: 6:15; simplify to 2:5.

3: $180:45=4:1$.

4: 60 and 180.

5: 9 parts; one=2,000. Shares=4,000;6,000;8,000.

6: Unit=250; cost=J\$2,000.

7: One ratio part= $600 \div 4 = 150$ g sugar.

8: $3x=60$, so $x=20$.

9: Both simplify to 2:3, so yes.

10: $6 \times 50,000 = 300,000$ cm=3 km.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A drink mixture uses syrup:water=2:7. Make 4.5 L. Find each amount. [5 marks]

Working space: _____

2. Three students share J\$24,000 in ratio 3:4:5. Find each share and verify. [6 marks]

Working space: _____

3. Eight identical tiles cover 1.6 m^2 . How many cover 5 m^2 at the same rate? [5 marks]

Working space: _____

4. A photo width:height is 4:3. If height becomes 27 cm, find width. Then explain whether 40 cm by 30 cm has the same ratio. [6 marks]

Working space: _____

5. A scale drawing uses 1 cm:2.5 m. A rectangular yard measures 7.2 cm by 4.8 cm. Find actual dimensions and perimeter. [7 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 5 marks

Model solution: Total parts=9; one= $4.5 \div 9 = 0.5$ L. Syrup=1.0 L; water=3.5 L. (5)

Question 2 — 6 marks

Model solution: Total parts=12; one=2,000. Shares=6,000;8,000;10,000. Sum=24,000. (6)

Question 3 — 5 marks

Model solution: Tiles per $m^2 = 8 \div 1.6 = 5$. For 5 m^2 : 25 tiles. (5)

Question 4 — 6 marks

Model solution: Width= $27 \times 4/3 = 36$ cm. 40:30=4:3, so it is proportional. (6)

Question 5 — 7 marks

Model solution: Dimensions=18 m by 12 m. Perimeter= $2(18+12) = 60$ m. (7)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Simplify all ratio parts by the same factor.

Remember: Use the same units first.

Remember: Sharing: total parts → one part → each share.

Remember: Direct proportion: same scale factor or unit rate.

Remember: Check shares add to the total.

Final Self-Check

- ☐ I can simplify and form ratios in the same units.
- ☐ I can share a quantity in a given ratio.
- ☐ I can solve a direct proportion using unit rate or equivalent ratios.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 7

Bank Accounts, Deposit and Withdrawal Forms, and Cheques

Unit 4: Numbers | Consumer Arithmetic

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 4: Numbers
Strand	Consumer Arithmetic
Lesson	7: Bank Accounts, Deposit and Withdrawal Forms, and Cheques
NSC objective(s)	Identify bank-account types and their features. • Complete withdrawal and deposit forms. • Complete information accurately on a cheque leaf.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Banking literacy helps students save safely, read forms, avoid errors and understand everyday financial transactions in Jamaica.

Learning Objectives — I Can

- I can compare common account types.
- I can identify the required fields on deposit and withdrawal forms.
- I can complete and check the main parts of a cheque safely.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Savings account	An account mainly used to store money and earn some interest.
Current account	An account designed for frequent payments and withdrawals, often with cheque access.
Fixed deposit	Money kept for a stated time, usually for a higher interest rate.
Payee	The person or organisation receiving payment.
Drawer	The account holder who writes and signs a cheque.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

Account features differ by access, fees, interest, minimum balance and purpose. Always compare current bank terms because rates and fees can change.

A deposit form normally needs date, account name/number, cash and cheque amounts, total, and depositor details. A withdrawal form needs account details, amount, date and authorised signature.

A cheque needs the date, payee, amount in figures, amount in words, and drawer's signature. The words and figures must agree; blank spaces should be closed to reduce alteration risk.

Small Steps to Follow

1. Choose the account or form that matches the transaction.
2. Write names and account numbers exactly as held by the institution.
3. Enter money with dollars and cents in the correct boxes.
4. Add cash and cheque amounts and verify the total.
5. For a cheque, match words and figures, cross unused space and sign only after checking.

Notebook visual: Draw a safe sample cheque with arrows to DATE, PAYEE, AMOUNT IN WORDS, AMOUNT IN FIGURES and SIGNATURE. Use fictional details only.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Choose an account for emergency savings with easy access.

1. The goal is saving, not daily business payments.
2. Money must remain accessible.
3. A fixed deposit may restrict access.

Answer: A regular savings account is usually the best starting match; compare actual fees and rates.

Worked Example 2

Problem: Complete the total on a deposit slip: cash J\$12,500; cheque J\$8,750.

1. Write each amount in its correct section.
2. Add $12,500 + 8,750$.
3. Check the column addition.

Answer: Total deposit=J\$21,250.

Worked Example 3

Problem: Write J\$14,506.25 in cheque words.

1. Write the dollar part: Fourteen thousand five hundred and six dollars.
2. Write cents clearly: twenty-five cents.
3. Close unused space with a line.

Answer: Fourteen thousand five hundred and six dollars and twenty-five cents only.

Common Mistakes and Fixes

Common mistake	How to fix it
Using real account details in practice	Use fictional information in learning materials.
Leaving cheque spaces blank	Close unused space to reduce alteration risk.
Not reconciling totals	Add every cash/cheque entry and verify before signing.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Which account is mainly for frequent business payments?

Check: A current account.

2. Which account commonly holds money for a fixed term?

Check: A fixed-deposit account.

3. Cash J\$7,000 plus cheque J\$3,450 gives what deposit total?

Check: J\$10,450.

4. Who is the payee on a cheque?

Check: The person or organisation receiving the payment.

5. Why must cheque words and figures match?

Check: A mismatch may cause rejection or create fraud/error risk.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. State one usual purpose of a savings account.

Working: _____

2. State one usual feature of a current account.

Working: _____

3. Which account may offer higher interest for restricted access?

Working: _____

4. List four common fields on a deposit slip.

Working: _____

5. Deposit J\$9,825 cash and J\$2,175 cheque. Find total.

Working: _____

6. Withdrawal balance is J\$30,000. After J\$7,500 withdrawal, what remains before fees?

Working: _____

7. Write J\$2,040.00 in words.

Working: _____

8. Name the person who writes and signs a cheque.

Working: _____

9. Should a cheque be signed while blank? Explain.

Working: _____

10. A cheque says J\$5,600 in figures and 'five thousand sixty dollars' in words. What should happen?

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: To store savings securely and possibly earn interest.
- 2: It supports frequent transactions and may allow cheque payments.
- 3: A fixed deposit.
- 4: Date, account name, account number, cash/cheque amounts, total, depositor details; any four.
- 5: J\$12,000.
- 6: J\$22,500.
- 7: Two thousand and forty dollars only.
- 8: The drawer.
- 9: No. A signed blank cheque can be altered and creates serious loss/fraud risk.
- 10: Do not issue it. Void/rewrite correctly because the amounts do not agree.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Compare savings, current and fixed-deposit accounts using purpose, access and likely interest. [7 marks]

Working space: _____

2. A deposit contains cash J\$18,400 and cheques J\$7,250 and J\$3,350. Calculate the deposit total and state two checks before submission. [6 marks]

Working space: _____

3. Prepare the content for a cheque dated 10 August 2026 to Island Books Ltd for J\$27,845.50 using fictional account details. [7 marks]

Working space: _____

4. Explain four security practices when using cheques or banking forms. [6 marks]

Working space: _____

5. A student wants high interest but needs the money for school next week. Evaluate whether a long fixed deposit is suitable. [5 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 7 marks

Model solution: Savings: general saving, accessible, some interest. Current: frequent transactions, high access, usually low/no interest and possible fees. Fixed deposit: money locked for a term, limited early access, usually higher interest. Actual terms vary by institution. (7)

Question 2 — 6 marks

Model solution: Cheque total=10,600; deposit total=29,000. Check account number/name, arithmetic, dates and cheque details; any two. (6)

Question 3 — 7 marks

Model solution: Date: 10 Aug 2026; payee: Island Books Ltd; figures J\$27,845.50; words: Twenty-seven thousand eight hundred and forty-five dollars and fifty cents only; drawer signs after checking. (7)

Question 4 — 6 marks

Model solution: Do not sign blanks; use permanent ink; close blank spaces; verify words/figures and account number; keep records; protect PIN/account information; any four with explanation. (6)

Question 5 — 5 marks

Model solution: Probably not. A long fixed term may restrict access or charge an early-withdrawal penalty. An accessible savings option better matches the immediate need, even if interest is lower. (5)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Savings=store money; current=frequent transactions; fixed deposit=money held for a term.

Remember: Deposit total=cash+all cheques.

Remember: Cheque: date, payee, words, figures, signature.

Remember: Words and figures must match.

Remember: Never sign a blank cheque.

Final Self-Check

- ☐ I can compare common account types.
- ☐ I can identify the required fields on deposit and withdrawal forms.
- ☐ I can complete and check the main parts of a cheque safely.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 8

Simple Interest on Loans and Deposits

Unit 4: Numbers | Consumer Arithmetic

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 4: Numbers
Strand	Consumer Arithmetic
Lesson	8: Simple Interest on Loans and Deposits
NSC objective(s)	Use proportions involving principal, rate and time to develop simple interest. • Calculate simple interest on loans and deposits.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Simple interest helps students understand the cost of borrowing and the money earned by saving. It supports better financial choices in Jamaica.

Learning Objectives — I Can

- I can identify principal, rate and time.
- I can calculate simple interest and final amount.
- I can rearrange the formula to find a missing value.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Principal	The starting amount borrowed or invested.
Rate	The percentage charged or earned each year.
Time	How long the money is borrowed or invested, measured in years for this formula.
Interest	The extra money paid or earned.
Amount	The principal plus the interest.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

Simple interest is calculated only on the original principal. The same interest is added each year.

Use $I = P \times R \times T \div 100$, where P is principal, R is the annual percentage rate and T is time in years. Then $A = P + I$.

Convert months to years before substituting: 6 months = $6/12 = 0.5$ year. Money answers should normally be written to two decimal places.

Small Steps to Follow

1. Write P, R and T from the question.
2. Convert time to years if needed.
3. Substitute in $I = P \times R \times T \div 100$.
4. Calculate carefully and label the answer as interest.
5. If the total amount is required, add $A = P + I$ and check that it is reasonable.

Notebook visual: Draw a four-column table headed Year, Starting Principal, Interest for Year and Total. For simple interest, shade the principal column to show that it stays unchanged.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Find the simple interest on J\$20,000 at 6% per year for 3 years.

1. $P=20,000$, $R=6$ and $T=3$.
2. $I = P \times R \times T \div 100$.
3. $I = 20,000 \times 6 \times 3 \div 100$.
4. $20,000 \times 18 \div 100 = 3,600$.

Answer: Interest=J\$3,600; final amount=J\$23,600.

Worked Example 2

Problem: A J\$45,000 loan earns simple interest at 8% for 9 months. Find the interest.

1. Convert time: $T = 9/12 = 0.75$ year.
2. $I = 45,000 \times 8 \times 0.75 \div 100$.

3. $8 \times 0.75 = 6$, so $I = 45,000 \times 6 \div 100$.

Answer: Interest = J\$2,700.

Worked Example 3

Problem: Simple interest is J\$4,800 on J\$30,000 for 2 years. Find the annual rate.

1. $I = P \times R \times T \div 100$.

2. $4,800 = 30,000 \times R \times 2 \div 100$.

3. $R = 4,800 \times 100 \div (30,000 \times 2)$.

4. $R = 8$.

Answer: Rate = 8% per year.

Common Mistakes and Fixes

Common mistake	How to fix it
Using months as years	Divide the number of months by 12.
Writing 6% as 6 and also dividing by 100 twice	Use either 6 with $\div 100$ or 0.06, not both.
Giving interest when amount is asked	Read the final instruction and use $A = P + I$ when needed.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Find I on J\$10,000 at 5% for 2 years.

Check: $I = 10,000 \times 5 \times 2 \div 100 = \text{J\$}1,000$.

2. Find the amount after simple interest on J\$8,000 at 4% for 3 years.

Check: $I = 960$; $A = 8,000 + 960 = \text{J\$}8,960$.

3. Convert 15 months to years.

Check: $15/12 = 1.25$ years.

4. Find I on J\$24,000 at 7.5% for 6 months.

Check: $T = 0.5$; $I = 24,000 \times 7.5 \times 0.5 \div 100 = \text{J\$}900$.

5. $I = \text{J\$}2,000$, $P = \text{J\$}20,000$, $T = 2$ years. Find R.

Check: $R = 2,000 \times 100 \div (20,000 \times 2) = 5\%$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. In the formula $I = PRT/100$, what does P mean?

Working: _____

2. Find I on J\$15,000 at 4% for 5 years.

Working: _____

3. Find A on J\$25,000 at 6% simple interest for 2 years.

Working: _____

4. Find I on J\$12,500 at 8% for 9 months.

Working: _____

5. Which is the final amount? A. $P - I$ B. $P + I$ C. $I \div P$ D. $P \times 100$

Working: _____

6. $P = \text{J\$}40,000$, $I = \text{J\$}7,200$ and $T = 3$ years. Find R.

Working: _____

7. $I = \text{J\$}5,400$ at 9% for 2 years. Find P.

Working: _____

8. I=J\$2,400 on J\$16,000 at 5%. Find T.

Working: _____

9. Compare interest on J\$10,000 at 5% for 1 year and 3 years.

Working: _____

10. A savings account starts with J\$75,000 at 3.5% simple interest for 4 years. Find amount.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: Principal.

2: J\$3,000.

3: $I=3,000$; $A=J\$28,000$.

4: $T=0.75$; $I=J\$750$.

5: B. $P+I$.

6: $R=7,200 \times 100 \div 120,000 = 6\%$.

7: $P=5,400 \times 100 \div 18 = J\$30,000$.

8: $T=2,400 \times 100 \div (16,000 \times 5) = 3$ years.

9: One year: J\$500; three years: J\$1,500. It triples because simple interest grows evenly with time.

10: $I=75,000 \times 3.5 \times 4 \div 100 = 10,500$; $A=J\$85,500$.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A credit union pays 4.5% simple interest on J\$120,000 for 30 months. Find interest and final amount. [6 marks]

Working space: _____

2. Two offers are made for J\$80,000: Plan A is 7% simple interest for 2 years; Plan B charges J\$10,400 total interest. Which costs less and by how much? [6 marks]

Working space: _____

3. A borrower repays J\$58,800 after 3 years on a J\$49,000 simple-interest loan. Find the interest and annual rate. [7 marks]

Working space: _____

4. How long will J\$36,000 take to earn J\$5,400 at 5% simple interest? Give years and months. [6 marks]

Working space: _____

5. Explain why calculating 8% of the growing balance each year would not be simple interest. [5 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 6 marks

Model solution: $T = 30/12 = 2.5$. $I = 120,000 \times 4.5 \times 2.5 \div 100 = \text{J\$}13,500$. $A = \text{J\$}133,500$. (6)

Question 2 — 6 marks

Model solution: Plan A interest $= 80,000 \times 7 \times 2 \div 100 = 11,200$. Plan B $= 10,400$, so B costs J\$800 less. (6)

Question 3 — 7 marks

Model solution: $I = 58,800 - 49,000 = 9,800$. $R = 9,800 \times 100 \div (49,000 \times 3) = 6.666\ldots\%$, about 6.67% per year. (7)

Question 4 — 6 marks

Model solution: $T = 5,400 \times 100 \div (36,000 \times 5) = 3 \text{ years} = 36 \text{ months}$. (6)

Question 5 — 5 marks

Model solution: Simple interest is always based on the original principal. Using the growing balance makes later interest earn interest, which is compound interest. (5)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: $I = P \times R \times T \div 100$.

Remember: $A = P + I$.

Remember: T must be in years: $\text{months} \div 12$.

Remember: Simple interest uses the original principal every year.

Remember: Show P, R and T before calculating.

Final Self-Check

- ☐ I can identify principal, rate and time.
- ☐ I can calculate simple interest and final amount.
- ☐ I can rearrange the formula to find a missing value.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 9

Compound Interest by the Recursive Method

Unit 4: Numbers | Consumer Arithmetic

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 4: Numbers
Strand	Consumer Arithmetic
Lesson	9: Compound Interest by the Recursive Method
NSC objective(s)	Calculate compound interest using a calculator and a recursive method.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Compound interest is used for many savings products and loans. Understanding it shows why balances grow faster over time.

Learning Objectives — I Can

- I can grow a balance one period at a time.
- I can calculate compound interest and final amount.
- I can compare simple and compound interest.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Compound interest	Interest calculated on the current balance, including earlier interest.
Recursive method	A method that uses one answer as the starting value for the next period.
Balance	The amount in the account at a given time.
Multiplier	The number 1 plus the decimal interest rate.
Compounding period	The time between interest calculations.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

With annual compounding, calculate one year, add that interest, then use the new balance for the next year. The interest therefore increases when the balance increases.

For rate $r\%$, the yearly multiplier is $1+r/100$. Repeat: new balance = old balance $\times$ multiplier. The compact check is $A = P(1+r/100)^n$.

Compound interest earned is $A - P$. Follow the rounding instruction; keeping full calculator values until the end usually gives the most accurate result.

Small Steps to Follow

1. Write the starting principal and rate.
2. Change the rate to a multiplier: $1+r/100$.
3. Multiply the balance by that multiplier once for each compounding period.
4. Record each new balance in a year-by-year table.
5. Subtract the original principal if the question asks for interest only.

Notebook visual: Draw a balance ladder: Year 0 $\rightarrow$ Year 1 $\rightarrow$ Year 2 $\rightarrow$ Year 3. Above each arrow write $\times(1+r/100)$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Invest J\$20,000 at 5% compounded annually for 3 years.

1. Multiplier = 1.05.
2. Year 1: $20,000 \times 1.05 = 21,000$.
3. Year 2: $21,000 \times 1.05 = 22,050$.
4. Year 3: $22,050 \times 1.05 = 23,152.50$.
5. Interest = $23,152.50 - 20,000$.

Answer: Amount = J\$23,152.50; compound interest = J\$3,152.50.

Worked Example 2

Problem: Find the amount on J\$50,000 at 8% compounded annually for 2 years.

1. Multiplier = 1.08.

2. Year 1 balance = $50,000 \times 1.08 = 54,000$.

3. Year 2 balance = $54,000 \times 1.08 = 58,320$.

Answer: Amount = J\$58,320; interest = J\$8,320.

Worked Example 3

Problem: Compare simple and compound interest on J\$10,000 at 10% for 2 years.

1. Simple: $I = 10,000 \times 10 \times 2 \div 100 = 2,000$.

2. Compound: Year 1 = 11,000; Year 2 = 12,100.

3. Compound interest = $12,100 - 10,000 = 2,100$.

4. Difference = $2,100 - 2,000$.

Answer: Compound interest is J\$100 more.

Common Mistakes and Fixes

Common mistake	How to fix it
Adding the same interest every year	Recalculate using the new balance each period.
Subtracting principal too early	First find final amount; subtract principal only for interest earned.
Rounding every intermediate step too roughly	Keep calculator accuracy and round money at the end unless instructed otherwise.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. What multiplier represents 6% annual growth?

Check: 1.06.

2. J\$12,000 grows by 5% for one year. Find balance.

Check: $12,000 \times 1.05 = \text{J\$}12,600$.

3. J\$10,000 at 10% compounded annually for 2 years: find amount.

Check: $10,000 \times 1.10 \times 1.10 = \text{J\$}12,100$.

4. If the final amount is J\$26,244 and principal is J\$24,000, find compound interest.

Check: $26,244 - 24,000 = \text{J\$}2,244$.

5. Why is Year 2 interest usually larger than Year 1 interest?

Check: Year 2 interest is calculated on a balance that already includes Year 1 interest.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Write the multiplier for 3%.

Working: _____

2. Find one-year balance on J\$40,000 at 7%.

Working: _____

3. Find amount on J\$30,000 at 4% for 2 years, annually compounded.

Working: _____

4. Find compound interest in the previous question.

Working: _____

5. Which balance is used for the second year? A. original only B. Year 1 closing balance C. interest only D. zero

Working: _____

6. Find amount on J\$25,000 at 6% for 3 years.

Working: _____

7. Find compound interest on J\$80,000 at 2.5% for 2 years.

Working: _____

8. J\$15,000 becomes J\$16,537.50 in 2 years. Verify a 5% annual compound rate.

Working: _____

9. At 10%, compare Year 1 and Year 2 interest on J\$50,000.

Working: _____

10. Explain the meaning of 1.08^3 .

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: 1.03.

2: J\$42,800.

3: $30,000 \times 1.04^2 = \text{J\$}32,448$.

4: $32,448 - 30,000 = \text{J\$}2,448$.

5: B. Year 1 closing balance.

6: $25,000 \times 1.06^3 = \text{J\$}29,775.40$.

7: $A = 80,000 \times 1.025^2 = 84,050$; interest = J\$4,050.

8: $15,000 \times 1.05 = 15,750$; $\times 1.05 = 16,537.50$, so verified.

9: Year 1 = 5,000. New balance = 55,000; Year 2 = 5,500. Year 2 is J\$500 higher.

10: Multiply by 1.08 three times, representing three annual increases of 8%.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. J\$75,000 is invested at 4% compounded annually for 3 years. Create a recursive table and find interest. [8 marks]

Working space: _____

2. A J\$120,000 balance compounds annually at 6% for 2 years. Find amount and compare with 6% simple interest for 2 years. [7 marks]

Working space: _____

3. An account has J\$50,000 and earns 5% annually. A student adds J\$10,000 after the first year's interest. Find the balance after the second year's interest. [7 marks]

Working space: _____

4. A calculator shows J\$31,762.98 after compounding J\$28,000 at 4.3% for 3 years. Explain how to verify it recursively. [7 marks]

Working space: _____

5. Explain, without only quoting a formula, why compound and simple interest are equal after one year but can differ after two years. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 8 marks

Model solution: Year 0=75,000; Year 1=78,000; Year 2=81,120; Year 3=84,364.80.
Interest=84,364.80–75,000=J\$9,364.80. (8)

Question 2 — 7 marks

Model solution: Compound amount= $120,000 \times 1.06^2 = 134,832$; compound interest=14,832. Simple interest=14,400; compound is J\$432 more. (7)

Question 3 — 7 marks

Model solution: After Year 1: $50,000 \times 1.05 = 52,500$. Add 10,000 → 62,500. After Year 2: $62,500 \times 1.05 = \text{J\$}65,625$. (7)

Question 4 — 7 marks

Model solution: Use multiplier 1.043: Year 1=29,204; Year 2=30,459.772; Year 3=31,769.541... Thus J\$31,762.98 is not correct; correct rounded amount is J\$31,769.54. (7)

Question 5 — 6 marks

Model solution: Both use the original principal in Year 1. In Year 2, simple interest still uses the original principal, while compound interest uses principal plus Year 1 interest, so it earns interest on interest. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Multiplier = $1 + r/100$.

Remember: Recursive rule: new balance = old balance $\times$ multiplier.

Remember: Repeat once per compounding period.

Remember: Compound interest = final amount – principal.

Remember: Use a year-by-year table to avoid missed periods.

Final Self-Check

- ☐ I can grow a balance one period at a time.
- ☐ I can calculate compound interest and final amount.
- ☐ I can compare simple and compound interest.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 10

Hire Purchase Cost and Comparison

Unit 4: Numbers | Consumer Arithmetic

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 4: Numbers
Strand	Consumer Arithmetic
Lesson	10: Hire Purchase Cost and Comparison
NSC objective(s)	Calculate total cost in a hire-purchase agreement. • Compare hire-purchase price and cash price.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Hire purchase is offered for furniture, appliances and other goods. Calculating every payment reveals the true cost before an agreement is made.

Learning Objectives — I Can

- I can calculate the balance after a deposit.
- I can calculate total hire-purchase price.
- I can compare hire purchase with cash price.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Cash price	The price paid in full at once.
Deposit	The first payment made before instalments begin.
Instalment	One regular part-payment.
Hire-purchase price	The deposit plus all instalments.
Finance charge	The extra amount paid above the cash price.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A hire-purchase offer normally has a deposit and a stated number of instalments. Total instalments = number $\times$ one instalment. Hire-purchase price = deposit + total instalments.

Finance charge = hire-purchase price – cash price. It shows the extra cost of paying over time.

If the deposit is a percentage, calculate it from the cash price first. Read carefully for fees, payment frequency and whether the question asks for the outstanding balance or the total paid.

Small Steps to Follow

1. List the cash price, deposit, instalment amount and number of instalments.
2. Calculate a percentage deposit if necessary.
3. Multiply instalment amount by number of instalments.
4. Add the deposit to get hire-purchase price.
5. Subtract cash price to find the finance charge and compare offers.

Notebook visual: Draw a payment strip: DEPOSIT + instalment + instalment + ... = HIRE-PURCHASE PRICE. Under it write HP price – cash price = finance charge.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: A cooker costs J\$90,000 cash or J\$15,000 deposit plus 12 payments of J\$7,000. Find HP price and finance charge.

1. Total instalments = $12 \times 7,000 = 84,000$.
2. HP price = $15,000 + 84,000 = 99,000$.
3. Finance charge = $99,000 - 90,000 = 9,000$.

Answer: HP price = J\$99,000; finance charge = J\$9,000.

Worked Example 2

Problem: A laptop costs J\$120,000 cash. HP terms: 20% deposit and 18 monthly payments of J\$6,400.

1. Deposit = 20% of $120,000 = 24,000$.
2. Instalments total = $18 \times 6,400 = 115,200$.
3. HP price = $24,000 + 115,200 = 139,200$.

4. Extra cost = $139,200 - 120,000$.

Answer: HP price = J\$139,200; extra cost = J\$19,200.

Worked Example 3

Problem: Compare Offer A: J\$10,000 deposit + $10 \times$ J\$5,500; Offer B: J\$16,000 deposit + $8 \times$ J\$6,000.

1. A total = $10,000 + 55,000 = 65,000$.

2. B total = $16,000 + 48,000 = 64,000$.

3. Difference = 1,000.

Answer: Offer B costs J\$1,000 less overall, though its deposit is higher.

Common Mistakes and Fixes

Common mistake	How to fix it
Ignoring the deposit	Add it to all instalments for the total.
Comparing monthly payments only	Calculate each complete offer.
Calling HP price the finance charge	Finance charge is HP price minus cash price.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Find total of 12 instalments at J\$4,500.

Check: J\$54,000.

2. Deposit J\$8,000 plus instalments totalling J\$42,000: find HP price.

Check: J\$50,000.

3. HP price J\$72,000; cash price J\$65,000: find finance charge.

Check: J\$7,000.

4. Find a 15% deposit on J\$80,000.

Check: J\$12,000.

5. Why can the offer with the smallest monthly payment still cost more?

Check: It may have more payments, a larger deposit or added fees, so the total can be higher.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. What is the hire-purchase price?

Working: _____

2. 9 payments of J\$6,200 total how much?

Working: _____

3. Deposit J\$20,000 and $10 \times \text{J\$8,500}$: find HP price.

Working: _____

4. Cash J\$98,000, HP J\$105,000: find extra cost.

Working: _____

5. Which formula is correct? A. $\text{HP} - \text{cash} = \text{finance charge}$ B. $\text{cash} - \text{HP} = \text{deposit}$ C. $\text{deposit} \times \text{cash} = \text{HP}$ D. $\text{HP} \div \text{cash} = \text{instalments}$

Working: _____

6. Find a 25% deposit on J\$140,000.

Working: _____

7. Cash J\$140,000; deposit 25%; 15 payments of J\$8,000. Find HP price.

Working: _____

8. For the previous offer, find finance charge as a percentage of cash price.

Working: _____

9. Offer X: $J\$5,000 + 20 \times J\$3,000$. Offer Y: $J\$11,000 + 12 \times J\$4,400$. Which is cheaper?

Working: _____

10. State two non-price factors to consider.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: Deposit plus all instalments and any stated required fees.

2: J\$55,800.

3: J\$105,000.

4: J\$7,000.

5: A.

6: J\$35,000.

7: $35,000 + 120,000 = \text{J\$}155,000$.

8: $15,000 \div 140,000 \times 100 \approx 10.71\%$.

9: $X = 65,000$; $Y = 63,800$. Y is J\$1,200 cheaper.

10: Affordability of deposit/monthly payment, contract fees, late penalties, warranty, reliability or repayment length; any two.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A refrigerator costs J\$135,000 cash. HP terms are 10% deposit and 24 monthly payments of J\$5,900. Find deposit, HP price and finance charge. [8 marks]

Working space: _____

2. A television has cash price J\$84,000. Plan A: $\text{J\$}12,000 + 12 \times \text{J\$}6,600$. Plan B: 20% deposit + $18 \times \text{J\$}4,100$. Compare total costs and deposits. [9 marks]

Working space: _____

3. A customer has paid a J\$18,000 deposit and 7 of 15 instalments of J\$6,500. Find amount paid, instalment balance remaining and full HP price. [7 marks]

Working space: _____

4. An advertisement says 'Only J\$3,999 monthly' for 36 months plus J\$25,000 deposit. Cash price is J\$150,000. Find the true HP price and explain the advertisement's limitation. [7 marks]

Working space: _____

5. A family can pay J\$8,000 monthly but has only J\$10,000 now. Compare cash and HP choices using both total cost and affordability; state what information is still needed. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 8 marks

Model solution: Deposit=13,500. Instalments=141,600. HP price=155,100. Finance charge=20,100. (8)

Question 2 — 9 marks

Model solution: A=12,000+79,200=91,200. B deposit=16,800; total=16,800+73,800=90,600. B is J\$600 cheaper overall but needs J\$4,800 more deposit. (9)

Question 3 — 7 marks

Model solution: Paid=18,000+7×6,500=63,500. Remaining=8×6,500=52,000. Full HP price=115,500. (7)

Question 4 — 7 marks

Model solution: HP=25,000+36×3,999=168,964. Extra=18,964. The monthly figure hides the deposit, duration and total extra cost. (7)

Question 5 — 6 marks

Model solution: HP may be immediately affordable if deposit≤10,000 and instalment≤8,000, but cash usually costs less overall. Need exact deposit, number/payments, fees, income stability and cash price before deciding. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Total instalments = number $\times$ instalment.

Remember: HP price = deposit + total instalments.

Remember: Finance charge = HP price – cash price.

Remember: Percentage deposit = rate $\times$ cash price.

Remember: Compare both total cost and affordability.

Final Self-Check

- ☐ I can calculate the balance after a deposit.
- ☐ I can calculate total hire-purchase price.
- ☐ I can compare hire purchase with cash price.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 11

Discounts and Taxes from Given Instructions

Unit 4: Numbers | Consumer Arithmetic

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 4: Numbers
Strand	Consumer Arithmetic
Lesson	11: Discounts and Taxes from Given Instructions
NSC objective(s)	Calculate discounts and taxes from given instructions.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Shops, bills and receipts use percentages. Careful discount-and-tax calculations help students check what they should pay.

Learning Objectives — I Can

- I can calculate a discount and sale price.
- I can calculate tax from the amount named in the question.
- I can follow the correct order when both discount and tax apply.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Marked price	The price shown before a discount.
Discount	An amount taken off a price.
Sale price	The price after discount but before any later tax stated.
Tax	A required percentage charge calculated according to the instruction.
Final price	The amount paid after all stated changes.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

Discount = discount rate $\times$ marked price. Sale price = marked price – discount.

Tax = tax rate $\times$ the taxable price named in the problem. Final price = taxable price + tax.

When a question says 'discount, then tax on the discounted price,' the order matters. Use the rate given; real tax rules and rates can change.

Small Steps to Follow

1. Underline the original or taxable amount named in the instruction.
2. Change the percentage to a decimal or divide by 100.
3. Calculate the discount and subtract it.
4. Calculate tax on the stated price—often the discounted price—and add it.
5. Estimate and check that a discount lowers while a tax raises the price.

Notebook visual: Draw a price flow: MARKED PRICE $\rightarrow$ subtract discount $\rightarrow$ SALE PRICE $\rightarrow$ add tax $\rightarrow$ FINAL PRICE.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: A school bag marked J\$8,000 has 15% discount. Find sale price.

1. Discount = 15% of 8,000.
2. $0.15 \times 8,000 = 1,200$.
3. Sale price = $8,000 - 1,200$.

Answer: Discount = J\$1,200; sale price = J\$6,800.

Worked Example 2

Problem: Tax of 12% is added to a J\$25,000 service. Find final bill.

1. Tax = $0.12 \times 25,000 = 3,000$.
2. Final = $25,000 + 3,000$.

Answer: Tax = J\$3,000; final bill = J\$28,000.

Worked Example 3

Problem: A J\$20,000 item gets 10% discount, then 15% tax on the discounted price.

1. Discount = $0.10 \times 20,000 = 2,000$.
2. Discounted price = 18,000.
3. Tax = $0.15 \times 18,000 = 2,700$.
4. Final = $18,000 + 2,700$.

Answer: Final price = J\$20,700.

Common Mistakes and Fixes

Common mistake	How to fix it
Adding a discount	A discount is subtracted.
Using the original price for every later percentage	Use the exact base named in each instruction.
Adding percentage rates automatically	Successive percentages act on changing amounts and may not simply add.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Find 20% discount on J\$5,000.

Check: J\$1,000.

2. Find sale price after 25% off J\$12,000.

Check: Discount=3,000; sale price=J\$9,000.

3. Add 10% tax to J\$7,500.

Check: Tax=750; total=J\$8,250.

4. J\$10,000 gets 20% discount, then 5% tax on sale price. Find final.

Check: Sale=8,000; tax=400; final=J\$8,400.

5. Why must you identify the taxable amount?

Check: Tax may be based on the original price or the discounted price, depending on the instruction.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Find a 30% discount on J\$9,000.

Working: _____

2. Find price after 12% discount on J\$15,000.

Working: _____

3. Find 8% tax on J\$40,000.

Working: _____

4. Find total after 7.5% tax on J\$16,000.

Working: _____

5. A discount should make a price: A. larger B. smaller C. unchanged D. negative

Working: _____

6. J\$50,000 is reduced by 18%. Find sale price.

Working: _____

7. J\$24,000 gets 25% discount then 10% tax. Find final.

Working: _____

8. The sale price after 20% discount is J\$32,000. Find marked price.

Working: _____

9. A J\$6,000 item has two successive 10% discounts. Is total discount 20%? Calculate.

Working: _____

10. A receipt has subtotal J\$18,500 and tax J\$2,775. Find tax rate.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: J\$2,700.

2: $15,000 - 1,800 = \text{J\$}13,200$.

3: J\$3,200.

4: Tax=1,200; total=J\$17,200.

5: B. Smaller.

6: Discount=9,000; sale=J\$41,000.

7: Sale=18,000; tax=1,800; final=J\$19,800.

8: 80% is 32,000; marked= $32,000 \div 0.8 = \text{J\$}40,000$.

9: After first=5,400; after second=4,860. Discount=1,140=19%, not 20%.

10: $2,775 \div 18,500 \times 100 = 15\%$.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A tablet marked J\$72,000 is discounted 12.5%, then taxed 10% on the discounted price. Find discount, tax and final price. [8 marks]

Working space: _____

2. Shop A offers 20% off J\$45,000. Shop B sells the same item for J\$37,500 then gives J\$1,000 off. Which is cheaper and by how much? [7 marks]

Working space: _____

3. A bill before tax is J\$28,800. After tax it is J\$33,120. Find the tax amount and rate. [6 marks]

Working space: _____

4. A store mistakenly adds 10% tax before applying a 20% discount. Compare with discount then tax on J\$50,000, assuming both percentages apply to the updated price. [8 marks]

Working space: _____

5. An item costs J\$18,400 after a 20% discount and then 15% tax. Find the marked price. [8 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 8 marks

Model solution: Discount=9,000; sale=63,000; tax=6,300; final=J\$69,300. (8)

Question 2 — 7 marks

Model solution: A=36,000. B=36,500. Shop A is J\$500 cheaper. (7)

Question 3 — 6 marks

Model solution: Tax=4,320. Rate= $4,320 \div 28,800 \times 100 = 15\%$. (6)

Question 4 — 8 marks

Model solution: Tax then discount: $50,000 \times 1.10 \times 0.80 = 44,000$. Discount then tax: $50,000 \times 0.80 \times 1.10 = 44,000$. They match because multiplication is commutative when each rate applies to the updated price; the intermediate amounts differ. (8)

Question 5 — 8 marks

Model solution: Let sale before tax=S. $S \times 1.15 = 18,400$, so $S = 16,000$. $\text{Marked} \times 0.80 = 16,000$, so marked=J\$20,000. (8)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: $\text{Discount} = \text{rate} \times \text{marked price}$.

Remember: $\text{Sale price} = \text{marked price} - \text{discount}$.

Remember: $\text{Tax} = \text{rate} \times \text{stated taxable price}$.

Remember: $\text{Final price} = \text{taxable price} + \text{tax}$.

Remember: Follow the order written in the question.

Final Self-Check

- ☐ I can calculate a discount and sale price.
- ☐ I can calculate tax from the amount named in the question.
- ☐ I can follow the correct order when both discount and tax apply.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 12

Areas of Parallelograms, Triangles and Trapezia

Unit 5: Measurement | Measurement

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 5: Measurement
Strand	Measurement
Lesson	12: Areas of Parallelograms, Triangles and Trapezia
NSC objective(s)	Derive and use formulae for the area of parallelograms, triangles and trapezia.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Area is needed for flooring, painting, farming plots, road signs and construction. The formulas help calculate material and cost without counting squares one by one.

Learning Objectives — I Can

- I can explain where each area formula comes from.
- I can calculate areas using perpendicular height.
- I can find a missing dimension from a known area.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Area	The amount of flat surface inside a shape.
Base	The chosen side used in an area calculation.
Perpendicular height	The shortest distance from the base at a right angle.
Parallel	Lines that stay the same distance apart and never meet.
Trapezium	A quadrilateral with one pair of parallel sides in this lesson.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A parallelogram can be cut and rearranged into a rectangle, so $A=b \times h$. The sloping side is not the perpendicular height.

Two matching triangles form a parallelogram, so triangle area is half: $A=\frac{1}{2}bh$.

Two matching trapezia form a parallelogram whose base is the sum of the parallel sides, so $A=\frac{1}{2}(a+b)h$. Always write square units.

Small Steps to Follow

1. Identify the shape and write its formula.
2. Mark the perpendicular height and the correct base or parallel sides.
3. Substitute values with matching units.
4. Calculate using brackets before multiplication when needed.
5. Write cm^2 , m^2 or another correct square unit and check reasonableness.

Notebook visual: Draw each shape beside a rearranged rectangle/parallelogram: move a triangular end of a parallelogram; pair two triangles; pair two trapezia. Label right-angle height marks.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

1. Use $A=bh$.
2. Substitute $b=12$ and $h=7$.
3. $A=12 \times 7=84$.

Answer: 84 cm^2 .

Worked Example 2

Problem: Find the area of a triangle with base 15 m and height 8 m.

1. Use $A=\frac{1}{2}bh$.
2. $A=\frac{1}{2} \times 15 \times 8$.
3. Half of 8 is 4; $15 \times 4=60$.

Answer: 60 m^2 .

Worked Example 3

Problem: A trapezium has parallel sides 9 cm and 15 cm and height 6 cm.

1. Use $A = \frac{1}{2}(a+b)h$.
2. $A = \frac{1}{2}(9+15) \times 6$.
3. $9+15=24$; half of 24 is 12.
4. $12 \times 6 = 72$.

Answer: 72 cm^2 .

Common Mistakes and Fixes

Common mistake	How to fix it
Using a sloping side as height	Find the segment perpendicular to the base.
Forgetting one-half	Triangles and trapezia require $\frac{1}{2}$ in these formulas.
Writing linear units	Area requires square units such as cm^2 .

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Parallelogram: $b=9$ cm, $h=5$ cm. Find area.

Check: $9 \times 5 = 45$ cm².

2. Triangle: $b=10$ m, $h=7$ m. Find area.

Check: $\frac{1}{2} \times 10 \times 7 = 35$ m².

3. Trapezium: $a=4$ cm, $b=8$ cm, $h=5$ cm. Find area.

Check: $\frac{1}{2}(4+8) \times 5 = 30$ cm².

4. Triangle area = 48 cm², base = 12 cm. Find height.

Check: $48 = \frac{1}{2} \times 12 \times h = 6h$, so $h=8$ cm.

5. Why is a sloping side not automatically the height?

Check: Height must meet the base at 90° and measure the perpendicular distance.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Find parallelogram area for $b=14$ cm, $h=9$ cm.

Working: _____

2. Find triangle area for $b=18$ cm, $h=11$ cm.

Working: _____

3. Find trapezium area for parallel sides 7 m and 13 m, height 4 m.

Working: _____

4. Which is the triangle formula? A. bh B. $2bh$ C. $\frac{1}{2}bh$ D. $(a+b)h$

Working: _____

5. A parallelogram has area 96 cm² and height 8 cm. Find base.

Working: _____

6. A triangle has area 75 m² and height 10 m. Find base.

Working: _____

7. A trapezium has area 66 cm², parallel sides 8 cm and 14 cm. Find height.

Working: _____

8. A triangular flag has base 1.2 m and height 0.8 m. Find area.

Working: _____

9. A parallelogram has base 10 cm, sloping side 7 cm and perpendicular height 5 cm. Find area.

Working: _____

10. Explain why all areas use square units.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: 126 cm^2 .
- 2: 99 cm^2 .
- 3: 40 m^2 .
- 4: C. $\frac{1}{2}bh$.
- 5: $b=96\div8=12 \text{ cm}$.
- 6: $75=5b$, so $b=15 \text{ m}$.
- 7: $66=\frac{1}{2}(22)h=11h$, so $h=6 \text{ cm}$.
- 8: $\frac{1}{2}\times1.2\times0.8=0.48 \text{ m}^2$.
- 9: $10\times5=50 \text{ cm}^2$; the 7 cm sloping side is not used.
- 10: Area counts two-dimensional unit squares covering a surface.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A trapezium-shaped garden has parallel sides 18 m and 30 m and perpendicular height 12 m. Find area and the cost of grass at J\$850 per m^2 . [7 marks]

Working space: _____

2. A parallelogram and triangle share base 16 cm and height 9 cm. Find both areas and explain the relationship. [6 marks]

Working space: _____

3. A composite sign consists of a 20 cm by 12 cm parallelogram and a triangle of base 20 cm, height 7 cm attached without overlap. Find total area. [7 marks]

Working space: _____

4. A trapezium has area 150 cm^2 , height 10 cm and one parallel side 8 cm. Find the other parallel side. [7 marks]

Working space: _____

5. A student calculates triangle area as $12\times9=108 \text{ cm}^2$. Diagnose the error and correct it. [5 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 7 marks

Model solution: $\text{Area} = \frac{1}{2}(18+30) \times 12 = 288 \text{ m}^2$. $\text{Cost} = 288 \times 850 = \text{J\$}244,800$. (7)

Question 2 — 6 marks

Model solution: $\text{Parallelogram} = 16 \times 9 = 144 \text{ cm}^2$. $\text{Triangle} = \frac{1}{2} \times 16 \times 9 = 72 \text{ cm}^2$. Triangle is half because two congruent triangles make the parallelogram. (6)

Question 3 — 7 marks

Model solution: $\text{Parallelogram} = 240 \text{ cm}^2$. $\text{Triangle} = 70 \text{ cm}^2$. $\text{Total} = 310 \text{ cm}^2$. (7)

Question 4 — 7 marks

Model solution: $150 = \frac{1}{2}(8+b)10 = 5(8+b)$. $30 = 8+b$, so $b = 22 \text{ cm}$. (7)

Question 5 — 5 marks

Model solution: The student used parallelogram formula and forgot the half. $\text{Correct area} = \frac{1}{2} \times 12 \times 9 = 54 \text{ cm}^2$. (5)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Parallelogram: $A=bh$.

Remember: Triangle: $A=\frac{1}{2}bh$.

Remember: Trapezium: $A=\frac{1}{2}(a+b)h$.

Remember: h is perpendicular to the base.

Remember: Area uses square units.

Final Self-Check

- ☐ I can explain where each area formula comes from.
- ☐ I can calculate areas using perpendicular height.
- ☐ I can find a missing dimension from a known area.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 13

Total Surface Area of Prisms and Cylinders

Unit 5: Measurement | Measurement

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 5: Measurement
Strand	Measurement
Lesson	13: Total Surface Area of Prisms and Cylinders
NSC objective(s)	Compute total surface area of cubes, cuboids, cylinders and triangular prisms.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Surface area tells how much material covers a box, tank, package or column. It is used for paint, wrapping, labels and construction.

Learning Objectives — I Can

- I can use a net to identify every outer face.
- I can calculate surface area of cubes, cuboids and triangular prisms.
- I can calculate total surface area of a closed cylinder.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Surface area	The total area of all outside surfaces of a solid.
Net	A flat pattern that folds to make a solid.
Face	One flat surface of a solid.
Circumference	The distance around a circle, $2\pi r$.
Lateral surface	The side surface, excluding the bases.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

Surface area is found by adding the areas of every exposed face. A cube has six equal square faces: $SA=6s^2$. A cuboid has three matching pairs: $SA=2(lw+lh+wh)$.

For any right prism, $SA=2 \times \text{area of end} + \text{perimeter of end} \times \text{length}$. For a triangular prism, the ends are triangles and the three rectangles use each triangle side times prism length.

A closed cylinder has two circular ends and a curved rectangle: $SA=2\pi r^2+2\pi rh$. Use the stated value of π and square units.

Small Steps to Follow

1. Sketch or imagine the net and list all exposed surfaces.
2. Label dimensions and match equal faces.
3. Calculate each face or use the correct compact formula.
4. Add once—do not omit or double-count a face.
5. Round only as instructed and attach square units.

Notebook visual: Draw nets for a cube, cuboid, triangular prism and cylinder. For the cylinder, show two circles plus one rectangle of length $2\pi r$ and height h .

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Find surface area of a cube with side 5 cm.

1. One face $= 5^2 = 25 \text{ cm}^2$.
2. A cube has 6 faces.
3. $SA = 6 \times 25 = 150$.

Answer: 150 cm^2 .

Worked Example 2

Problem: Find surface area of a cuboid 8 cm by 5 cm by 3 cm.

1. Use $2(lw+lh+wh)$.
2. $lw=40$, $lh=24$, $wh=15$.
3. Sum $= 79$; multiply by 2.

Answer: 158 cm^2 .

Worked Example 3

Problem: Find total surface area of a closed cylinder with $r=3 \text{ cm}$, $h=10 \text{ cm}$, using $\pi \approx 3.14$.

1. Two ends $= 2\pi r^2 = 2 \times 3.14 \times 9 = 56.52$.
2. Curved area $= 2\pi rh = 2 \times 3.14 \times 3 \times 10 = 188.4$.
3. Total $= 56.52 + 188.4$.

Answer: 244.92 cm^2 .

Common Mistakes and Fixes

Common mistake	How to fix it
Using volume units	Surface area is measured in square units.
Forgetting hidden/end faces	Use a net and count every exposed surface.
Using πr^2 for the curved side	Curved area is circumference $\times$ height $= 2\pi rh$.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Cube side 4 cm: find SA.

Check: $6 \times 4^2 = 96 \text{ cm}^2$.

2. Cuboid $6 \times 4 \times 2$ cm: find SA.

Check: $2(24 + 12 + 8) = 88 \text{ cm}^2$.

3. What flat shape forms a cylinder's curved surface in its net?

Check: A rectangle.

4. Closed cylinder $r=2$ cm, $h=5$ cm: write, but do not simplify, the SA expression.

Check: $2\pi(2^2) + 2\pi(2)(5) = 8\pi + 20\pi$.

5. Why does a closed triangular prism have two triangular terms?

Check: It has two congruent triangular end faces.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Find SA of cube side 7 m.

Working: _____

2. Find SA of cuboid 10 cm by 7 cm by 4 cm.

Working: _____

3. Which unit is appropriate for surface area? A. cm B. cm^2 C. cm^3 D. L

Working: _____

4. A closed cylinder has $r=4$ cm and $h=6$ cm. Give exact SA in terms of π .

Working: _____

5. Using $\pi=3.14$, evaluate 80π .

Working: _____

6. An open-top cube-shaped box has side 5 cm. Find exposed material area for its five faces.

Working: _____

7. A triangular prism has a 3-4-5 right-triangle end and length 10 cm. Find SA.

Working: _____

8. A cuboid SA is 148 cm^2 with $l=6 \text{ cm}$ and $w=4 \text{ cm}$. Find h .

Working: _____

9. Explain the difference between area and surface area.

Working: _____

10. A label covers only the curved side of a can $r=3 \text{ cm}$, $h=8 \text{ cm}$. Write its area.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $6 \times 49 = 294 \text{ m}^2$.

2: $2(70+40+28) = 276 \text{ cm}^2$.

3: B. cm^2 .

4: $2\pi(16)+2\pi(4)(6) = 80\pi \text{ cm}^2$.

5: 251.2 cm^2 .

6: $5 \times 25 = 125 \text{ cm}^2$.

7: Ends = $2 \times \frac{1}{2} \times 3 \times 4 = 12$. Rectangles = $(3+4+5) \times 10 = 120$. Total = 132 cm^2 .

8: $148 = 2(24+6h+4h)$; $74 = 24+10h$; $h = 5 \text{ cm}$.

9: Area measures one flat region; surface area totals all exposed surfaces of a 3D solid.

10: $2\pi rh = 48\pi \text{ cm}^2$, about 150.72 cm^2 with $\pi = 3.14$.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A closed rectangular package is $30 \text{ cm} \times 20 \text{ cm} \times 12 \text{ cm}$. Find cardboard area, then add 8% for overlaps and waste. [7 marks]

Working space: _____

2. A cylindrical tank is closed, radius 1.5 m and height 4 m . Find total exterior area using $\pi = 3.14$ and painting cost at J\$1,200 per m^2 . [8 marks]

Working space: _____

3. A triangular prism tent has triangular ends with base 6 m , height 4 m and equal sloping sides 5 m ; length 8 m . Find total closed surface area. [8 marks]

Working space: _____

4. A cube and cuboid both have volume 216 cm^3 . Cube side is 6 cm ; cuboid is $9 \times 6 \times 4 \text{ cm}$. Compare surface areas. [7 marks]

Working space: _____

5. A student uses $2\pi r^2$ only for a closed cylinder. Identify what is missing and correct the formula. [5 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 7 marks

Model solution: $SA=2(600+360+240)=2,400 \text{ cm}^2$. Extra= 192 cm^2 . Required= $2,592 \text{ cm}^2$. (7)

Question 2 — 8 marks

Model solution: $SA=2\pi r^2+2\pi rh=14.13+37.68=51.81 \text{ m}^2$. Cost= $51.81 \times 1,200=\text{J\$}62,172$. (8)

Question 3 — 8 marks

Model solution: Ends= $2 \times \frac{1}{2} \times 6 \times 4=24 \text{ m}^2$. Rectangles= $(6+5+5) \times 8=128 \text{ m}^2$. Total= 152 m^2 . (8)

Question 4 — 7 marks

Model solution: Cube $SA=6 \times 36=216 \text{ cm}^2$. Cuboid $SA=2(54+36+24)=228 \text{ cm}^2$. Cuboid has 12 cm^2 more surface area. (7)

Question 5 — 5 marks

Model solution: $2\pi r^2$ covers only the two circles. The curved surface $2\pi rh$ is missing. Total $SA=2\pi r^2+2\pi rh$. (5)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Cube: $SA=6s^2$.

Remember: Cuboid: $SA=2(lw+lh+wh)$.

Remember: Right prism: $SA=2B+\text{perimeter of end} \times \text{length}$.

Remember: Closed cylinder: $SA=2\pi r^2+2\pi rh$.

Remember: Surface area uses square units.

Final Self-Check

- ☐ I can use a net to identify every outer face.
- ☐ I can calculate surface area of cubes, cuboids and triangular prisms.
- ☐ I can calculate total surface area of a closed cylinder.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 14

Converting Square and Cubic Units

Unit 5: Measurement | Measurement

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 5: Measurement
Strand	Measurement
Lesson	14: Converting Square and Cubic Units
NSC objective(s)	Perform conversions within units up to cubed units.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Correct unit conversion is essential when plans, containers and material prices use different metric units.

Learning Objectives — I Can

- I can convert metric length units.
- I can square a conversion factor for area.
- I can cube a conversion factor for volume.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Linear unit	A unit measuring one dimension, such as cm.
Square unit	A unit measuring two-dimensional area, such as cm^2 .
Cubic unit	A unit measuring three-dimensional volume, such as cm^3 .
Conversion factor	The number used to change one unit into another.
Dimension	One direction of measurement.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

If $1\text{ m}=100\text{ cm}$, then $1\text{ m}^2=(100\text{ cm})^2=10,000\text{ cm}^2$. The factor is squared because area has two dimensions.

Likewise $1\text{ m}^3=(100\text{ cm})^3=1,000,000\text{ cm}^3$. The factor is cubed because volume has three dimensions.

Moving from a larger unit to a smaller unit multiplies by the appropriate factor; moving back divides. The number changes greatly even though the physical size stays the same.

Small Steps to Follow

1. Write the basic length fact, such as $1\text{ m}=100\text{ cm}$.
2. Decide whether the measurement is linear, square or cubic.
3. Raise the length factor to power 1, 2 or 3.
4. Multiply when moving to smaller units; divide when moving to larger units.
5. Write the new unit and reverse-convert to check.

Notebook visual: Draw three tiles: a 1 m line split into 100 cm; a 1 m square labelled 100×100 ; a 1 m cube labelled $100\times 100\times 100$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Convert 3.2 m^2 to cm^2 .

1. $1\text{ m}=100\text{ cm}$, so $1\text{ m}^2=100^2=10,000\text{ cm}^2$.
2. $3.2\times 10,000=32,000$.

Answer: $32,000\text{ cm}^2$.

Worked Example 2

Problem: Convert $450,000\text{ cm}^3$ to m^3 .

1. $1\text{ m}^3=1,000,000\text{ cm}^3$.
2. Move from smaller cm^3 to larger m^3 , so divide.
3. $450,000\div 1,000,000=0.45$.

Answer: 0.45 m^3 .

Worked Example 3

Problem: Convert 2.5 L to cm^3 .

1. Use the capacity-volume fact $1 \text{ L} = 1,000 \text{ cm}^3$.
2. $2.5 \times 1,000 = 2,500$.

Answer: $2,500 \text{ cm}^3$.

Common Mistakes and Fixes

Common mistake	How to fix it
Using the linear factor for area	Square the conversion factor.
Using the area factor for volume	Cube the conversion factor.
Multiplying in the wrong direction	Larger to smaller units multiply; smaller to larger divide.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Convert 4 m^2 to cm^2 .

Check: $4 \times 10,000 = 40,000 \text{ cm}^2$.

2. Convert $25,000 \text{ cm}^2$ to m^2 .

Check: $25,000 \div 10,000 = 2.5 \text{ m}^2$.

3. Convert 0.8 m^3 to cm^3 .

Check: $0.8 \times 1,000,000 = 800,000 \text{ cm}^3$.

4. Convert $3,500 \text{ cm}^3$ to litres.

Check: $3,500 \div 1,000 = 3.5 \text{ L}$.

5. Why is the m^2 -to- cm^2 factor not 100?

Check: Both length and width change by 100, so area changes by $100^2 = 10,000$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Convert 6.5 m^2 to cm^2 .

Working: _____

2. Convert $72,000 \text{ cm}^2$ to m^2 .

Working: _____

3. Convert 2 m^3 to cm^3 .

Working: _____

4. Convert $125,000 \text{ cm}^3$ to m^3 .

Working: _____

5. 1 m^3 equals: A. 100 cm^3 B. $10,000 \text{ cm}^3$ C. $1,000,000 \text{ cm}^3$ D. $1,000 \text{ cm}^3$

Working: _____

6. Convert 4.2 km^2 to m^2 .

Working: _____

7. Convert $350,000 \text{ m}^2$ to km^2 .

Working: _____

8. Convert 7.5 L to cm^3 .

Working: _____

9. Convert 820 mL to cm^3 .

Working: _____

10. Convert 0.006 m^3 to litres.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $65,000 \text{ cm}^2$.

2: 7.2 m^2 .

3: $2,000,000 \text{ cm}^3$.

4: 0.125 m^3 .

5: C. $1,000,000 \text{ cm}^3$.

6: $1 \text{ km}^2 = 1,000,000 \text{ m}^2$, so $4,200,000 \text{ m}^2$.

7: 0.35 km^2 .

8: $7,500 \text{ cm}^3$.

9: 820 cm^3 because $1 \text{ mL} = 1 \text{ cm}^3$.

10: $0.006 \times 1,000 = 6 \text{ L}$ because $1 \text{ m}^3 = 1,000 \text{ L}$.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A floor is 4.5 m by 3.2 m . Find its area in m^2 and cm^2 . [6 marks]

Working space: _____

2. A box volume is 0.035 m^3 . Express it in cm^3 and litres. [6 marks]

Working space: _____

3. A student says $2 \text{ m}^3 = 200 \text{ cm}^3$ because $1 \text{ m} = 100 \text{ cm}$. Explain and correct the error. [5 marks]

Working space: _____

4. A rectangular board area is $48,000 \text{ cm}^2$ and length is 3 m . Find width in metres. [7 marks]

Working space: _____

5. A tank holds 2.4 m^3 . Water is removed in 15 L buckets. How many full buckets represent the whole capacity? [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 6 marks

Model solution: $\text{Area} = 4.5 \times 3.2 = 14.4 \text{ m}^2$. In cm^2 : $14.4 \times 10,000 = 144,000 \text{ cm}^2$. (6)

Question 2 — 6 marks

Model solution: $0.035 \times 1,000,000 = 35,000 \text{ cm}^3$. Since $1,000 \text{ cm}^3 = 1 \text{ L}$, capacity = 35 L. (6)

Question 3 — 5 marks

Model solution: Volume has three dimensions, so cube the factor: $2 \times 100^3 = 2,000,000 \text{ cm}^3$. (5)

Question 4 — 7 marks

Model solution: $48,000 \text{ cm}^2 = 4.8 \text{ m}^2$. Width = $\text{area} \div \text{length} = 4.8 \div 3 = 1.6 \text{ m}$. (7)

Question 5 — 6 marks

Model solution: $2.4 \text{ m}^3 = 2,400 \text{ L}$. $2,400 \div 15 = 160$ buckets. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: $1 \text{ m}^2 = 10,000 \text{ cm}^2$.

Remember: $1 \text{ m}^3 = 1,000,000 \text{ cm}^3$.

Remember: $1 \text{ m}^3 = 1,000 \text{ L}$.

Remember: $1 \text{ L} = 1,000 \text{ cm}^3$; $1 \text{ mL} = 1 \text{ cm}^3$.

Remember: Area squares the factor; volume cubes it.

Final Self-Check

- ☐ I can convert metric length units.
- ☐ I can square a conversion factor for area.
- ☐ I can cube a conversion factor for volume.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 2 • LESSON 15

Volume and Capacity of Prisms, Cylinders and Composite Objects

Unit 5: Measurement | Measurement

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 2
Unit	Unit 5: Measurement
Strand	Measurement
Lesson	15: Volume and Capacity of Prisms, Cylinders and Composite Objects
NSC objective(s)	Establish volume formulae. • Estimate and calculate volume and capacity of cubes, cuboids, prisms, cylinders and composite objects.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Volume and capacity are used for water tanks, packaging, concrete, storage and cooking. They help answer how much space an object occupies or can hold.

Learning Objectives — I Can

- I can explain volume as layers of unit cubes.
- I can calculate volume of prisms and cylinders.
- I can split composite solids and connect volume to capacity.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Volume	The amount of three-dimensional space an object occupies.
Capacity	The amount a container can hold.
Cross-section	A flat end shape that stays the same through a prism.
Prism	A solid with a constant cross-section along its length.
Composite solid	A solid made by joining or removing simpler solids.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A cuboid made of unit cubes has length $\times$ width cubes in each layer and height layers, so $V=lwh$. A cube has $V=s^3$.

Every prism follows $V=\text{area of cross-section}\times\text{length}$. A cylinder is a circular prism, so $V=\pi r^2h$.

For composite objects, add joined volumes or subtract removed volumes. Cubic centimetres connect to capacity: $1,000\text{ cm}^3=1\text{ L}$ and $1\text{ cm}^3=1\text{ mL}$.

Small Steps to Follow

1. Identify the solid or split a composite object into simple parts.
2. Write each volume formula and mark perpendicular dimensions.
3. Use matching units before calculating.
4. Add or subtract component volumes carefully.
5. Write cubic units, then convert to litres or millilitres only if capacity is requested.

Notebook visual: Sketch a cuboid as layers of cubes, then a prism labelled cross-section area B and length L. Write $V=B\times L$ beneath both.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Find volume of a cuboid $12\text{ cm}\times 5\text{ cm}\times 4\text{ cm}$.

1. Use $V=lwh$.
2. $V=12\times 5\times 4$.
3. $12\times 5=60$; $60\times 4=240$.

Answer: 240 cm^3 .

Worked Example 2

Problem: Find volume of a cylinder with $r=3\text{ cm}$, $h=10\text{ cm}$, using $\pi=3.14$.

1. Use $V=\pi r^2h$.
2. $V=3.14\times 3^2\times 10$.
3. $3^2=9$; $3.14\times 9\times 10=282.6$.

Answer: 282.6 cm^3 .

Worked Example 3

Problem: A tank measures $80 \text{ cm} \times 50 \text{ cm} \times 40 \text{ cm}$. Find capacity in litres.

1. $\text{Volume} = 80 \times 50 \times 40 = 160,000 \text{ cm}^3$.
2. $1,000 \text{ cm}^3 = 1 \text{ L}$.
3. $160,000 \div 1,000 = 160$.

Answer: Capacity = 160 L.

Common Mistakes and Fixes

Common mistake	How to fix it
Using surface-area formulas	Volume measures inside space and needs cubic units.
Using diameter as radius	Radius is half the diameter.
Converting cm^3 to litres by 100	Divide by 1,000 because $1 \text{ L} = 1,000 \text{ cm}^3$.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Cube side 6 cm: find volume.

Check: $6^3=216 \text{ cm}^3$.

2. Cuboid $8 \times 3 \times 5$ m: find volume.

Check: 120 m^3 .

3. A prism has cross-section area 14 cm^2 and length 9 cm. Find volume.

Check: $14 \times 9 = 126 \text{ cm}^3$.

4. Cylinder $r=2 \text{ cm}$, $h=7 \text{ cm}$: write exact volume.

Check: $\pi \times 2^2 \times 7 = 28\pi \text{ cm}^3$.

5. Convert $6,500 \text{ cm}^3$ to litres.

Check: 6.5 L.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Find volume of a cube side 9 cm.

Working: _____

2. Find volume of cuboid $15 \times 8 \times 4$ cm.

Working: _____

3. Which formula gives cylinder volume? A. $2\pi r$ B. $\pi r^2 h$ C. $2\pi r h$ D. lwh^2

Working: _____

4. Cylinder $r=5 \text{ cm}$, $h=8 \text{ cm}$: give exact volume.

Working: _____

5. Using $\pi=3.14$, evaluate the previous volume.

Working: _____

6. A triangular prism has triangle area 24 cm^2 and length 11 cm. Find volume.

Working: _____

7. A cuboid volume is 360 cm^3 , $l=12 \text{ cm}$, $w=5 \text{ cm}$. Find height.

Working: _____

8. A container holds 18 L. Express in cm^3 .

Working: _____

9. Estimate then calculate $49 \text{ cm} \times 21 \text{ cm} \times 10 \text{ cm}$.

Working: _____

10. Two cuboids of volumes 120 cm^3 and 75 cm^3 are joined without overlap. Find total.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: 729 cm^3 .
- 2: 480 cm^3 .
- 3: B. $\pi r^2 h$.
- 4: $200\pi \text{ cm}^3$.
- 5: 628 cm^3 .
- 6: 264 cm^3 .
- 7: $h = 360 \div 60 = 6 \text{ cm}$.
- 8: $18,000 \text{ cm}^3$.
- 9: Estimate $50 \times 20 \times 10 = 10,000 \text{ cm}^3$. Exact $= 49 \times 21 \times 10 = 10,290 \text{ cm}^3$.
- 10: 195 cm^3 .

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A cylindrical water drum has radius 35 cm and height 90 cm. Using $\pi = 22/7$, find volume in cm^3 and capacity in litres. [8 marks]

Working space: _____

2. A swimming pool is $8 \text{ m} \times 4 \text{ m} \times 1.5 \text{ m}$. Find volume and litres of water when full. [6 marks]

Working space: _____

3. A composite solid consists of cuboids $10 \times 6 \times 4 \text{ cm}$ and $5 \times 4 \times 3 \text{ cm}$ joined without overlap. Find total volume. [6 marks]

Working space: _____

4. A 12 cm cube has a 4 cm cube cut out. Find remaining volume and percentage removed. [8 marks]

Working space: _____

5. A rectangular tank $1.2 \text{ m} \times 0.8 \text{ m} \times 0.5 \text{ m}$ is filled at 24 L per minute. Find capacity and filling time. [7 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 8 marks

Model solution: $V = 22/7 \times 35^2 \times 90 = 346,500 \text{ cm}^3$. Capacity = 346.5 L. (8)

Question 2 — 6 marks

Model solution: $V = 8 \times 4 \times 1.5 = 48 \text{ m}^3$. Since $1 \text{ m}^3 = 1,000 \text{ L}$, capacity = 48,000 L. (6)

Question 3 — 6 marks

Model solution: First = 240 cm^3 . Second = 60 cm^3 . Total = 300 cm^3 . (6)

Question 4 — 8 marks

Model solution: Original = $12^3 = 1,728 \text{ cm}^3$. Removed = $4^3 = 64 \text{ cm}^3$. Remaining = $1,664 \text{ cm}^3$. Percentage removed = $64/1,728 \times 100 \approx 3.70\%$. (8)

Question 5 — 7 marks

Model solution: Volume = $0.48 \text{ m}^3 = 480 \text{ L}$. Time = $480 \div 24 = 20$ minutes. (7)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Cube: $V=s^3$.

Remember: Cuboid: $V=lwh$.

Remember: Prism: $V=\text{cross-section area} \times \text{length}$.

Remember: Cylinder: $V=\pi r^2 h$.

Remember: $1,000 \text{ cm}^3 = 1 \text{ L}$; composite volumes add or subtract.

Final Self-Check

- ☐ I can explain volume as layers of unit cubes.
- ☐ I can calculate volume of prisms and cylinders.
- ☐ I can split composite solids and connect volume to capacity.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3

Graphs, Constructions, Sets and Logic

11 complete lessons

Term goal: Build confidence through clear models, repeated practice and cumulative exam preparation.

Term 3 Lesson Map

The table below shows the lesson sequence and its direct curriculum objective coverage.

No.	Unit / strand	Lesson	NSC objective coverage
T3.1	Unit 1: Algebra Relations, Functions and Graphs	Mappings, Ordered Pairs and Coordinate Graphs	Plot the ordered pairs of a mapping as a graph.
T3.2	Unit 1: Algebra Relations, Functions and Graphs	Tables and Straight-Line Graphs	Draw straight-line graphs by plotting points and by using gradient and intercept.
T3.3	Unit 1: Algebra Relations, Functions and Graphs	Gradient and Intercepts of Straight Lines	Determine gradients and intercepts of straight-line graphs.
T3.4	Unit 1: Algebra Relations, Functions and Graphs	Gradient as a Rate of Change	Relate the gradient of a graph to the rate of change of quantities.
T3.5	Unit 1: Algebra Relations, Functions and Graphs	Finding the Equation of a Straight Line	Find the equation of any straight line.
T3.6	Unit 1: Algebra Relations, Functions and Graphs	Solving Two Linear Equations by Graphing	Plot two linear equations on the same axes and interpret their point or points of intersection, if any.
T3.7	Unit 1: Algebra Relations, Functions and Graphs	Graphing Linear Inequalities on the Coordinate Plane	Graph linear inequalities on the coordinate plane and identify regions.
T3.8	Unit 2: Geometry Constructions	Geometric Constructions with Ruler and Compasses	Construct angle bisectors, angles of 90° , 45° , 60° and 30° , and triangles using appropriate geometric instruments.
T3.9	Unit 3: Numbers Sets	Sets, Subsets and the Number of Subsets	Determine the number of subsets of a given set. • List every possible subset when the original set has no more than four elements.
T3.10	Unit 3: Numbers Sets	Two-Set Venn Diagrams and Problem Solving	Solve simple problems involving at most two subsets of the universal set.
T3.11	Unit 3: Numbers Logic	Propositions, Logical Connectives and Truth Tables	Understand propositions and use negation, conjunction, disjunction, conditional and equivalence. • Differentiate simple and compound propositions. • Express propositions symbolically and in words.

TERM 3 • LESSON 1

Mappings, Ordered Pairs and Coordinate Graphs

Unit 1: Algebra | Relations, Functions and Graphs

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 1: Algebra
Strand	Relations, Functions and Graphs
Lesson	1: Mappings, Ordered Pairs and Coordinate Graphs
NSC objective(s)	Plot the ordered pairs of a mapping as a graph.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Mappings connect inputs to outputs. They help describe conversions, prices, distances and many other real-life relationships.

Learning Objectives — I Can

- I can read a mapping diagram and write ordered pairs.
- I can identify domain and range.
- I can plot mapping pairs on a coordinate plane.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Mapping	A rule or diagram that connects each input to one or more stated outputs.
Ordered pair	Two numbers written (x,y), with x first and y second.
Domain	The set of input or x-values.
Range	The set of output or y-values actually used.
Coordinate plane	A grid formed by a horizontal x-axis and vertical y-axis.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

Every arrow $x \rightarrow y$ can be written as the ordered pair (x,y) . The order matters: the input is first and the output is second.

To plot (x,y) , begin at the origin, move x units horizontally, then y units vertically. Negative x moves left and negative y moves down.

The domain lists distinct inputs. The range lists distinct outputs; repeated outputs are written only once when using set notation.

Small Steps to Follow

1. Read each mapping arrow from input to output.
2. Write one ordered pair (input, output) for each arrow.
3. Draw and label equal scales on x - and y -axes.
4. For each pair, move along x first and then along y .
5. Plot, label and check each point against the mapping.

Notebook visual: Draw two ovals labelled INPUT x and OUTPUT y with arrows $1 \rightarrow 3$, $2 \rightarrow 5$, $3 \rightarrow 7$. Beside them draw axes and plot $(1,3)$, $(2,5)$, $(3,7)$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: A mapping has $-1 \rightarrow 1$, $0 \rightarrow 3$ and $2 \rightarrow 7$. Write the ordered pairs, domain and range.

1. Each arrow becomes (input, output).
2. Pairs are $(-1,1)$, $(0,3)$, $(2,7)$.
3. Inputs form domain $\{-1,0,2\}$.
4. Outputs form range $\{1,3,7\}$.

Answer: Pairs $\{(-1,1),(0,3),(2,7)\}$; domain $\{-1,0,2\}$; range $\{1,3,7\}$.

Worked Example 2

Problem: Plot the point $(-3,2)$.

1. The first value is $x=-3$, so move 3 units left from the origin.
2. The second is $y=2$, so move 2 units up.

- Place the point where those grid lines meet.

Answer: The point lies in Quadrant II.

Worked Example 3

Problem: Use $y=2x+1$ for $x=-1,0,1,2$ and create mapping pairs.

- $x=-1: y=2(-1)+1=-1.$
- $x=0: y=1.$
- $x=1: y=3.$
- $x=2: y=5.$

Answer: Pairs: $(-1,-1), (0,1), (1,3), (2,5).$

Common Mistakes and Fixes

Common mistake	How to fix it
Reversing coordinates	Always write and move x first, then y.
Listing repeated outputs many times in a set	A set lists each distinct element once.
Using uneven or unlabelled scales	Mark equal intervals and label both axes.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Write the pair for the mapping $4 \rightarrow 9$.

Check: (4,9).

2. For pairs $\{(1,5), (2,5), (3,7)\}$, state the domain.

Check: {1,2,3}.

3. For the same pairs, state the range.

Check: {5,7}; write the repeated 5 once.

4. Describe how to plot (2,-4).

Check: Move 2 right and 4 down from the origin.

5. Find y when $x=3$ in $y=x+4$ and write the pair.

Check: $y=7$, so (3,7).

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Write the ordered pair for $-2 \rightarrow 6$.

Working: _____

2. State the x -coordinate of (5,-3).

Working: _____

3. State the y -coordinate of (5,-3).

Working: _____

4. Which point is on the y -axis? A. (3,0) B. (0,3) C. (3,3) D. (-3,3)

Working: _____

5. Give the domain of $\{(-2,4), (0,1), (2,4)\}$.

Working: _____

6. Give the range of those pairs.

Working: _____

7. Which quadrant contains (-4,-1)?

Working: _____

8. Use $y=3x-2$ to find the output for $x=4$.

Working: _____

9. List pairs for inputs 0,1,2 under $y=5-x$.

Working: _____

10. Explain why (2,7) and (7,2) are different.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $(-2,6)$.

2: 5.

3: -3 .

4: B. $(0,3)$.

5: $\{-2,0,2\}$.

6: $\{1,4\}$.

7: Quadrant III.

8: $y=12-2=10$; pair $(4,10)$.

9: $(0,5)$, $(1,4)$, $(2,3)$.

10: Their x- and y-values are exchanged, so they name different locations and mappings.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A taxi model maps distance x km to cost $y=250x+500$ for $x=0,2,4,6$. Create the pairs and identify domain and range. [8 marks]

Working space: _____

2. A mapping has pairs $(-2,5), (-1,3), (0,1), (1,-1)$. Identify the rule and predict the pair for $x=4$. [7 marks]

Working space: _____

3. A student plots $(-3,4)$ by moving 3 right and 4 down. Diagnose and correct both errors. [5 marks]

Working space: _____

4. Pairs are $(1,2), (1,4), (3,6)$. Is every input connected to exactly one output? Explain using the graph. [6 marks]

Working space: _____

5. A relation has domain $\{-1,0,2\}$ and rule $y=x^2-1$. Find pairs, range and the point on the y-axis. [8 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 8 marks

Model solution: Pairs: $(0,500), (2,1,000), (4,1,500), (6,2,000)$. Domain $\{0,2,4,6\}$; range $\{500,1,000,1,500,2,000\}$. (8)

Question 2 — 7 marks

Model solution: As x rises by 1, y falls by 2; $y = -2x + 1$. At $x = 4$, $y = -7$, so $(4, -7)$. (7)

Question 3 — 5 marks

Model solution: Negative x means 3 left, not right; positive y means 4 up, not down. Correct point is in Quadrant II. (5)

Question 4 — 6 marks

Model solution: No. Input $x = 1$ has two outputs, $y = 2$ and $y = 4$. The graph has two points directly above $x = 1$. (6)

Question 5 — 8 marks

Model solution: Pairs $(-1,0), (0,-1), (2,3)$. Range $\{-1,0,3\}$. Point on y -axis is $(0,-1)$. (8)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Arrow $x \rightarrow y$ becomes (x, y) .

Remember: Domain=input/x-values; range=output/y-values.

Remember: Plot x horizontally first, then y vertically.

Remember: $x=0$ lies on y-axis; $y=0$ lies on x-axis.

Remember: Check every plotted point against its pair.

Final Self-Check

- ☐ I can read a mapping diagram and write ordered pairs.
- ☐ I can identify domain and range.
- ☐ I can plot mapping pairs on a coordinate plane.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3 • LESSON 2

Tables and Straight-Line Graphs

Unit 1: Algebra | Relations, Functions and Graphs

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 1: Algebra
Strand	Relations, Functions and Graphs
Lesson	2: Tables and Straight-Line Graphs
NSC objective(s)	Draw straight-line graphs by plotting points and by using gradient and intercept.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Straight-line graphs model steady changes such as a fixed cost plus a price per item, or distance travelled at constant speed.

Learning Objectives — I Can

- I can complete a value table for a linear rule.
- I can plot points and draw a straight line.
- I can sketch a line from its gradient and intercept.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Linear equation	An equation whose graph is a straight line.
Value table	A table of matching x- and y-values.
Gradient	How steep a line is and the direction it rises or falls.
y-intercept	The y-value where a line crosses the y-axis.
Scale	The value represented by each grid interval.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

For $y=mx+c$, choose x -values and substitute to find y . Plot at least three accurate points, then use a ruler to draw one straight line through them.

The number c gives the point $(0,c)$. The gradient m tells the step pattern: $m=2$ means rise 2 for run 1; $m=-\frac{1}{2}$ means fall 1 for run 2.

A mathematical line continues beyond the plotted points, so add arrowheads when appropriate. Label the line and axes clearly.

Small Steps to Follow

1. Choose or use the given x -values.
2. Substitute each x carefully to calculate y .
3. Select scales that show all points and label axes.
4. Plot every (x,y) pair accurately.
5. Draw one thin straight line through the points and verify a point in the equation.

Notebook visual: Create a table with x values $-2, -1, 0, 1, 2$ and a y row. Beside it show a coordinate grid with a ruler line through the five points.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Complete a table and plot $y=2x+1$ for $x=-2, -1, 0, 1, 2$.

1. Substitute each x .
2. y values are $-3, -1, 1, 3, 5$.
3. Plot $(-2, -3), (-1, -1), (0, 1), (1, 3), (2, 5)$.
4. Draw one line through them.

Answer: The line crosses the y -axis at 1 and rises 2 for each 1 right.

Worked Example 2

Problem: Sketch $y=-x+3$ using intercept and gradient.

1. $c=3$, so begin at $(0,3)$.
2. $m=-1=-1/1$, so move 1 right and 1 down.

3. Points include $(1,2), (2,1), (3,0)$.
4. Join with a straight line.

Answer: A descending line through $(0,3)$ and $(3,0)$.

Worked Example 3

Problem: Does $(4,11)$ lie on $y=3x-1$?

1. Substitute $x=4$ into the rule.
2. $3(4)-1=12-1=11$.
3. The calculated y matches the point's y .

Answer: Yes, $(4,11)$ lies on the line.

Common Mistakes and Fixes

Common mistake	How to fix it
Joining points dot-to-dot with bends	A linear graph is one straight ruler line.
Starting gradient steps at the origin automatically	Start from the y -intercept $(0,c)$.
Forgetting negative substitution	Use brackets around negative x -values.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. For $y=x+2$, find y when $x=3$.

Check: 5.

2. For $y=4x-1$, find y when $x=0$.

Check: -1.

3. State the y -intercept of $y=2x+7$.

Check: 7, at $(0,7)$.

4. From $(0,1)$, use gradient 3 to find another point.

Check: Move 1 right and 3 up to $(1,4)$.

5. Does $(2,6)$ lie on $y=2x+2$?

Check: Yes: $2(2)+2=6$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Find y for $x=-2$ in $y=x-5$.

Working: _____

2. Complete y -values for $y=2x$, $x=-1,0,1,2$.

Working: _____

3. Which point lies on $y=3x+1$? A. $(1,3)$ B. $(1,4)$ C. $(2,5)$ D. $(0,0)$

Working: _____

4. State the intercept point of $y=-4x+6$.

Working: _____

5. Give two points on $y=5x$.

Working: _____

6. Sketch instructions for $y=\frac{1}{2}x-2$.

Working: _____

7. Does $(-1,4)$ lie on $y=-2x+2$?

Working: _____

8. Find x-intercept of $y=x-4$ by setting $y=0$.

Working: _____

9. A line table has x 0,1,2 and y 3,6,9. Write a rule.

Working: _____

10. Why should three or more table points be used when possible?

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: -7.

2: -2,0,2,4.

3: B. (1,4).

4: (0,6).

5: For example (0,0) and (1,5).

6: Start (0,-2); move 2 right and 1 up to (2,-1), then draw the line.

7: Yes: $-2(-1)+2=4$.

8: $0=x-4$, so $x=4$; point (4,0).

9: $y=3x+3$.

10: They help reveal a calculation or plotting error before the line is drawn.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. For $y=-2x+5$, complete a table for $x=-2$ to 3, state intercepts and describe the line. [9 marks]

Working space: _____

2. A delivery charge follows $C=180n+600$, where n is number of parcels. Plot-ready values are required for $n=0,2,4,6$. Find them and interpret 600. [8 marks]

Working space: _____

3. Two students graph $y=3x-2$. One starts at (0,3); the other starts at (0,-2). Who is correct, and how is a second point found? [6 marks]

Working space: _____

4. A graph passes through $(-2,-5), (0,-1), (2,3)$. Show that the points are consistent with $y=2x-1$. [7 marks]

Working space: _____

5. A line is drawn through (0,4) and (3,4). Write its equation and explain its shape. [5 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 9 marks

Model solution: y values 9,7,5,3,1,-1. y-intercept (0,5). x-intercept (2.5,0). It descends 2 for every 1 right. (9)

Question 2 — 8 marks

Model solution: Pairs (0,600),(2,960),(4,1,320),(6,1,680). The 600 is the fixed charge when no parcels are counted. (8)

Question 3 — 6 marks

Model solution: The second student: $c=-2$. From (0,-2), gradient 3 means move 1 right and 3 up to (1,1). (6)

Question 4 — 7 marks

Model solution: Substitution gives -5,-1,3 respectively. Each matches, and equal x increases of 2 give y increases of 4, gradient 2. (7)

Question 5 — 5 marks

Model solution: $y=4$. Every point has y-coordinate 4, gradient 0, so it is horizontal. (5)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: For $y=mx+c$: m =gradient, c =y-intercept.

Remember: Create pairs by substituting x .

Remember: Plot x first, then y .

Remember: Use a ruler and label the line.

Remember: Check a plotted point in the equation.

Final Self-Check

- ☐ I can complete a value table for a linear rule.
- ☐ I can plot points and draw a straight line.
- ☐ I can sketch a line from its gradient and intercept.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3 • LESSON 3

Gradient and Intercepts of Straight Lines

Unit 1: Algebra | Relations, Functions and Graphs

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 1: Algebra
Strand	Relations, Functions and Graphs
Lesson	3: Gradient and Intercepts of Straight Lines
NSC objective(s)	Determine gradients and intercepts of straight-line graphs.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Gradient describes steepness and direction, while intercepts show starting or break-even values in real-world graphs.

Learning Objectives — I Can

- I can calculate gradient from two points or a graph.
- I can identify x- and y-intercepts.
- I can compare positive, negative, zero and undefined gradients.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Rise	The vertical change between two points.
Run	The horizontal change between two points.
Gradient	Rise divided by run.
x-intercept	The point where a graph crosses the x-axis.
Undefined gradient	The gradient of a vertical line because its run is zero.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

For two points (x_1, y_1) and (x_2, y_2) , $m = (y_2 - y_1) / (x_2 - x_1)$. Use the same point order in numerator and denominator.

A line rising left to right has positive gradient; falling has negative gradient; horizontal has zero; vertical has undefined gradient.

At the y-intercept $x=0$. At the x-intercept $y=0$. Read graph scales carefully or substitute these zero values into an equation.

Small Steps to Follow

1. Choose two clear points on the line, preferably grid intersections.
2. Find vertical change $y_2 - y_1$.
3. Find matching horizontal change $x_2 - x_1$.
4. Divide rise by run and simplify.
5. Read or calculate intercepts, writing them as coordinate points.

Notebook visual: Draw four mini-lines: rising labelled positive, falling negative, horizontal zero, vertical undefined. Beside a rising line draw a right-angle gradient triangle.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Find gradient through $(1, 3)$ and $(5, 11)$.

1. $m = (11 - 3) / (5 - 1)$.
2. Rise=8 and run=4.
3. $8 \div 4 = 2$.

Answer: Gradient $m=2$.

Worked Example 2

Problem: Find gradient and y-intercept of $y = -3x + 6$.

1. Compare with $y = mx + c$.
2. $m = -3$.
3. $c = 6$, so the line crosses y-axis when $x = 0$.

Answer: Gradient -3 ; y-intercept $(0,6)$.

Worked Example 3

Problem: Find both intercepts of $2x+y=8$.

1. For y-intercept, set $x=0$: $y=8 \rightarrow (0,8)$.
2. For x-intercept, set $y=0$: $2x=8$, so $x=4 \rightarrow (4,0)$.

Answer: x-intercept $(4,0)$; y-intercept $(0,8)$.

Common Mistakes and Fixes

Common mistake	How to fix it
Mixing point order	Use the same order in both differences.
Reading grid boxes without checking scale	Use axis values, not just counted boxes.
Writing an intercept as one number only	When asked for a point, write $(0,c)$ or $(a,0)$.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Find gradient from (0,2) to (3,8).

Check: $(8-2)/(3-0)=6/3=2$.

2. Find gradient from (-1,4) to (2,-2).

Check: $(-2-4)/(2-(-1))=-6/3=-2$.

3. State gradient and y-intercept of $y=\frac{1}{2}x-5$.

Check: $m=\frac{1}{2}$; y-intercept (0,-5).

4. What is the gradient of $y=7$?

Check: 0.

5. Find x-intercept of $y=2x-10$.

Check: Set $y=0$: $x=5$, so (5,0).

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Find gradient between (2,1) and (6,9).

Working: _____

2. Find gradient between (-3,5) and (1,3).

Working: _____

3. A line falling left to right has: A. positive B. negative C. zero D. no intercept

Working: _____

4. State m and c for $y=4x+9$.

Working: _____

5. State y-intercept of $y=-x$.

Working: _____

6. Find intercepts of $x+2y=6$.

Working: _____

7. Find gradient of a horizontal line through (-2,4) and (5,4).

Working: _____

8. Why is a vertical line's gradient undefined?

Working: _____

9. A graph rises 9 units for a run of 6. Simplify gradient.

Working: _____

10. Two points are (4,7) and (1,1). Show point order does not change gradient.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $8/4=2$.

2: $(3-5)/(1+3)=-2/4=-\frac{1}{2}$.

3: B. Negative gradient.

4: $m=4$, $c=9$.

5: $(0,0)$.

6: x-intercept $(6,0)$; y-intercept $(0,3)$.

7: $0/7=0$.

8: Its horizontal change is zero, and division by zero is undefined.

9: $9/6=3/2$.

10: $(7-1)/(4-1)=6/3=2$; reverse gives $(1-7)/(1-4)=-6/-3=2$.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A line passes through $(-2,7)$ and $(4,-5)$. Find gradient and describe direction. [6 marks]

Working space: _____

2. Find gradient and both intercepts of $3x+2y=12$. [8 marks]

Working space: _____

3. A line crosses axes at $(0,-4)$ and $(8,0)$. Find its gradient. [5 marks]

Working space: _____

4. Two road profiles have gradients 0.08 and 0.12. Which is steeper, and what does each mean per 100 horizontal metres? [6 marks]

Working space: _____

5. A student calculates $m=(10-2)/(1-5)$ for points $(1,2)$, $(5,10)$ and gets -2 . Diagnose the error. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 6 marks

Model solution: $m = (-5 - 7) / (4 - (-2)) = -12 / 6 = -2$. It falls 2 units for each 1 right. (6)

Question 2 — 8 marks

Model solution: Rearrange $y = -3/2x + 6$, so $m = -3/2$. y-intercept (0,6); x-intercept (4,0). (8)

Question 3 — 5 marks

Model solution: $m = (0 - (-4)) / (8 - 0) = 4/8 = 1/2$. (5)

Question 4 — 6 marks

Model solution: 0.12 is steeper. 0.08 means 8 m rise per 100 m; 0.12 means 12 m rise per 100 m. (6)

Question 5 — 6 marks

Model solution: The numerator used second minus first but denominator used first minus second. Use consistent order: $(10 - 2) / (5 - 1) = 8/4 = 2$. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: $m = (y_2 - y_1) / (x_2 - x_1)$.

Remember: Positive rises; negative falls; horizontal 0; vertical undefined.

Remember: y-intercept: $x=0$.

Remember: x-intercept: $y=0$.

Remember: Use two exact points and simplify the fraction.

Final Self-Check

- ☐ I can calculate gradient from two points or a graph.
- ☐ I can identify x- and y-intercepts.
- ☐ I can compare positive, negative, zero and undefined gradients.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3 • LESSON 4

Gradient as a Rate of Change

Unit 1: Algebra | Relations, Functions and Graphs

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 1: Algebra
Strand	Relations, Functions and Graphs
Lesson	4: Gradient as a Rate of Change
NSC objective(s)	Relate the gradient of a graph to the rate of change of quantities.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Rates describe speed, cost per item, water-level change, temperature change and many other everyday relationships.

Learning Objectives — I Can

- I can calculate change in output per unit input.
- I can attach correct units to a gradient.
- I can interpret the gradient and intercept in context.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Rate of change	How much one quantity changes for each unit change in another.
Dependent variable	The output shown on the vertical axis.
Independent variable	The input shown on the horizontal axis.
Constant rate	The same rate of change throughout.
Initial value	The output when the input is zero.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

On a graph, $\text{gradient} = \text{change in vertical quantity} \div \text{change in horizontal quantity}$. Its units are vertical units per horizontal unit.

A straight line has a constant rate. A positive rate means the vertical quantity increases; a negative rate means it decreases.

In $y = mx + c$, m is the rate and c is the initial value. Always explain them using the quantities and units in the problem.

Small Steps to Follow

1. Name what each axis represents and record its unit.
2. Choose two points and calculate vertical change.
3. Calculate horizontal change over the same interval.
4. Divide and attach output units per input unit.
5. Write a sentence explaining the sign and meaning; interpret the intercept if useful.

Notebook visual: Draw a distance-time line from $(0,0)$ to $(3,180)$. Add a gradient triangle labelled $180 \text{ km} \div 3 \text{ h} = 60 \text{ km/h}$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Distance rises from 40 km at 1 h to 160 km at 3 h. Find rate.

1. Distance change $= 160 - 40 = 120 \text{ km}$.
2. Time change $= 3 - 1 = 2 \text{ h}$.
3. Rate $= 120 \div 2 = 60$.

Answer: 60 km/h.

Worked Example 2

Problem: A tank contains 500 L at minute 0 and 380 L at minute 4.

1. Volume change $= 380 - 500 = -120 \text{ L}$.
2. Time change $= 4 \text{ min}$.
3. Rate $= -120 \div 4 = -30 \text{ L/min}$.

Answer: The tank is losing 30 L each minute.

Worked Example 3

Problem: A taxi cost follows $C=250d+600$.

1. The coefficient 250 is the gradient.
2. Cost is vertical and distance is horizontal, so units are dollars per kilometre.
3. The intercept 600 is cost at $d=0$.

Answer: Rate=J\$250/km; initial charge=J\$600.

Common Mistakes and Fixes

Common mistake	How to fix it
Ignoring axis units	Gradient units are vertical unit per horizontal unit.
Removing the negative sign	Keep it and explain that the quantity decreases.
Calling the intercept a rate	The intercept is an initial or fixed value, not change per unit.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. A worker earns J\$6,000 for 8 hours. Find constant rate.

Check: $6,000 \div 8 = \text{J\$}750$ per hour.

2. Temperature changes from 28°C to 22°C in 3 hours. Find average rate.

Check: $(22-28) \div 3 = -2^{\circ}\text{C}$ per hour.

3. In $C=90n+400$, interpret 90.

Check: Cost increases J\$90 for each additional item.

4. In $h=5t+12$, interpret 12 if h is centimetres and t is seconds.

Check: Initial height is 12 cm at $t=0$.

5. What does a zero gradient mean in a distance-time graph?

Check: Distance is unchanged during that time, so the object is stopped.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. A car travels 150 km in 2.5 h. Find rate.

Working: _____

2. Cost changes from J\$1,200 at 3 items to J\$2,000 at 7 items. Find rate.

Working: _____

3. Which units suit a rainfall-depth/time gradient? A. h/mm B. mm/h C. mm^2 D. h

Working: _____

4. Water level falls 18 cm in 6 min. Find signed rate.

Working: _____

5. Interpret $m=4$ in $y=4x+2$.

Working: _____

6. A pay graph has intercept 3,000 and gradient 500 per day. Write an equation.

Working: _____

7. A plant height follows $h=1.5w+8$. Interpret both numbers.

Working: _____

8. A straight line goes through (2,10) and (8,34), with x in minutes and y in litres. Find rate.

Working: _____

9. Why must the units be written with a rate?

Working: _____

10. Can a negative rate mean an error? Explain.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: 60 km/h.
- 2: $800 \div 4 = \text{J\$}200$ per item.
- 3: B. mm/h.
- 4: -3 cm/min.
- 5: y increases by 4 units for every 1-unit increase in x.
- 6: $P = 500d + 3,000$.
- 7: It grows 1.5 cm per week and started at 8 cm.
- 8: $(34 - 10) / (8 - 2) = 24 / 6 = 4$ L/min.
- 9: They identify which quantity changes per unit of which other quantity.
- 10: Not necessarily; it correctly describes a quantity decreasing, such as cooling or draining.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. A water tank graph passes through (5 min, 120 L) and (17 min, 300 L). Find and interpret its gradient. Predict the increase over 30 min at that rate. [7 marks]

Working space: _____

2. A phone plan is $C = 18m + 1,200$, where m is call minutes. Interpret gradient and intercept, then find cost for 75 min. [7 marks]

Working space: _____

3. Town A cools from 30°C to 24°C in 4 h. Town B cools from 27°C to 19°C in 5 h. Compare average rates. [8 marks]

Working space: _____

4. A distance-time line begins at 20 km when $t = 0$ and has gradient 65 km/h. Write a model and find distance at 3.5 h. [7 marks]

Working space: _____

5. A graph has segments with gradients 40 km/h, 0 km/h and -30 km/h. Explain a possible journey. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 7 marks

Model solution: Gradient = $(300 - 120) / (17 - 5) = 180 / 12 = 15$ L/min. In 30 min, increase = $15 \times 30 = 450$ L. (7)

Question 2 — 7 marks

Model solution: Gradient J\$18/min; intercept/base fee J\$1,200. $C = 18 \times 75 + 1,200 = \text{J\$}2,550$. (7)

Question 3 — 8 marks

Model solution: $A = (24 - 30) / 4 = -1.5^\circ\text{C/h}$. $B = (19 - 27) / 5 = -1.6^\circ\text{C/h}$. B cools faster by 0.1°C/h in magnitude. (8)

Question 4 — 7 marks

Model solution: $d = 65t + 20$. At 3.5 h, $d = 227.5 + 20 = 247.5$ km. (7)

Question 5 — 6 marks

Model solution: First the traveller moves away at 40 km/h, then stops, then returns toward the starting point at 30 km/h. Negative gradient indicates decreasing distance from the reference point. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Rate = vertical change $\div$ horizontal change.

Remember: Units: y-units per x-unit.

Remember: Positive = increase; negative = decrease; zero = no change.

Remember: In $y = mx + c$, m = rate and c = initial value.

Remember: Interpret numbers in a complete context sentence.

Final Self-Check

- ☐ I can calculate change in output per unit input.
- ☐ I can attach correct units to a gradient.
- ☐ I can interpret the gradient and intercept in context.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3 • LESSON 5

Finding the Equation of a Straight Line

Unit 1: Algebra | Relations, Functions and Graphs

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 1: Algebra
Strand	Relations, Functions and Graphs
Lesson	5: Finding the Equation of a Straight Line
NSC objective(s)	Find the equation of any straight line.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: An equation is a compact model of every point on a line. It can be used to predict costs, distances and other steady changes.

Learning Objectives — I Can

- I can write $y=mx+c$ from gradient and intercept.
- I can find an equation from a point and gradient or two points.
- I can identify equations of horizontal and vertical lines.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Slope-intercept form	The form $y=mx+c$, showing gradient m and y -intercept c .
Point-slope idea	Using a known point and gradient to build a line equation.
Substitute	Replace a variable with a known value.
Horizontal line	A line with equation $y=k$ and gradient zero.
Vertical line	A line with equation $x=k$ and undefined gradient.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

If gradient m and y -intercept c are known, the equation is immediately $y=mx+c$.

If m and a point (x,y) are known, substitute them in $y=mx+c$ to solve for c . If two points are known, find m first, then find c .

Horizontal lines have constant y , so $y=k$. Vertical lines have constant x , so $x=k$ and cannot be written as $y=mx+c$.

Small Steps to Follow

1. Identify what is given: graph, gradient/intercept, one point or two points.
2. Calculate gradient if it is not given.
3. Write $y=mx+c$ and substitute m .
4. Use a known point to solve for c .
5. Write the equation, simplify it and verify both a known point and the gradient.

Notebook visual: Draw a decision tree: m and c known $\rightarrow$ write equation; m and point $\rightarrow$ solve c ; two points $\rightarrow$ find m then c ; horizontal/vertical $\rightarrow$ use $y=k$ or $x=k$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Find the equation with gradient 3 and y -intercept -4 .

1. Use $y=mx+c$.
2. Replace m with 3 and c with -4 .
3. Write subtraction clearly.

Answer: $y=3x-4$.

Worked Example 2

Problem: Find the line with gradient 2 through $(4,11)$.

1. Write $y=2x+c$.
2. Substitute $(4,11)$: $11=2(4)+c$.
3. $11=8+c$, so $c=3$.

Answer: $y=2x+3$.

Worked Example 3

Problem: Find the equation through (1,5) and (4,14).

1. $m=(14-5)/(4-1)=9/3=3$.
2. Use point (1,5): $5=3(1)+c$.
3. $c=2$.
4. Check (4,14): $3(4)+2=14$.

Answer: $y=3x+2$.

Common Mistakes and Fixes

Common mistake	How to fix it
Swapping m and c	m multiplies x; c is the y-value at $x=0$.
Finding gradient but not intercept	Use a known point to solve for c.
Forcing a vertical line into $y=mx+c$	Write $x=k$ for a vertical line.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Write the equation with $m=4$ and $c=-1$.

Check: $y=4x-1$.

2. Find equation with $m=-2$ through $(0,5)$.

Check: Since the point is the y -intercept, $y=-2x+5$.

3. Find c if $y=3x+c$ passes through $(2,10)$.

Check: $10=6+c$, so $c=4$.

4. Write equation of the horizontal line through $y=7$.

Check: $y=7$.

5. Write equation of the vertical line through $x=-3$.

Check: $x=-3$.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Write equation for $m=\frac{1}{2}$, $c=6$.

Working: _____

2. Find equation with $m=5$ through $(1,8)$.

Working: _____

3. Which line has gradient -3 and intercept 2 ? A. $y=2x-3$ B. $y=-3x+2$ C. $y=3x-2$ D. $x=-3$

Working: _____

4. Find equation through $(0,-4)$ and $(2,2)$.

Working: _____

5. Find equation through $(-1,1)$ and $(3,9)$.

Working: _____

6. Find equation of a line parallel to $y=4x-7$ with y -intercept 5 .

Working: _____

7. Write the horizontal line through $(6,-2)$.

Working: _____

8. Write the vertical line through $(6, -2)$.

Working: _____

9. Rearrange $2x + y = 9$ into $y = mx + c$.

Working: _____

10. Verify that $(3, 7)$ lies on $y = 2x + 1$.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $y = \frac{1}{2}x + 6$.

2: $8 = 5 + c$, $c = 3$; $y = 5x + 3$.

3: B. $y = -3x + 2$.

4: $m = 6/2 = 3$; $c = -4$; $y = 3x - 4$.

5: $m = 8/4 = 2$; $1 = -2 + c$, $c = 3$; $y = 2x + 3$.

6: Parallel lines have same gradient: $y = 4x + 5$.

7: $y = -2$.

8: $x = 6$.

9: $y = -2x + 9$.

10: $2(3) + 1 = 7$, so it lies on the line.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Find the equation through $(-2, 8)$ and $(4, -4)$. Show a check. [8 marks]

Working space: _____

2. A line has x-intercept $(6, 0)$ and y-intercept $(0, 3)$. Find its equation. [7 marks]

Working space: _____

3. A repair cost is J\$9,500 at 2 hours and J\$17,000 at 5 hours. Assuming a linear model, find equation $C = mh + c$ and interpret both constants. [9 marks]

Working space: _____

4. Convert $3x - 2y = 12$ to slope-intercept form, then state gradient and intercepts. [8 marks]

Working space: _____

5. Explain why no equation $y = mx + c$ can describe the line through $(5, -2)$ and $(5, 9)$. Give its correct equation. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 8 marks

Model solution: $m = (-4 - 8)/(4 + 2) = -12/6 = -2$. Use $(-2, 8)$: $8 = 4 + c$, $c = 4$. Equation $y = -2x + 4$. Check $x = 4$ gives -4 . (8)

Question 2 — 7 marks

Model solution: $m = (0 - 3)/(6 - 0) = -3/6 = -\frac{1}{2}$. Since $c = 3$, $y = -\frac{1}{2}x + 3$. (7)

Question 3 — 9 marks

Model solution: $m = (17,000 - 9,500)/(5 - 2) = 2,500$ J\$/h. $9,500 = 2,500(2) + c$, $c = 4,500$. $C = 2,500h + 4,500$; hourly charge 2,500 and fixed fee 4,500. (9)

Question 4 — 8 marks

Model solution: $-2y = 12 - 3x$; $y = 1.5x - 6$. Gradient $3/2$; y-intercept $(0, -6)$; x-intercept $(4, 0)$. (8)

Question 5 — 6 marks

Model solution: Both points have $x = 5$, so run = 0 and gradient is undefined. It is vertical, with equation $x = 5$. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Line form: $y=mx+c$.

Remember: Two points: find m first, then substitute to find c .

Remember: Horizontal: $y=k$; vertical: $x=k$.

Remember: Parallel lines have equal gradients.

Remember: Verify by substituting a known point.

Final Self-Check

- ☐ I can write $y=mx+c$ from gradient and intercept.
- ☐ I can find an equation from a point and gradient or two points.
- ☐ I can identify equations of horizontal and vertical lines.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3 • LESSON 6

Solving Two Linear Equations by Graphing

Unit 1: Algebra | Relations, Functions and Graphs

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 1: Algebra
Strand	Relations, Functions and Graphs
Lesson	6: Solving Two Linear Equations by Graphing
NSC objective(s)	Plot two linear equations on the same axes and interpret their point or points of intersection, if any.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: An intersection gives values that satisfy two conditions at once, such as when two pricing plans cost the same.

Learning Objectives — I Can

- I can graph two equations on the same axes.
- I can read the intersection as a simultaneous solution.
- I can recognise one, no or infinitely many solutions.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Simultaneous equations	Two equations that must be true for the same x and y.
Intersection	A point where two graphs meet.
Solution	Values that make every equation true.
Parallel lines	Lines with the same gradient that never meet.
Coincident lines	Two equations that describe the same line.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

Create at least two or three points for each line and plot both on the same coordinate plane using different colours or labels.

The intersection (x,y) lies on both lines, so it is the graphical solution. Read it carefully and verify by substitution.

Different gradients give one intersection. Equal gradients with different intercepts give parallel lines and no solution. Equivalent equations give the same line and infinitely many solutions.

Small Steps to Follow

1. Rewrite each equation in a graph-friendly form if necessary.
2. Make a small value table for each line.
3. Plot both lines accurately on the same labelled axes.
4. Read the coordinate where the lines meet.
5. Substitute the coordinate into both equations and state the number of solutions.

Notebook visual: Draw two lines labelled $y=x+1$ and $y=-x+5$ meeting at $(2,3)$. Add a callout: one point, two true equations.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Solve $y=x+1$ and $y=-x+5$ graphically.

1. For $y=x+1$ use points $(0,1), (2,3), (4,5)$.
2. For $y=-x+5$ use $(0,5), (2,3), (4,1)$.
3. Draw both lines; they meet at $(2,3)$.
4. Check: $3=2+1$ and $3=-2+5$.

Answer: $x=2, y=3$.

Worked Example 2

Problem: Determine the solutions of $y=2x+1$ and $y=2x-4$.

1. Both gradients are 2.
2. Their y -intercepts 1 and -4 differ.

3. The lines are parallel and never meet.

Answer: No solution.

Worked Example 3

Problem: Compare $y=3x-2$ and $2y=6x-4$.

1. Divide the second equation by 2: $y=3x-2$.
2. Both equations describe exactly the same line.
3. Every point on that line satisfies both.

Answer: Infinitely many solutions.

Common Mistakes and Fixes

Common mistake	How to fix it
Reading the nearest grid point without checking	Substitute the estimated intersection into both equations.
Using separate axes or scales	Plot both lines on the same coordinate system.
Assuming every pair intersects	Parallel distinct lines have no solution; coincident lines have infinitely many.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. What does an intersection represent?

Check: The ordered pair satisfying both equations.

2. Solve $y=x$ and $y=-x+4$.

Check: They meet at $(2,2)$.

3. How many solutions for $y=5x+1$ and $y=5x+6$?

Check: None; same gradient, different intercepts.

4. How many solutions for $y=-2x+3$ and $2y=-4x+6$?

Check: Infinitely many; the equations are equivalent.

5. Why should a graph solution be checked algebraically?

Check: Graph reading and drawing can introduce small errors; substitution confirms the point.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Solve $y=x+2$ and $y=-x+6$.

Working: _____

2. Solve $y=2x$ and $y=x+3$.

Working: _____

3. Two distinct parallel lines have: A. one solution B. no solution C. two solutions D. infinitely many

Working: _____

4. Solve $y=3x-1$ and $y=x+3$.

Working: _____

5. State number of solutions: $y=-x+4$ and $x+y=4$.

Working: _____

6. State number of solutions: $y=\frac{1}{2}x+1$ and $2y=x+8$.

Working: _____

7. Verify $(1,3)$ for $y=2x+1$ and $y=-x+4$.

Working: _____

8. A graph shows lines meet near (2.1,4.9). What should you report if axes only allow tenths?

Working: _____

9. Why must both lines use the same axis scales?

Working: _____

10. Plans cost $A=300n+1,000$ and $B=200n+1,800$. Find break-even n and cost.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: Intersection (2,4).
- 2: $2x=x+3$, $x=3$; $y=6$, so (3,6).
- 3: B. No solution.
- 4: $3x-1=x+3$, so $x=2$, $y=5$; (2,5).
- 5: Infinitely many; $x+y=4$ rearranges to $y=-x+4$.
- 6: Second is $y=\frac{1}{2}x+4$; lines are parallel, so none.
- 7: First: $2(1)+1=3$. Second: $-1+4=3$. It satisfies both.
- 8: Approximately $x=2.1$, $y=4.9$, clearly marked as an estimate.
- 9: Otherwise their geometric intersection would not represent the same coordinate system.
- 10: $300n+1,000=200n+1,800$; $n=8$; cost=J\$3,400.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Graph-ready tables are needed to solve $y=2x-3$ and $y=-x+6$ for $x=0,1,2,3$. Give both tables and the solution. [9 marks]

Working space: _____

2. Plan A costs J\$500 per week plus J\$2,000. Plan B costs J\$800 per week plus J\$800. Find when costs are equal and which is cheaper before and after. [9 marks]

Working space: _____

3. Classify the system $3x+y=7$ and $6x+2y=20$ without drawing every point. [7 marks]

Working space: _____

4. A student reports intersection (2,5) for $y=2x+1$ and $y=-x+7$. Test and correct the answer. [5 marks]

Working space: _____

5. Explain the difference among one, none and infinitely many graphical solutions, including the gradient/intercept conditions. [8 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 9 marks

Model solution: First y : $-3, -1, 1, 3$. Second y : $6, 5, 4, 3$. Lines meet at $(3, 3)$. (9)

Question 2 — 9 marks

Model solution: $A = 500w + 2,000$; $B = 800w + 800$. Equal: $1,200 = 300w$, $w = 4$; cost = 4,000. Before 4 weeks B is cheaper; after 4 weeks A is cheaper. (9)

Question 3 — 7 marks

Model solution: First $y = -3x + 7$; second $y = -3x + 10$. Same gradient, different intercepts: parallel, no solution. (7)

Question 4 — 5 marks

Model solution: At $(2, 5)$, first is true and second gives 5, so it is actually true: $-2 + 7 = 5$. The report is correct. (5)

Question 5 — 8 marks

Model solution: One: lines with different gradients meet once. None: same gradient, different intercepts, so parallel. Infinitely many: same gradient and intercept/equivalent equations, so coincident. (8)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Intersection= (x,y) satisfying both lines.

Remember: Plot both on the same axes and label them.

Remember: Different $m \rightarrow$ one solution.

Remember: Same m , different $c \rightarrow$ none; same line $\rightarrow$ infinitely many.

Remember: Verify by substitution.

Final Self-Check

- ☐ I can graph two equations on the same axes.
- ☐ I can read the intersection as a simultaneous solution.
- ☐ I can recognise one, no or infinitely many solutions.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3 • LESSON 7

Graphing Linear Inequalities on the Coordinate Plane

Unit 1: Algebra | Relations, Functions and Graphs

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 1: Algebra
Strand	Relations, Functions and Graphs
Lesson	7: Graphing Linear Inequalities on the Coordinate Plane
NSC objective(s)	Graph linear inequalities on the coordinate plane and identify regions.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Inequalities show a whole region of possible solutions, useful for limits such as budgets, capacity and combinations of quantities.

Learning Objectives — I Can

- I can draw the correct boundary line.
- I can choose solid or dashed boundaries.
- I can test and shade the solution region.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Inequality	A comparison using $<$, $>$, $\leq$ or $\geq$.
Boundary line	The line that separates solution and non-solution regions.
Solution region	All points that make the inequality true.
Solid line	A boundary included in the solution for $\leq$ or $\geq$.
Dashed line	A boundary not included for $<$ or $>$.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

First replace the inequality sign with $=$ and graph that boundary line. The boundary divides the plane into two regions.

Use a solid boundary for $\leq$ or $\geq$ because equality is included. Use a dashed boundary for $<$ or $>$ because points on the line are excluded.

Test a point not on the boundary, often $(0,0)$. If it makes the inequality true, shade its side; otherwise shade the opposite side.

Small Steps to Follow

1. Rearrange into y compared with $mx+c$ when helpful.
2. Draw the equality boundary using a table or gradient and intercept.
3. Choose solid for inclusive signs or dashed for strict signs.
4. Substitute a convenient test point not on the boundary.
5. Shade the correct half-plane and check one shaded point and one unshaded point.

Notebook visual: Draw $y=2$ as a solid horizontal line. Shade above for $y\geq 2$. Beside it draw $x=-1$ dashed and shade left for $x<-1$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Graph $y\leq x+2$.

1. Boundary is $y=x+2$ through $(0,2)$ and $(-2,0)$.
2. Use a solid line because $\leq$ includes equality.
3. Test $(0,0)$: $0\leq 2$ is true.
4. Shade the side containing $(0,0)$, below the line.

Answer: Solid $y=x+2$; shade on and below it.

Worked Example 2

Problem: Graph $y>-2x+1$.

1. Boundary $y=-2x+1$ is dashed because $>$ excludes equality.
2. Test $(0,0)$: $0>1$ is false.

3. Shade the opposite side, above the line.

Answer: Dashed boundary; shade above.

Worked Example 3

Problem: Graph $2x + y \geq 6$.

1. Rearrange: $y \geq -2x + 6$.
2. Boundary crosses (0,6) and (3,0), and is solid.
3. Test (0,0): $0 \geq 6$ is false.
4. Shade the side not containing the origin.

Answer: Solid line $2x + y = 6$; shade above it.

Common Mistakes and Fixes

Common mistake	How to fix it
Shading from the sign alone without considering rearrangement	Draw and test a point.
Using the wrong boundary style	$\leq$ or $\geq$ solid; $<$ or $>$ dashed.
Testing a boundary point	Choose a point away from the line, such as the origin when possible.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. For $y < 3$, is the boundary solid or dashed?

Check: Dashed.

2. For $y \geq -x + 4$, which side is shaded?

Check: On and above the line.

3. Does $(0,0)$ satisfy $y \leq 2x + 1$?

Check: Yes: $0 \leq 1$.

4. Graph $x > 2$ in words.

Check: Dashed vertical line $x = 2$; shade right.

5. Why must a test point not lie on the boundary?

Check: A boundary point gives equality and may not distinguish which side satisfies the inequality.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. State boundary for $y > 3x - 2$.

Working: _____

2. State boundary type for $2x + y \leq 8$.

Working: _____

3. Which sign uses a dashed line? A. $\leq$ B. $\geq$ C. $<$ D. $=$

Working: _____

4. Does $(1,1)$ satisfy $y \geq x - 2$?

Working: _____

5. Does $(3,0)$ satisfy $x + y < 2$?

Working: _____

6. Describe graph of $y \leq -1$.

Working: _____

7. Describe graph of $x \geq 4$.

Working: _____

8. Rearrange $x+2y>6$ into y form.

Working: _____

9. For $y<-x+5$, test $(0,0)$ and choose region.

Working: _____

10. Explain what a shaded point means.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: $y=3x-2$, dashed.

2: Solid because equality is included.

3: C. $<$.

4: Yes: $1 \geq -1$.

5: No: $3 < 2$ is false.

6: Solid horizontal line $y=-1$; shade below.

7: Solid vertical line $x=4$; shade right.

8: $2y > 6-x$, so $y > -\frac{1}{2}x + 3$.

9: $0 < 5$ is true, so shade the side containing origin, below the dashed line.

10: Its coordinates make the original inequality true.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Describe the complete graph of $3x+y \leq 9$, including intercepts, boundary and test. [8 marks]

Working space: _____

2. A budget condition is $500x+1,000y \leq 10,000$ for quantities $x, y \geq 0$. Find axis intercepts and explain the feasible region. [9 marks]

Working space: _____

3. A student graphs $y < 2x-4$ with a solid line and shades above. Correct both choices using a test point. [7 marks]

Working space: _____

4. Determine whether points A(0,5), B(2,1), C(4,-3) satisfy $y \geq -2x+4$. [7 marks]

Working space: _____

5. Two regions $y \geq x$ and $y \leq -x+4$ overlap. Give three integer solution points and explain how both boundaries are drawn. [8 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 8 marks

Model solution: Boundary $y = -3x + 9$ crosses $(0, 9), (3, 0)$, solid. Test origin: $0 \leq 9$ true, so shade on/below the line including origin. (8)

Question 2 — 9 marks

Model solution: With $y = 0$, $x = 20$; with $x = 0$, $y = 10$. Use solid line through $(20, 0), (0, 10)$. Feasible points are in first quadrant on/below line, including axes. (9)

Question 3 — 7 marks

Model solution: Use dashed because $<$ excludes equality. Test $(0, 0)$: $0 < -4$ false, so origin is not shaded; shade below the line. (7)

Question 4 — 7 marks

Model solution: A: $5 \geq 4$ yes. B: $1 \geq 0$ yes. C: $-3 \geq -4$ yes. All three satisfy; none lies on boundary because comparisons are strict greater in each case. (7)

Question 5 — 8 marks

Model solution: Examples $(0, 0), (1, 2), (2, 2)$. Both boundaries are solid. Shade above $y = x$ and below $y = -x + 4$; their overlap is the solution region. (8)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Replace the sign with = to draw the boundary.

Remember: $\leq, \geq$: solid; $<, >$: dashed.

Remember: Test a point not on the line.

Remember: True $\rightarrow$ shade test-point side; false $\rightarrow$ shade opposite.

Remember: Every shaded point satisfies the inequality.

Final Self-Check

- ☐ I can draw the correct boundary line.
- ☐ I can choose solid or dashed boundaries.
- ☐ I can test and shade the solution region.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3 • LESSON 8

Geometric Constructions with Ruler and Compasses

Unit 2: Geometry | Constructions

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 2: Geometry
Strand	Constructions
Lesson	8: Geometric Constructions with Ruler and Compasses
NSC objective(s)	Construct angle bisectors, angles of 90° , 45° , 60° and 30° , and triangles using appropriate geometric instruments.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Exact constructions are used in technical drawing, design, building layouts and geometric proofs. They create lengths and angles without guessing from a protractor.

Learning Objectives — I Can

- I can bisect an angle using arcs.
- I can construct 90° , 45° , 60° and 30° angles.
- I can construct triangles from sufficient measurements.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Construct	Draw exactly using permitted geometric instruments and steps.
Compasses	An instrument for drawing arcs and transferring distances.
Bisect	Divide into two equal parts.
Arc	Part of a circle drawn with compasses.
Congruent	Exactly the same size and shape.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A construction keeps evidence: arcs and intersection marks must remain visible. A straightedge draws lines; compasses transfer equal distances.

An angle bisector is found by making equal arcs from both arms, then joining the vertex to the arc intersection. A perpendicular creates 90° ; bisecting it gives 45° .

An equilateral triangle creates 60° because all three angles are 60° . Bisecting 60° gives 30° . Triangle construction requires enough information such as SSS, SAS or ASA.

Small Steps to Follow

1. Use a sharp pencil, labelled points, a straightedge and compasses.
2. Set the compass opening from the stated length or a chosen convenient radius.
3. Draw light arcs accurately without changing the opening when equal distances are required.
4. Mark arc intersections and join the correct points.
5. Keep construction arcs, label the result and verify with geometric properties.

Notebook visual: Draw a four-panel notebook guide: angle bisector; perpendicular at a point; equilateral triangle for 60° ; SSS triangle with two intersecting arcs.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Bisect angle ABC.

1. With centre B, draw an arc cutting BA at P and BC at Q.
2. Using the same compass opening, draw arcs from P and Q that meet at R inside the angle.
3. Draw ray BR.
4. Triangles BPR and BQR have matching radii, so the two angles at B are equal.

Answer: BR bisects $\angle ABC$.

Worked Example 2

Problem: Construct 60° and then 30° at point A on ray AB.

1. Choose a compass radius and draw an arc centred at A cutting AB at P.
2. With the same radius centred at P, cut the arc at Q.

3. Join A to Q; equilateral triangle APQ gives 60° .
4. Bisect $\angle QAB$ using equal intersecting arcs.

Answer: The first ray makes 60° ; its bisector makes 30° .

Worked Example 3

Problem: Construct triangle ABC with $AB=7$ cm, $AC=5$ cm, $BC=6$ cm.

1. Draw $AB=7$ cm.
2. Centre A, radius 5 cm, draw an arc.
3. Centre B, radius 6 cm, draw an arc crossing the first at C.
4. Join AC and BC; retain arcs.

Answer: A unique SSS triangle ABC (up to reflection).

Common Mistakes and Fixes

Common mistake	How to fix it
Erasing arcs	Keep construction evidence visible.
Changing radius for equal arcs	Lock the compass opening until both matching arcs are drawn.
Using thick or blunt lines	Use a sharp pencil and light, precise arcs.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. What does it mean to bisect a 70° angle?

Check: Divide it into two 35° angles.

2. How can 45° be constructed from 90° ?

Check: Construct a right angle and bisect it.

3. Why must the compass opening stay the same for equal-radius arcs?

Check: Changing it would remove the equal-distance condition that guarantees accuracy.

4. For SSS sides 4, 5, 6 cm, which length can be the first base?

Check: Any one can; for example draw the 6 cm side first, then arcs of 4 and 5 cm.

5. Should construction arcs be erased?

Check: No. Keep them as evidence of the method unless specifically told otherwise.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. An angle bisector divides 96° into what angles?

Working: _____

2. Bisecting 60° produces what angle?

Working: _____

3. Bisecting 90° produces: A. 30° B. 40° C. 45° D. 60°

Working: _____

4. Name the instruments normally used for an exact classical construction.

Working: _____

5. Describe the first two steps in constructing a perpendicular at a point P on a line.

Working: _____

6. Why does an equilateral triangle create 60° ?

Working: _____

7. Can sides 2 cm, 3 cm and 6 cm form a triangle? Explain.

Working: _____

8. Describe SSS construction for sides 5,7,8 cm.

Working: _____

9. What does the intersection of two compass arcs represent in triangle construction?

Working: _____

10. State one accuracy check after a construction.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: 48° and 48° .

2: 30° .

3: C. 45° .

4: A straightedge/ruler, compasses and pencil; measurement tasks may also specify a protractor, but the standard arc construction demonstrates exactness.

5: Mark equal points A and B on the line using an arc centred P; then draw equal arcs from A and B to intersect above/below the line.

6: All three sides are equal, so all three angles are equal and total 180° ; each is 60° .

7: No. The two shorter sides total 5 cm, which is less than 6 cm, so their arcs cannot meet.

8: Draw 8 cm base; draw arcs of radius 5 and 7 cm from its endpoints; join their intersection to both endpoints.

9: A point at both required distances from the two arc centres.

10: Measure the result, verify equal lengths/angles geometrically, ensure arcs meet sharply and labels match; any one.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Write an exact construction for a 45° angle at A on line AB, naming all arc centres and the final bisector. [9 marks]

Working space: _____

2. Construct triangle PQR with $PQ=8$ cm, $PR=6$ cm and $QR=7$ cm. Describe the steps and state why the point R is fixed. [8 marks]

Working space: _____

3. Explain how to construct 30° without using a protractor and why the result is exact. [7 marks]

Working space: _____

4. A student draws an angle bisector by measuring both arms from the vertex but then uses visibly different compass radii from the marked points. Explain why the construction may fail. [6 marks]

Working space: _____

5. Given $AB=5$ cm, $\angle A=60^\circ$ and $AC=7$ cm, describe an SAS triangle construction and an independent check. [8 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 9 marks

Model solution: Construct a perpendicular AC at A: mark equal P,Q on line AB; equal arcs from P,Q meet at C; join AC to make 90° . Use an arc centred A to cut AB and AC, then equal arcs from those cuts meet at D; join AD. $\angle DAB=45^\circ$. (9)

Question 2 — 8 marks

Model solution: Draw PQ=8. Arc centre P radius 6; arc centre Q radius 7; intersection is R; join PR,QR. R is fixed because it must be exactly 6 cm from P and 7 cm from Q. (8)

Question 3 — 7 marks

Model solution: Construct an equilateral triangle to obtain 60° , then bisect that angle with equal-radius arcs. The properties of equal radii and angle bisectors guarantee 30° . (7)

Question 4 — 6 marks

Model solution: The intersection is not necessarily equidistant from both arms when radii differ, so the joining ray need not split the angle equally. Use the same radius from the two marked points. (6)

Question 5 — 8 marks

Model solution: Draw AC=7 cm. Construct a 60° ray at A. On it mark B 5 cm from A. Join B to C. Check AB=5 cm, AC=7 cm and $\angle A=60^\circ$ or measure BC as a derived check. (8)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Bisector: one vertex arc, then equal crossing arcs, then join vertex.

Remember: 90° =perpendicular; 45° =half of 90° .

Remember: 60° =equilateral triangle; 30° =half of 60° .

Remember: SSS: base, two radius arcs, join intersection.

Remember: Label points and retain arcs.

Final Self-Check

- ☐ I can bisect an angle using arcs.
- ☐ I can construct 90° , 45° , 60° and 30° angles.
- ☐ I can construct triangles from sufficient measurements.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3 • LESSON 9

Sets, Subsets and the Number of Subsets

Unit 3: Numbers | Sets

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 3: Numbers
Strand	Sets
Lesson	9: Sets, Subsets and the Number of Subsets
NSC objective(s)	Determine the number of subsets of a given set. • List every possible subset when the original set has no more than four elements.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Sets organise information and choices. Subsets describe every possible selection from a group, including choosing none or all.

Learning Objectives — I Can

- I can use correct set notation.
- I can list all subsets of a set with up to four elements.
- I can calculate the number of subsets using 2^n .

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Set	A well-defined collection of distinct objects.
Element	One object in a set.
Subset	A set whose every element belongs to another set.
Empty set	The set with no elements, written $\emptyset$ or $\{\}$.
Proper subset	A subset that is not equal to the whole original set.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A set uses braces, such as $A=\{a,b\}$. Order does not matter and repeated elements are listed once.

For each element, a subset makes one of two choices: include it or do not include it. Therefore a set with n elements has 2^n subsets.

The empty set and the entire set are always subsets. A proper subset excludes the whole set itself; conventions about whether $\emptyset$ is called proper should follow the class definition, but here every subset unequal to A is proper.

Small Steps to Follow

1. Count distinct elements n .
2. Calculate 2^n to know how many subsets to expect.
3. List $\emptyset$ first, then one-element, two-element and larger subsets in an organised order.
4. Include the whole set last.
5. Count the list and check that every entry uses only original elements and totals 2^n .

Notebook visual: Draw a subset tree for $\{a,b\}$: at each branch write include or leave out. The four ends are $\emptyset, \{a\}, \{b\}, \{a,b\}$.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: List every subset of $A=\{p,q\}$.

1. There are $n=2$ elements, so expect $2^2=4$ subsets.
2. Choose none: $\emptyset$.
3. Choose one: $\{p\}, \{q\}$.
4. Choose both: $\{p,q\}$.

Answer: $P(A)=\{\emptyset, \{p\}, \{q\}, \{p,q\}\}$; 4 subsets.

Worked Example 2

Problem: How many subsets does $B=\{2,4,6,8\}$ have?

1. There are 4 distinct elements.
2. Use 2^n .

3. $2^4=16$.

Answer: 16 subsets.

Worked Example 3

Problem: List subsets of $C=\{x,y,z\}$ by size.

1. Size 0: $\emptyset$.
2. Size 1: $\{x\},\{y\},\{z\}$.
3. Size 2: $\{x,y\},\{x,z\},\{y,z\}$.
4. Size 3: $\{x,y,z\}$.
5. Count: $1+3+3+1=8=2^3$.

Answer: Eight subsets as listed.

Common Mistakes and Fixes

Common mistake	How to fix it
Forgetting the empty set	Begin every subset list with $\emptyset$.
Treating order as different	$\{a,b\}=\{b,a\}$.
Omitting the whole set	Every set is a subset of itself.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. How many subsets has a one-element set?

Check: $2^1=2$: $\emptyset$ and the whole set.

2. Is $\emptyset$ a subset of $\{1,2\}$?

Check: Yes; the empty set is a subset of every set.

3. List subsets of $\{m\}$.

Check: $\emptyset$ and $\{m\}$.

4. How many subsets has a 5-element set?

Check: $2^5=32$.

5. Is $\{1,3\}$ a subset of $\{1,2,3\}$?

Check: Yes; both 1 and 3 belong to the larger set.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Write in set notation the vowels in the word CAT.

Working: _____

2. How many subsets has the empty set?

Working: _____

3. How many subsets has a 3-element set?

Working: _____

4. Which is not a subset of $\{a,b,c\}$? A. $\emptyset$ B. $\{a,c\}$ C. $\{b\}$ D. $\{a,d\}$

Working: _____

5. List every subset of $\{4,5\}$.

Working: _____

6. List all one-element subsets of $\{r,s,t,u\}$.

Working: _____

7. List all three-element subsets of $\{1,2,3,4\}$.

Working: _____

8. A set has 64 subsets. How many elements?

Working: _____

9. How many proper subsets does a 4-element set have under this lesson's definition?

Working: _____

10. Explain why $\{a,a,b\}$ is the same set as $\{a,b\}$.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: {A}; if using lowercase, {a}.

2: $2^0=1$; its only subset is $\emptyset$.

3: 8.

4: D. {a,d}.

5: $\emptyset, \{4\}, \{5\}, \{4,5\}$.

6: {r},{s},{t},{u}.

7: {1,2,3},{1,2,4},{1,3,4},{2,3,4}.

8: $2^6=64$, so 6 elements.

9: $2^4-1=15$, excluding the whole set.

10: Sets list distinct membership; repetition does not create a new element.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. List all subsets of {J,A,M} systematically and identify the proper subsets. [9 marks]

Working space: _____

2. List all 16 subsets of {1,2,3,4} grouped by number of elements. [12 marks]

Working space: _____

3. A set has twice as many subsets as a set with 7 elements. How many elements does it have? [6 marks]

Working space: _____

4. A student lists {a,b}, {b,a} as two different subsets. Diagnose the mistake and state how it affects the count. [5 marks]

Working space: _____

5. Set D={red,blue,green}. How many selections contain red? List them and explain the pattern. [7 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 9 marks

Model solution: Subsets: $\emptyset, \{J\}, \{A\}, \{M\}, \{J,A\}, \{J,M\}, \{A,M\}, \{J,A,M\}$. Proper subsets are the first seven, all except $\{J,A,M\}$. (9)

Question 2 — 12 marks

Model solution: 0: $\emptyset$. 1: $\{1\}, \{2\}, \{3\}, \{4\}$. 2: $\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}$. 3: $\{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\}$. 4: $\{1,2,3,4\}$. (12)

Question 3 — 6 marks

Model solution: A 7-element set has 2^7 . Twice this is $2 \times 2^7 = 2^8$, so the new set has 8 elements. (6)

Question 4 — 5 marks

Model solution: Set order does not matter, so they are the same subset. Counting both creates a duplicate and makes the list too large. (5)

Question 5 — 7 marks

Model solution: $\{\text{red}\}, \{\text{red}, \text{blue}\}, \{\text{red}, \text{green}\}, \{\text{red}, \text{blue}, \text{green}\}$: 4. Once red is included, blue and green each have two choices, giving $2^2 = 4$. (7)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: A subset uses only elements from the original set.

Remember: n elements $\rightarrow 2^n$ subsets.

Remember: Always include $\emptyset$ and the whole set.

Remember: List by size to avoid repeats.

Remember: Order and repetition do not change a set.

Final Self-Check

- ☐ I can use correct set notation.
- ☐ I can list all subsets of a set with up to four elements.
- ☐ I can calculate the number of subsets using 2^n .
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3 • LESSON 10

Two-Set Venn Diagrams and Problem Solving

Unit 3: Numbers | Sets

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 3: Numbers
Strand	Sets
Lesson	10: Two-Set Venn Diagrams and Problem Solving
NSC objective(s)	Solve simple problems involving at most two subsets of the universal set.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Venn diagrams organise overlapping groups, such as students who play football, netball, both or neither.

Learning Objectives — I Can

- I can place elements in a two-set Venn diagram.
- I can calculate intersection, union, only and neither.
- I can solve word problems without double-counting.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Universal set	All objects being considered, shown by a rectangle.
Intersection	Elements common to both sets, $A \cap B$.
Union	Elements in A or B or both, $A \cup B$.
Complement	Elements in the universal set but outside the named set.
Neither	Elements outside both circles.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

In a two-circle diagram, fill the overlap first. Then subtract it from each total to find the 'A only' and 'B only' regions.

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$. The overlap is subtracted once because adding both set totals counts it twice.

Neither = universal total - union. In mathematics, 'A or B' is usually inclusive, so it includes both unless the problem says 'but not both.'

Small Steps to Follow

1. Draw a rectangle U and two overlapping circles A and B.
2. Place or calculate $A \cap B$ in the overlap first.
3. Find A only = $n(A) - \text{intersection}$ and B only = $n(B) - \text{intersection}$.
4. Add the three circle regions for the union.
5. Subtract union from U for neither, then check all four regions total $n(U)$.

Notebook visual: Draw a labelled two-circle Venn diagram with four numbered regions: A only, both, B only, neither. Add arrows showing 'overlap first'.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: In a class of 30, 18 play football, 12 play netball and 5 play both. Find each region.

1. Both = 5.
2. Football only = $18 - 5 = 13$.
3. Netball only = $12 - 5 = 7$.
4. Union = $13 + 5 + 7 = 25$.
5. Neither = $30 - 25 = 5$.

Answer: Football only 13; both 5; netball only 7; neither 5.

Worked Example 2

Problem: Given $n(A) = 24$, $n(B) = 17$, $n(A \cup B) = 33$, find $n(A \cap B)$.

1. Use union = A + B - intersection.

2. $33 = 24 + 17 - \text{intersection}$.

3. $\text{Intersection} = 41 - 33 = 8$.

Answer: $n(A \cap B) = 8$.

Worked Example 3

Problem: Place $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 4\}$, $B = \{2, 3, 4\}$.

1. Both A and B: $\{2, 4\}$.

2. A only: $\{1\}$.

3. B only: $\{3\}$.

4. Outside both: $\{5, 6\}$.

Answer: $A \cap B = \{2, 4\}$; $A \cup B = \{1, 2, 3, 4\}$; $\text{neither} = \{5, 6\}$.

Common Mistakes and Fixes

Common mistake	How to fix it
Adding set totals without subtracting overlap	Use $A + B - \text{both}$.
Subtracting overlap twice from the union	It should appear once in the final union.
Putting neither outside the rectangle	Neither is outside both circles but inside universal set U.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. $n(A)=10$, $n(B)=8$, $n(A \cap B)=3$. Find union.

Check: $10+8-3=15$.

2. In $U=20$, union=15. Find neither.

Check: 5.

3. A total 14 includes 4 in both. Find A only.

Check: 10.

4. What does $A \cap B$ mean?

Check: Elements belonging to both A and B.

5. Why is the overlap filled first?

Check: It is included in both set totals, so it must be removed before finding each only region.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. $n(A)=16$, $n(B)=12$, intersection=5. Find union.

Working: _____

2. $U=40$ and union=31. Find neither.

Working: _____

3. A has 20 total and 7 in both. Find A only.

Working: _____

4. Which notation means union? A. $A \cap B$ B. $A \cup B$ C. A' D. U'

Working: _____

5. If A and B have no common elements, what is $n(A \cap B)$?

Working: _____

6. $n(A)=25$, $n(B)=18$, union=35. Find intersection.

Working: _____

7. $U=50$, A only=14, both=9, B only=16. Find neither.

Working: _____

8. Using the previous regions, find $n(A)$.

Working: _____

9. Using the previous regions, find $n(B)$.

Working: _____

10. Explain inclusive 'or' in $A \cup B$.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

1: 23.

2: 9.

3: 13.

4: B. AUB.

5: 0; the sets are disjoint.

6: $25+18-35=8$.

7: $50-(14+9+16)=11$.

8: A only+both= $14+9=23$.

9: B only+both= $16+9=25$.

10: It includes elements in A only, B only and both sets.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Of 60 students, 38 study Spanish, 29 study French and 14 study both. Find Spanish only, French only, at least one and neither. [8 marks]

Working space: _____

2. In a survey of 80 households, 52 have internet, 31 have cable and 9 have neither. Find both. [7 marks]

Working space: _____

3. A Venn diagram has A only= $x+3$, both= $2x$, B only= $x-1$ and neither= 6 in $U=40$. Find x and all regions. [9 marks]

Working space: _____

4. $U=\{\text{whole numbers 1 to 12}\}$, $A=\{\text{multiples of 2}\}$, $B=\{\text{multiples of 3}\}$. List all four regions. [9 marks]

Working space: _____

5. A student uses $n(A \cup B) = n(A) + n(B)$ and gets 46 when $n(A)=27, n(B)=19, \text{both}=6$. Explain and correct. [6 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 8 marks

Model solution: Spanish only=24; French only=15; union= $38+29-14=53$; neither=7. (8)

Question 2 — 7 marks

Model solution: Union= $80-9=71$. Both= $52+31-71=12$. (7)

Question 3 — 9 marks

Model solution: $(x+3)+2x+(x-1)+6=40 \rightarrow 4x+8=40 \rightarrow x=8$. Regions: A only 11, both 16, B only 7, neither 6. (9)

Question 4 — 9 marks

Model solution: Both {6,12}; A only {2,4,8,10}; B only {3,9}; neither {1,5,7,11}. (9)

Question 5 — 6 marks

Model solution: The 6 in both were counted twice. Subtract once: union= $27+19-6=40$. (6)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: Fill intersection first.

Remember: A only = A total - both; B only = B total - both.

Remember: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Remember: Neither = U - union.

Remember: All four regions must total U.

Final Self-Check

- ☐ I can place elements in a two-set Venn diagram.
- ☐ I can calculate intersection, union, only and neither.
- ☐ I can solve word problems without double-counting.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

TERM 3 • LESSON 11

Propositions, Logical Connectives and Truth Tables

Unit 3: Numbers | Logic

3 worked	5 guided	10 independent	5 exam-style
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Lesson Alignment and Purpose

Lesson detail	Information
Subject	Mathematics
Grade / term	Grade 8 / Term 3
Unit	Unit 3: Numbers
Strand	Logic
Lesson	11: Propositions, Logical Connectives and Truth Tables
NSC objective(s)	Understand propositions and use negation, conjunction, disjunction, conditional and equivalence. • Differentiate simple and compound propositions. • Express propositions symbolically and in words.
Suggested pacing	2–4 sessions of 45–60 minutes, adjusted for learner readiness

Why this matters: Logic helps students test arguments, follow conditions and communicate precisely in mathematics, computing and everyday decisions.

Learning Objectives — I Can

- I can decide whether a sentence is a proposition.
- I can translate simple and compound statements into symbols.
- I can complete truth tables for common connectives.

Success Check

- I can explain the key idea in my own words.
- I can complete a familiar question without help.
- I can show enough working for another person to follow.
- I can find and correct a mistake using the cheat sheet.

Quick Prerequisite Check

Directions: Try these first. They warm up useful skills; they do not decide whether you can learn today's topic.

1. What is 7×8 ?
2. What is the value of the digit 4 in 45,230?
3. Why should you show your working?

Prerequisite Answers

- 1: $7 \times 8 = 56$.
- 2: The 4 is worth 40,000.
- 3: Working helps you find mistakes and can earn method marks.

Words to Know

Word	Simple meaning
Proposition	A statement that is either true or false.
Negation	The opposite truth claim, written $\neg p$.
Conjunction	The statement p and q, written $p \wedge q$.
Disjunction	The inclusive statement p or q, written $p \vee q$.
Conditional	The statement if p then q, written $p \rightarrow q$.
Equivalence	The statement p if and only if q, written $p \leftrightarrow q$.

The Main Idea

Slow down: Read one paragraph at a time. After each paragraph, say the idea aloud without looking.

A proposition must make a definite claim with a truth value. Questions, commands and open sentences such as $x+2=5$ without a stated x are not propositions.

$\neg p$ reverses p . $p \wedge q$ is true only when both are true. Inclusive $p \vee q$ is false only when both are false.

$p \rightarrow q$ is false only when p is true and q is false. $p \leftrightarrow q$ is true when p and q have matching truth values. A compound proposition joins or modifies simpler ones.

Small Steps to Follow

1. Name each simple proposition with a letter such as p or q .
2. Identify words such as not, and, or, if...then, and if and only if.
3. Translate in the original order using $\neg, \wedge, \vee, \rightarrow$ or $\leftrightarrow$.
4. For two variables, list TT, TF, FT, FF.
5. Apply the connective rule row by row, then translate the result back into clear words.

Notebook visual: Draw a five-column truth-rule card: $\neg$ flips; $\wedge$ needs both; $\vee$ needs at least one; $\rightarrow$ fails only $T \rightarrow F$; $\leftrightarrow$ needs matching values.

Let's Try Together — Fully Worked Examples

Learning routine: Cover the answer. Read one step, copy it, and explain why it is allowed before moving to the next step.

Worked Example 1

Problem: Let p : '8 is even' and q : '8 is greater than 10'. Find truth values of $p \wedge q$ and $p \vee q$.

1. p is True.
2. q is False.
3. $T \wedge F$ is False because both are not true.
4. $T \vee F$ is True because at least one is true.

Answer: $p \wedge q$ is False; $p \vee q$ is True.

Worked Example 2

Problem: Write the negation of 'All Grade 8 students are present.'

1. Negation must deny the entire universal claim.
2. It is enough for at least one Grade 8 student to be absent.

3. Do not write 'All are absent,' which is stronger than needed.

Answer: At least one Grade 8 student is not present.

Worked Example 3

Problem: Complete $p \rightarrow q$ for rows TT, TF, FT, FF.

1. $T \rightarrow T$ is True.
2. $T \rightarrow F$ is False: the condition happened but the conclusion failed.
3. $F \rightarrow T$ is True under the logical rule.
4. $F \rightarrow F$ is True under the logical rule.

Answer: $p \rightarrow q$ column: T, F, T, T.

Common Mistakes and Fixes

Common mistake	How to fix it
Treating questions as propositions	A proposition is a declarative claim with a truth value.
Using exclusive everyday 'or'	Mathematical $\vee$ is inclusive unless stated otherwise.
Making every false antecedent conditional false	$p \rightarrow q$ is false only for $T \rightarrow F$.

Your Turn — Guided Practice

Directions: Try each question, then check immediately. If it is wrong, correct the first wrong step before continuing.

1. Is 'Close the door' a proposition?

Check: No; it is a command, not a true-or-false claim.

2. If p is True, what is $\neg p$?

Check: False.

3. Evaluate $T \wedge F$.

Check: False.

4. Evaluate FVT using inclusive or.

Check: True.

5. When is $p \leftrightarrow q$ true?

Check: When p and q have the same truth value.

Show What You Know — Independent Practice

Exam habit: Do all ten questions before reading the answer key. Show working even when you can calculate mentally.

1. Is 'Kingston is in Jamaica' a proposition? Give truth value.

Working: _____

2. Is 'What time is it?' a proposition?

Working: _____

3. If $p=F$, evaluate $\neg p$.

Working: _____

4. Which symbol means conjunction? A. $\vee$ B. $\wedge$ C. $\neg$ D. $\rightarrow$

Working: _____

5. Evaluate $T \vee F$.

Working: _____

6. Evaluate $T \rightarrow F$.

Working: _____

7. Evaluate $F \rightarrow F$.

Working: _____

8. Evaluate $T \leftrightarrow T$ and $T \leftrightarrow F$.

Working: _____

9. Let p ='It rains' and q ='School is open.' Translate $\neg p \vee q$.

Working: _____

10. Is $p \wedge (q \vee \neg p)$ simple or compound? Explain.

Working: _____

Independent Practice — Full Answer Key

For a one-step fact or definition, the direct statement is the complete response. For calculations, follow the displayed method and units.

- 1: Yes; True.
- 2: No; it is a question.
- 3: T.
- 4: B. $\wedge$.
- 5: T.
- 6: F.
- 7: T.
- 8: $T \leftrightarrow T$ is T; $T \leftrightarrow F$ is F.
- 9: It does not rain, or school is open (inclusive or).
- 10: Compound; it uses several propositions/connectives.

Test Prep — Exam-Style Questions

Timing: Work without notes first. Read command words, underline given values, show every method and check units.

1. Let p be True and q be False. Evaluate $\neg p$, $p \wedge q$, $p \vee q$, $p \rightarrow q$ and $p \leftrightarrow q$, with one rule for each. [10 marks]

Working space: _____

2. Construct the full truth table for $(p \wedge q) \rightarrow p$. [10 marks]

Working space: _____

3. Let p = 'A number is divisible by 4' and q = 'It is even.' Express 'If a number is divisible by 4, then it is even,' give symbols and assess with a counterexample search. [8 marks]

Working space: _____

4. Negate 'If it rains, then the match is cancelled' accurately. [7 marks]

Working space: _____

5. Compare ' p or q ' with ' p if and only if q ' by completing their output columns for TT,TF,FT,FF. [8 marks]

Working space: _____

Exam-Style Marking Guide and Reasoning

Award method marks for correct setup and logical working even if a later arithmetic slip occurs.

Question 1 — 10 marks

Model solution: $\neg p = F$ (flip); $p \wedge q = F$ (both needed); $p \vee q = T$ (at least one); $p \rightarrow q = F$ (only failing case $T \rightarrow F$); $p \leftrightarrow q = F$ (values differ). (10)

Question 2 — 10 marks

Model solution: Rows p, q : TT, TF, FT, FF. $p \wedge q$: T, F, F, F. Final implication: T, T, T, T. It is always true because whenever $p \wedge q$ is true, p is true; when antecedent is false, implication is true. (10)

Question 3 — 8 marks

Model solution: Symbol $p \rightarrow q$. The statement is true for whole numbers: every multiple of 4 is even. A counterexample would need divisible by 4 but odd; none exists. (8)

Question 4 — 7 marks

Model solution: A conditional $p \rightarrow q$ is false exactly when $p \wedge \neg q$. Negation: It rains and the match is not cancelled. (7)

Question 5 — 8 marks

Model solution: $p \vee q$: T, T, T, F. $p \leftrightarrow q$: T, F, F, T. Or needs at least one true; equivalence needs matching values. (8)

One-Page Summary and Cheat Sheet

Use it well: Read this page before a quiz, then close it and write the key rules from memory.

Remember: $\neg$ flips truth.

Remember: $\wedge$ true only TT; $\vee$ false only FF.

Remember: $\rightarrow$ false only T $\rightarrow$ F.

Remember: $\leftrightarrow$ true when values match.

Remember: Two-variable row order: TT, TF, FT, FF.

Final Self-Check

- ☐ I can decide whether a sentence is a proposition.
- ☐ I can translate simple and compound statements into symbols.
- ☐ I can complete truth tables for common connectives.
- ☐ I corrected every missed question. ☐ I can teach one example to someone else.

Encouragement: Mistakes show what to practise next. Review one step, try one similar question, and keep going—you can do this!

Curriculum Sources and Editorial Note

Objective coverage and term placement were reconciled against the sources below on 10 August 2026. This resource is independently authored and is not an official Ministry publication or endorsement.

[Ministry of Education and Youth — Secondary Curriculum](#)

[Ministry of Education and Youth — Grade 7–9 Mathematics NSC listing](#)

[Grade 8 Mathematics Curriculum Guide / objective outline](#)

[IXL — Jamaica NSC Grade 8 APSE I skill-plan cross-check](#)

Editorial Principles

Language is intentionally direct, examples are scaffolded, Jamaican-dollar and familiar local contexts are used where appropriate, and assessment difficulty increases gradually. Financial examples use fictional terms and rates supplied inside each question; real fees, rates and tax rules can change.

End of volume: Return to missed questions, not just the notes. Retrieval and correction build exam readiness.